

Casual Document for Probabilistic Principal Component Analysis MLE Asymptotic Normality

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Chapter 1

Low Dimensional Asymptotic Normality

1.1 Basic Setup

Given a "high dimensional" data set $\{\mathbf{x}_i\}_{i=1}^n$ where $\mathbf{x}_i \in \mathbb{R}^p$ for all $i = 1, \dots, n$, the statistician fixes a positive integer $q < p$ and wishes to find the q principal components. Assuming that the data is centred (that is, with mean 0), the probabilistic principal component analysis model is a statistical model that treats $\mathbf{x}_i$'s as realizations of independent random variables $\mathbf{X}_i = \mathbf{W}\mathbf{Z}_i + \boldsymbol{\epsilon}_i$, where $\mathbf{W}$ is an unknown $p \times q$ loading matrix of full rank, $\mathbf{Z}_i \sim N_q(\mathbf{0}, \mathbf{I}_q)$ is the latent normal variable, and $\boldsymbol{\epsilon}_i \sim N_p(\mathbf{0}, \sigma^2 \mathbf{I}_p)$ is the Gaussian noise with unknown standard deviation $\sigma > 0$, and $(\mathbf{Z}_1, \dots, \mathbf{Z}_n, \boldsymbol{\epsilon}_1, \dots, \boldsymbol{\epsilon}_n)$ are all independent (for now). Put $\theta = (\mathbf{W}_\theta, \sigma_\theta^2)$ for the unknown parameter to estimate and Θ for all possible choices of θ . Θ is thus the collection of all tuples $(\mathbf{W}, \sigma^2)$ such that $\mathbf{W}$ is a real $p \times q$ matrix with $\text{rank}(\mathbf{W}) = q$ and $\sigma^2 \in (0, \infty)$. Given that the collection of all real $p \times q$ matrix with rank q is an open subset of the Euclidean space $\mathbb{R}^{pq}$, we consider Θ as an open subset of the Euclidean space $\mathbb{R}^{pq+1}$ and our statistical model is

$$\mathcal{P} = \{N_p(\mathbf{0}, \mathbf{W}_\theta \mathbf{W}_\theta^T + \sigma_\theta^2 \mathbf{I}_p)\}_{\theta \in \Theta}. \quad (1.1)$$

To resolve the non-identifiability of $\mathcal{P}$, we introduce the quotient manifold structure. For $(\mathbf{A}, s) \in \Theta$ and $(\mathbf{B}, t) \in \Theta$, we define $(\mathbf{A}, s) \sim (\mathbf{B}, t)$ to mean that they give rise to the same covariance matrix, namely when $\mathbf{A}\mathbf{A}^T + s\mathbf{I}_p = \mathbf{B}\mathbf{B}^T + t\mathbf{I}_p$. The equivalence classes are denoted by $[(\mathbf{A}, s)]$ or more briefly $[\mathbf{A}, s]$. Unless otherwise mentioned, $(\mathbf{A}, s) \in \Theta$ is the default representative we pick when we write an equivalence class as $[\mathbf{A}, s]$. Writing $O(q)$ for the collection of all $q \times q$ real orthogonal matrices, it was shown in Datta's thesis that $[\mathbf{A}, s] = \{(\mathbf{A}\mathbf{Q}, s) \mid \mathbf{Q} \in O(q)\}$. Therefore, $M := \Theta / \sim$ can be identified as the orbits of the right action of $O(q)$ on Θ and, given that this Lie group action is smooth, proper, and free, the quotient manifold theorem (see [Lee, 2013](#), Theorem 21.10) shows that there is a unique smooth structure making M a manifold of dimension

$$d = pq + 1 - \frac{q(q-1)}{2} \quad (1.2)$$

and the map $\pi : \Theta \rightarrow M$ defined by

$$\pi((\mathbf{A}, s)) = [\mathbf{A}, s] \quad (1.3)$$

is a smooth submersion. Moreover, this action is vertical, transitive on fibers and is isometric with respect to the Euclidean metric. Thus, there exists a unique metric g on M making (M, g) a Riemannian manifold and π a Riemannian submersion (see [Lee, 2018](#), Theorem 2.28). In more concrete terms, if g^Θ is the standard Euclidean metric given by

$$g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)) = \text{tr}(\mathbf{V}\mathbf{U}^T) + cb, \quad (1.4)$$

whenever $(\mathbf{A}, s) \in \Theta$, $(\mathbf{V}, c), (\mathbf{U}, b) \in T_{(\mathbf{A}, s)}\Theta \cong \mathbb{R}^{p \times q} \times \mathbb{R}$, then

$$g_{[\mathbf{A}, s]}(\mathbf{v}, \mathbf{h}) = g_{[\mathbf{A}, s]}(d\pi_{(\mathbf{A}, s)}((\mathbf{V}, c)), d\pi_{(\mathbf{A}, s)}((\mathbf{U}, b))) = g_{(\mathbf{A}, s)}^\Theta((\mathbf{V}, c), (\mathbf{U}, b)), \quad (1.5)$$

whenever $\mathbf{v}, \mathbf{h} \in T_{[\mathbf{A}, s]}M$ lifting to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ respectively. For later convenience, let us define two functions $\Phi : \Theta \rightarrow \mathbb{R}^{p \times p}$ and $\Sigma : M \rightarrow \mathbb{R}^{p \times p}$ by setting, for every $[\mathbf{A}, s] \in M$:

$$\Sigma([\mathbf{A}, s]) := \Phi([\mathbf{A}, s]) := \mathbf{A}\mathbf{A}^T + s\mathbf{I}_p. \quad (1.6)$$

In other words, $\Sigma \circ \pi = \Phi$ gives the covariance matrix. (1.1) is then reparametrized as

$$\mathcal{P} = \{N_p(\mathbf{0}, \Sigma([\mathbf{A}, s]))\}_{[\mathbf{A}, s] \in M} \quad (1.7)$$

and this is the identifiable model we shall work with.

Finally, throughout the entire document, our default norm for a matrix $\mathbf{A}$ will be the spectral norm, denoted by $\|\mathbf{A}\|$. We will also make use of the Frobenius norm $\|\mathbf{A}\|_F$ as well. The eigenvalues and the singular values of a matrix will always be labelled in decreasing order. In symbols, if $\mathbf{A}$ is a matrix of size $p \times q$, then we denote its q singular values as

$$\|\mathbf{A}\| = \sigma_1(\mathbf{A}) \geq \sigma_2(\mathbf{A}) \geq \cdots \geq \sigma_q(\mathbf{A}) = \sigma_{\min}(\mathbf{A}) \geq 0$$

and, if $\mathbf{S}$ is a symmetric matrix of size $p \times p$, then we denote its p eigenvalues as

$$\lambda_1(\mathbf{S}) \geq \lambda_2(\mathbf{S}) \geq \cdots \geq \lambda_p(\mathbf{S}).$$

1.2 Basic Facts about (M, g)

Working with Riemannian submersion, we shall as usual start with computations regarding the horizontal and vertical tangent spaces.

Proposition 1.

1. The map Σ defined in (1.6) is a smooth injective immersion.
2. At every $(\mathbf{A}, s) \in \Theta$, the vertical tangent space

$$V_{(\mathbf{A}, s)} := \ker(d\pi_{(\mathbf{A}, s)}) = \ker(d\Phi_{(\mathbf{A}, s)}) \quad (1.8)$$

equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A}, s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

3. At every $(\mathbf{A}, s) \in \Theta$, the horizontal tangent space $H_{(\mathbf{A}, s)} := V_{(\mathbf{A}, s)}^\perp$ equals to the collection of all tangent vectors of the form $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$ ranged over all $c \in \mathbb{R}$ and all $p \times q$ matrices $\mathbf{V}$ satisfying $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$.

Proof. Fix a point $(\mathbf{A}, s) \in \Theta$ throughout this proof. Before we start the proof, let us first study the set $\ker(d\Phi_{(\mathbf{A}, s)})$, where Φ is given in (1.6). For $(\mathbf{V}, c) \in T_{(\mathbf{A}, s)}\Theta$, define a curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ by setting $\gamma(t) = (\mathbf{A} + t\mathbf{V}, s + tc)$. Then:

$$d\Phi_{(\mathbf{A}, s)}((\mathbf{V}, c)) = \left. \frac{d}{dt} \right|_{t=0} \Phi(\gamma(t)) = \mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p.$$

If $(\mathbf{V}, c) \in \ker(d\Phi_{(\mathbf{A}, s)})$, then

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p = \mathbf{0}. \quad (1.9)$$

If $\mathbf{u} \in \mathbb{R}^p$ is any nonzero vector orthogonal to all column vectors of $\mathbf{A}$, then $\mathbf{A}^T \mathbf{u} = \mathbf{0}$ and

$$0 = \mathbf{u}^T (\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T + c\mathbf{I}_p) \mathbf{u} = c \mathbf{u}^T \mathbf{u}.$$

Given that $\mathbf{u} \neq \mathbf{0}$, we must have $c = 0$ and (1.9) reduces to

$$\mathbf{A}\mathbf{V}^T + \mathbf{V}\mathbf{A}^T = \mathbf{0}. \quad (1.10)$$

Apply $\mathbf{u}$ to the right to get:

$$\mathbf{A}\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Given that $\mathbf{A}$ has full rank, rank nullity theorem forces

$$\mathbf{V}^T \mathbf{u} = \mathbf{0}.$$

Therefore, we have proven that every vector $\mathbf{u}$ orthogonal to columns vectors of $\mathbf{A}$ will also be orthogonal to column vectors of $\mathbf{V}$. This means $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some $q \times q$ matrix $\mathbf{\Omega}$. Substitute this back to (1.10) gives

$$\mathbf{A}(\mathbf{\Omega}^T + \mathbf{\Omega})\mathbf{A}^T = \mathbf{0}.$$

Applying $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$ to the left and $\mathbf{A}(\mathbf{A}^T \mathbf{A})^{-1}$ to the right yields $\mathbf{\Omega}^T + \mathbf{\Omega} = \mathbf{0}$. Therefore, $\mathbf{\Omega}$ is skew-symmetric. Conversely, (1.10) is satisfied by $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ with any skew-symmetric matrix $\mathbf{\Omega}$. We have proven that $\ker(\mathrm{d}\Phi_{(\mathbf{A},s)})$ equals to the collection of all tangent vectors of the form $(\mathbf{A}\mathbf{\Omega}, 0) \in T_{(\mathbf{A},s)}\Theta$, with $\mathbf{\Omega}$ ranged over all $q \times q$ skew symmetric matrices.

We now move back to prove 1 and 2 at the same time by showing (1.8). Clearly, Σ is well-defined and is injective. The smoothness of Σ is guaranteed by relation $\Sigma \circ \pi = \Phi$ (see for example Lee, 2013, Theorem 4.29). By the chain rule, we have

$$\mathrm{d}\Phi_{(\mathbf{A},s)} = \mathrm{d}\Sigma_{[\mathbf{A},s]} \circ \mathrm{d}\pi_{(\mathbf{A},s)}. \quad (1.11)$$

It follows immediately that $\ker(\mathrm{d}\pi_{(\mathbf{A},s)}) \subseteq \ker(\mathrm{d}\Phi_{(\mathbf{A},s)})$. The other direction will follow if we can prove that $\mathrm{d}\Sigma_{[\mathbf{A},s]}$ is injective.

Suppose that $\mathbf{v} \in T_{[\mathbf{A},s]}M$ is such that $\mathrm{d}\Sigma_{[\mathbf{A},s]}(\mathbf{v}) = \mathbf{0}$. We prove that $\mathbf{v} = \mathbf{0}$. Let $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$ be the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$, meaning that $(\mathbf{V}, c)$ is the unique horizontal tangent vector at $(\mathbf{A}, s)$ such that $\mathbf{v} = \mathrm{d}\pi_{(\mathbf{A},s)}((\mathbf{V}, c))$. (1.11) then implies that $(\mathbf{V}, c) \in \ker(\mathrm{d}\Phi_{(\mathbf{A},s)})$ and the preceding discussion shows that $c = 0$ and $\mathbf{V} = \mathbf{A}\mathbf{\Omega}$ for some skew-symmetric $\mathbf{\Omega}$. To proceed, we look at the curve $\gamma : (-\epsilon, \epsilon) \rightarrow \Theta$ defined by $\gamma(t) = (\mathbf{A}e^{t\mathbf{\Omega}}, s)$. Clearly $\gamma(0) = (\mathbf{A}, s)$ and $\gamma'(0) = (\mathbf{A}\mathbf{\Omega}, 0)$. Moreover, the matrix $e^{t\mathbf{\Omega}}$ is always orthogonal for any t and any skew-symmetric $\mathbf{\Omega}$ so that $\pi(\gamma(t)) = [\mathbf{A}, s]$ for all t . We have

$$\mathbf{v} = \mathrm{d}\pi_{(\mathbf{A},s)}((\mathbf{A}\mathbf{\Omega}, 0)) = \mathrm{d}\pi_{(\mathbf{A},s)}(\gamma'(0)) = \left. \frac{d}{dt} \right|_{t=0} \pi(\gamma(t)) = \mathbf{0} \in T_{[\mathbf{A},s]}M.$$

Therefore, $\mathrm{d}\Sigma_{[\mathbf{A},s]}$ is injective and the vertical tangent space is the same as $\ker(\mathrm{d}\Phi_{(\mathbf{A},s)})$. Since $[\mathbf{A}, s]$ is arbitrary, Σ is an immersion.

To compute the horizontal tangent space, any $(\mathbf{V}, c) \in V_{(\mathbf{A},s)}^\perp$ will satisfy

$$0 = g_{(\mathbf{A},s)}^\Theta((\mathbf{V}, c), (\mathbf{A}\mathbf{\Omega}, 0)) = \mathrm{tr}(\mathbf{A}^T \mathbf{V} \mathbf{\Omega}) \quad (1.12)$$

for all skew-symmetric $\mathbf{\Omega}$. Decomposing $\mathbf{A}^T \mathbf{V} = \mathbf{S} + \mathbf{K}$, where $\mathbf{S}$ is the symmetric part and $\mathbf{K}$ is the skew-symmetric part, we find out that

$$0 = \mathrm{tr}(\mathbf{S}\mathbf{\Omega}) + \mathrm{tr}(\mathbf{K}\mathbf{\Omega}) = 0 + \mathrm{tr}(\mathbf{K}\mathbf{\Omega}) = \mathrm{tr}(\mathbf{K}\mathbf{\Omega}).$$

Setting $\mathbf{\Omega} = \mathbf{K}^T$ will yield that $\mathbf{K} = \mathbf{0}$. Therefore, $\mathbf{A}^T \mathbf{V}$ is a symmetric matrix. Conversely, any $\mathbf{V}$ such that $\mathbf{A}^T \mathbf{V}$ is symmetric will satisfy (1.12). The proof is complete. $\square$

We now move to compute the exponential map and the logarithmic map. The following lemma is crucial for computing the orthogonal matrix based on given points in M .

Lemma 2. If L is an invertible $q \times q$ matrix and $Q \in O(q)$ are such that LQ is symmetric, then there exists matrices $R \in O(q)$, $P \in O(q)$, a $q \times q$ diagonal matrix D whose diagonal entries are 1 or -1 , and a $q \times q$ diagonal matrix S with diagonal entries $\sigma_1(L) \geq \dots \geq \sigma_q(L) > 0$ such that $L = RSP^T$ and $Q = PDR^T$.

If we add furthermore the assumption that L is symmetric positive definite, then we can achieve the results mentioned above with $P = R$. In particular, this implies $Q = Q^T$ and $\|Q - I_q\| = \|D - I_q\|$ can take only 0 or 2 as values.

Proof. Let us start by applying the singular value decomposition of L to get matrices $R' \in O(q)$, $P' \in O(q)$ and a diagonal matrix S with diagonal entries $\sigma_1(L) \geq \dots \geq \sigma_q(L) > 0$ such that $L = R'S(P')^T$. Note that all singular values are positive because L is invertible. Substituting the given condition that LQ is symmetric will yield that $SJ = J^T S$, where $J := (P')^T Q R'$.

Clearly $J \in O(q)$. The equation

$$S^2 = SJJ^T S = (SJ)(SJ)^T = (J^T S)(J^T S)^T = J^T S^2 J$$

shows that

$$JS^2 = S^2 J.$$

Looking at each entry of both side, we end up with

$$J_{ij}(\sigma_j(L)^2 - \sigma_i(L)^2) = 0$$

for $i, j = 1, \dots, q$. Given that all singular values are positive, we in fact have

$$J_{ij}(\sigma_j(L) - \sigma_i(L)) = 0 \quad (1.13)$$

for all i and all j so that $JS = SJ = J^T S$. Inverting S yields that J is actually symmetric. Being a symmetric orthogonal matrix, $J^2 = I_q$ and the eigenvalues of J must be 1 or -1 . Another look at (1.13) shows that J is a block-diagonal matrix, diagonal if all singular values are distinct. Furthermore, each block of J must still be orthogonal and symmetric.

We now diagonalize J based on the above information. Suppose that $S = \bigoplus_{i=1}^k s_i I_{m_i}$, where $s_1 > \dots > s_k$ are the unique diagonal entries of S and each repeats for m_i times. Thus, J has k blocks. Write the i th block by J_i so that $J = \bigoplus_{i=1}^k J_i$. The spectral theorem then allows us to orthogonally diagonalize $J_i = K_i D_i K_i^T$, where $K_i \in O(m_i)$ and D_i is an $m_i \times m_i$ diagonal matrix with all diagonal entries being 1 or -1 . Put $K = \bigoplus_{i=1}^k K_i$ and $D = \bigoplus_{i=1}^k D_i$. Then $K \in O(q)$, D is a $q \times q$ diagonal matrix with 1 or -1 diagonal entry, and $J = KDK^T$. This involved construction leads us to the outcome that K in fact commutes with S :

$$KS = \bigoplus_{i=1}^k s_i K_i = SK.$$

Put $P = P'K$, $R = R'K$. We have

$$Q = P'J(R')^T = P'KDK^T(R')^T = PDR^T$$

and

$$RSP^T = R'KSK^T(P')^T = R'SKK^T(P')^T = R'S(P')^T = L.$$

This completes the proof.

Finally, let us just note that, in case L is symmetric positive definite, the spectral theorem allows us to take $R' = P'$ from the start and it will yield in the end $L = PSP^T$ and $Q = PDP^T$. It then follows easily that $Q^2 = I_q$ and $Q^T = Q$. If $D = I_q$, then $\|Q - I_q\| = \|D - I_q\| = 0$. If any entry of D is -1 , $\|D - I_q\| = 2$. $\square$

We now state and prove all the results about the exponential and the logarithmic map in one single proposition.

Proposition 3. Given $[\mathbf{A}, s] \in M$ and $\mathbf{v} \in T_{[\mathbf{A}, s]}M$, geodesics in M starting at $[\mathbf{A}, s]$ with initial velocity $\mathbf{v}$ takes the format of curves $\gamma : I \rightarrow M$, where I is an open interval containing 0, defined by $\gamma(t) = [\mathbf{A} + t\mathbf{V}, s + tc]$, where $(\mathbf{V}, c) \in H_{(\mathbf{A}, s)}$ is the horizontal lift of $\mathbf{v}$ at $(\mathbf{A}, s)$. Consequently, the exponential map is computed as

$$\text{Exp}_{[\mathbf{A}, s]}(\mathbf{v}) = [\mathbf{A} + \mathbf{V}, s + c]. \quad (1.14)$$

Let us use $\mathcal{E}_{[\mathbf{A}, s]}$ to denote the open ball in $T_{[\mathbf{A}, s]}M \cong \mathbb{R}^d$ centred at $\mathbf{0}$ of radius $\min(\sigma_{\min}(\mathbf{A}), s)$. Given that (1.14) is well-defined at least over all $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$, we from now on treat $\mathcal{E}_{[\mathbf{A}, s]}$ as the domain of the exponential map. That is, $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$, where $U_{[\mathbf{A}, s]} := \text{Exp}_{[\mathbf{A}, s]}(\mathcal{E}_{[\mathbf{A}, s]})$ is an open subset of M , is defined by (1.14). Note the following facts:

1. For every $\mathbf{v} \in \mathcal{E}_{[\mathbf{A}, s]}$, $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$ is symmetric positive definite.
2. $\text{Exp}_{[\mathbf{A}, s]}$ is a diffeomorphism.
3. If $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$, then for every representative $(\mathbf{A}, s)$ of $[\mathbf{A}, s]$ and every representative $(\mathbf{B}, t)$ of $[\mathbf{B}, t]$, the matrix $\mathbf{A}^T \mathbf{B}$ will always be invertible. Given representative $(\mathbf{A}, s)$ of $[\mathbf{A}, s]$, some representative $(\mathbf{B}, t)$ of $[\mathbf{B}, t]$ will furthermore make $\mathbf{A}^T \mathbf{B}$ symmetric positive definite.
4. The logarithmic map $\text{Exp}_{[\mathbf{A}, s]}^{-1} : U_{[\mathbf{A}, s]} \rightarrow \mathcal{E}_{[\mathbf{A}, s]}$ is computed as follows: for every $[\mathbf{B}, t] \in U_{[\mathbf{A}, s]}$,

$$\text{Exp}_{[\mathbf{A}, s]}^{-1}([\mathbf{B}, t]) = d\pi_{(\mathbf{A}, s)}((\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}, t - s)), \quad (1.15)$$

where

$$\mathbf{Q}(\mathbf{A}, \mathbf{B}) = (\mathbf{B}^T \mathbf{A} \mathbf{A}^T \mathbf{B})^{-\frac{1}{2}} \mathbf{B}^T \mathbf{A} \in O(q). \quad (1.16)$$

5. $U_{[\mathbf{A}, s]}$ equals to the set of all $[\mathbf{B}, t] \in M$ such that $\mathbf{A}^T \mathbf{B}$ is invertible and

$$\|\mathbf{B}\mathbf{Q}(\mathbf{A}, \mathbf{B}) - \mathbf{A}\|_F^2 + (t - s)^2 < \min(\sigma_{\min}(\mathbf{A}), s)^2. \quad (1.17)$$

Proof. Before we start the proof, let us first check that all the statements are indeed well-defined. First of all, the curve γ is well defined. If $(\mathbf{A}\mathbf{Q}, s)$, where $\mathbf{Q} \in O(q)$, is another representative of $[\mathbf{A}, s]$, let $(\mathbf{V}', c')$ be the horizontal lift of $\mathbf{v}$ to $(\mathbf{A}\mathbf{Q}, s)$. With the function $R_{\mathbf{Q}} : \Theta \rightarrow \Theta$ defined by $R_{\mathbf{Q}}([\mathbf{A}, s]) = (\mathbf{A}\mathbf{Q}, s)$, we have

$$\mathbf{v} = d\pi_{(\mathbf{A}, s)}(\mathbf{V}, c) = d(\pi \circ R_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(dR_{\mathbf{Q}})_{(\mathbf{A}, s)}(\mathbf{V}, c) = d\pi_{(\mathbf{A}\mathbf{Q}, s)}(\mathbf{V}\mathbf{Q}, c).$$

Thus, $\mathbf{V}' = \mathbf{V}\mathbf{Q}$ and $c' = c$. Substitute this into the definition of γ will yield the exact same curve. Secondly, if $\mathbf{R} \in O(q)$ and $\mathbf{P} \in O(q)$, then

$$\mathbf{Q}(\mathbf{A}\mathbf{P}, \mathbf{B}\mathbf{R}) = \mathbf{R}^T \mathbf{Q}(\mathbf{A}, \mathbf{B}) \mathbf{P}. \quad (1.18)$$

Therefore, the same computation will show that (1.15) is indeed well-defined. Also, the contents of the inequality (1.17) is independent of representatives and the description of the set $U_{[\mathbf{A}, s]}$ makes sense.

A well-known property of geodesics in case of Riemannian submersion (see for example O'Neill, 1967, Section 2 Corollary 2 or Valencia, 2021, Theorem 3.6) is that any geodesic c in Θ with initial velocity being horizontal will always have horizontal velocities and $\gamma = \pi \circ c$ will remain a geodesic in M . Therefore, (1.14) is valid for all vectors $\mathbf{v} \in T_{[\mathbf{A}, s]}M$ of small enough norm making sure that $(\mathbf{A} + \mathbf{V}, s + c) \in \Theta$. To have a quantified upper bound of the norm, we use the classical result of Weyl's perturbation inequality (see for example Chapter 7.3 Problem 16 equation (7.3.15) of Horn and Johnson, 2013) to get

$$\sigma_{\min}(\mathbf{A} + \mathbf{V}) \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\| \geq \sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F \geq \sigma_{\min}(\mathbf{A}) - |\mathbf{v}| > 0,$$

whenever $|v| < \min(\sigma_{\min}(\mathbf{A}), s)$ or equivalently $v \in \mathcal{E}_{[\mathbf{A}, s]}$. We have made use of the two facts that $\|\mathbf{V}\| \leq \|\mathbf{V}\|_F$ for any matrix $\mathbf{V}$ and that $|v| = \sqrt{\|\mathbf{V}\|_F^2 + c^2}$ by isometry. Thus, for all $v \in \mathcal{E}_{[\mathbf{A}, s]}$, $\text{rank}(\mathbf{A} + \mathbf{V}) = q$ and the exponential map is well-defined.

We now start the proof of the five facts of the function $\text{Exp}_{[\mathbf{A}, s]} : \mathcal{E}_{[\mathbf{A}, s]} \rightarrow U_{[\mathbf{A}, s]}$.

The first fact can be quickly settled. Let $v \in \mathcal{E}_{[\mathbf{A}, s]}$ with horizontal lift $(\mathbf{V}, c)$ to $(\mathbf{A}, s)$ be given. As proved in Proposition 1, $\mathbf{V}^T \mathbf{A} = \mathbf{A}^T \mathbf{V}$ and thus $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$ is symmetric. If $x \in \mathbb{R}^q$ is any unit vector, then

$$x^T \mathbf{A}^T(\mathbf{A} + \mathbf{V})x = |\mathbf{A}x|^2 + (\mathbf{A}x)^T \mathbf{V}x \geq |\mathbf{A}x|(|\mathbf{A}x| - |\mathbf{V}x|) \geq \sigma_{\min}(\mathbf{A})(\sigma_{\min}(\mathbf{A}) - \|\mathbf{V}\|_F) > 0$$

because $\sigma_{\min}(\mathbf{A}) > |v| \geq \|\mathbf{V}\|_F$.

The second fact has a longer proof. Being a well-defined exponential map, $\text{Exp}_{[\mathbf{A}, s]}$ is automatically a smooth map. To prove that it is actually a diffeomorphism in $\mathcal{E}_{[\mathbf{A}, s]}$, we must first show that it is injective in $\mathcal{E}_{[\mathbf{A}, s]}$ so that the inverse function, the logarithmic map $\text{Exp}_{[\mathbf{A}, s]}^{-1}$, exists. Once this is done, we prove that $U_{[\mathbf{A}, s]}$ is open and that $\text{Exp}_{[\mathbf{A}, s]}^{-1}$ is smooth in one shot by proving that its differential $d(\text{Exp}_{[\mathbf{A}, s]})_v$ is an isomorphism for every given $v \in \mathcal{E}_{[\mathbf{A}, s]}$.

We thus move to show that $\text{Exp}_{[\mathbf{A}, s]}$ is injective in $\mathcal{E}_{[\mathbf{A}, s]}$. Suppose that v_1 and v_2 are two vectors in $\mathcal{E}_{[\mathbf{A}, s]}$ who lifts to $(\mathbf{V}_1, c_1)$ and $(\mathbf{V}_2, c_2)$ respectively at $(\mathbf{A}, s)$ such that $\text{Exp}_{[\mathbf{A}, s]}(v_1) = \text{Exp}_{[\mathbf{A}, s]}(v_2)$. We prove that $v_1 = v_2$. (1.14) immediately shows that $c_1 = c_2$ and

$$\mathbf{A} + \mathbf{V}_1 = (\mathbf{A} + \mathbf{V}_2)\mathbf{Q}$$

for some $\mathbf{Q} \in O(q)$. Lemma 2 with the symmetric positive definite matrix $\mathbf{L} = \mathbf{A}^T(\mathbf{A} + \mathbf{V}_2)$ shows that $\|\mathbf{Q} - \mathbf{I}_q\|$ can only equal to 0 or 2. Given that

$$\sigma_{\min}(\mathbf{A})\|\mathbf{Q} - \mathbf{I}_q\| \leq \|\mathbf{A}\mathbf{Q} - \mathbf{A}\| = \|\mathbf{V}_1 - \mathbf{V}_2\mathbf{Q}\| \leq \|\mathbf{V}_1\| + \|\mathbf{V}_2\| < 2\sigma_{\min}(\mathbf{A}),$$

we have proved that $\|\mathbf{Q} - \mathbf{I}_q\| < 2$ and $\mathbf{Q} = \mathbf{I}_q$. Therefore, $\mathbf{V}_1 = \mathbf{V}_2$ and $v_1 = v_2$. The exponential map is indeed injective in $\mathcal{E}_{[\mathbf{A}, s]}$.

Next, let $v \in \mathcal{E}_{[\mathbf{A}, s]}$ be given. We prove that $d(\text{Exp}_{[\mathbf{A}, s]})_v$ is an isomorphism because $\ker(d(\text{Exp}_{[\mathbf{A}, s]})_v)$ consists only of the zero vector. Suppose that v lifts to $(\mathbf{V}, c)$ at $(\mathbf{A}, s)$. To make the task notationally feasible, we implicitly canonically identify the Euclidean spaces together. Let us write $l : T_{[\mathbf{A}, s]}M \rightarrow H_{(\mathbf{A}, s)}$ for the horizontal lift map: $l(v) = (\mathbf{V}, c)$, which is a linear isometry. Define a map $F : l(\mathcal{E}_{[\mathbf{A}, s]}) \rightarrow M$ by setting, for any $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]}) \subseteq H_{(\mathbf{A}, s)}$, that $F((\mathbf{U}, b)) = \pi((\mathbf{A} + \mathbf{U}, s + b))$. In this notation, $\text{Exp}_{[\mathbf{A}, s]} = F \circ l$ and the chain rule shows that

$$d(\text{Exp}_{[\mathbf{A}, s]})_v = dF_{(\mathbf{V}, c)} \circ dl_v.$$

Therefore, the only thing left to show is that $\ker(dF_{(\mathbf{V}, c)})$ contains only the zero vector. For any $(\mathbf{U}, b) \in l(\mathcal{E}_{[\mathbf{A}, s]})$, we always have

$$dF_{(\mathbf{V}, c)}((\mathbf{U}, b)) = \left. \frac{d}{dt} \right|_{t=0} \pi((\mathbf{A} + \mathbf{V} + t\mathbf{U}, s + c + tb)) = d\pi_{(\mathbf{A} + \mathbf{V}, s + c)}((\mathbf{U}, b)).$$

Therefore, if $(\mathbf{U}, b) \in \ker(dF_{(\mathbf{V}, c)})$, then $(\mathbf{U}, b) \in V_{(\mathbf{A} + \mathbf{V}, s + c)}$ so that $b = 0$ and $\mathbf{U} = (\mathbf{A} + \mathbf{V})\mathbf{\Omega}$ for some $q \times q$ skew-symmetric matrix $\mathbf{\Omega}$. But $(\mathbf{U}, b) \in H_{(\mathbf{A}, s)}$ as well. Thus, $\mathbf{A}^T(\mathbf{A} + \mathbf{V})\mathbf{\Omega} = \mathbf{\Omega}^T(\mathbf{A} + \mathbf{V})^T \mathbf{A}$. Given that $\mathbf{A}^T(\mathbf{A} + \mathbf{V})$ is also symmetric, we arrive at

$$\mathbf{A}^T(\mathbf{A} + \mathbf{V})\mathbf{\Omega} + \mathbf{\Omega}\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{0}.$$

By the spectral theorem, choose $\mathbf{P} \in O(q)$ such that $\mathbf{A}^T(\mathbf{A} + \mathbf{V}) = \mathbf{P}\mathbf{S}\mathbf{P}^T$ where $\mathbf{S}$ is the $q \times q$ diagonal matrix with diagonal entries $\lambda_1(\mathbf{A}^T(\mathbf{A} + \mathbf{V})), \dots, \lambda_q(\mathbf{A}^T(\mathbf{A} + \mathbf{V}))$. Substitute this yields

$$S(P^T \Omega P) + (P^T \Omega P)S = 0.$$

Looking at each entry to get

$$(\lambda_i(A^T(A+V)) + \lambda_j(A^T(A+V)))(P^T \Omega P)_{ij} = 0,$$

for all i and all $j \in \{1, \dots, q\}$. Given that $A^T(A+V)$ is positive definite, this forces $P^T \Omega P = 0$ and hence $\Omega = 0$. We have proven that $\text{Exp}_{[A,s]}$ is a diffeomorphism in $\mathcal{E}_{[A,s]}$ and that $U_{[A,s]}$ is an open subset of M .

Now that the logarithmic map $\text{Exp}_{[A,s]}^{-1} : U_{[A,s]} \rightarrow \mathcal{E}_{[A,s]}$ is well-defined and is smooth, we will compute it and show the three last facts in one shot. Let $[B, t] \in U_{[A,s]}$ be given. By (1.14), $[B, t] = [A+V, s+c]$, where (V, c) is the horizontal lift of $v = \text{Exp}_{[A,s]}^{-1}([B, t])$ to (A, s) . It immediately follows that $c = t - s$. The first fact shows that, with $B_0 := A + V$, $A^T B_0$ is symmetric positive definite. Given that the value of $|\det(A^T B)|$ is independent of representatives, $A^T B$ is always invertible for any representative. The third fact is therefore established. Given that $Q(A, B_0) = I_q$, we immediately see that (1.15) holds with B_0 in place of B . However, (1.15) is independent of representatives so it is valid for every representative B . Finally, let us write C for the set of all points $[B, t] \in M$ such that $A^T B$ is invertible and satisfies (1.17). By using B_0 in place of B in (1.17), it is clear that $U_{[A,s]} \subseteq C$. Conversely, let $[B, t] \in C$ be given. Put $B_0 := BQ(A, B)$. Then $A^T B_0$ is again symmetric positive definite. In particular, $A^T(B_0 - A)$ is symmetric and $(V, c) := (B_0 - A, t - s)$ is a horizontal vector at (A, s) . If $v := d\pi_{(A,s)}((V, c))$, then (1.17) shows that $v \in \mathcal{E}_{[A,s]}$ and $[B, t] \in U_{[A,s]}$. We conclude that $U_{[A,s]} = C$. The proof is complete. $\square$

Remark 4. In the above proof, it might seem that the proof of (1.15) is trivialized while the amazing matrix (1.16) seems to come from nowhere. This is not at all the case. The matrix $Q(A, B)$ comes from considering the cases for a generic representative whose matrix $A^T B$ is only invertible, but not necessarily symmetric or positive definite. Given $[B, t] \in U_{[A,s]}$, (1.14) shows that

$$\text{Exp}_{[A,s]}^{-1}([B, t]) = d\pi_{(A,s)}((BQ - A, t - s))$$

for some $Q \in O(q)$. Given that $A^T(BQ - A)$ is symmetric, it follows that $A^T BQ$ is symmetric and Lemma 2 shows that there exists $R \in O(q)$, $P \in O(q)$ such that $A^T B = RSP^T$ and $Q = PDR^T$. Thus, by continuity of the logarithmic map, if $[B, t] \rightarrow [A, s]$ and for some representative $(B, t) \rightarrow (A, s)$, then it must be the case that $BQ - A \rightarrow 0$ or equivalently $Q \rightarrow I_q$. Therefore, heuristically, the only reasonable choice of D is I_q . This argument is not discarded: it is still in the proof of the injectivity of the exponential map but simplified by symmetric positive definiteness. So $Q = PR^T$. This value is actually uniquely determined by A and B and gives exactly $Q = Q(A, B)$ with $Q(A, B)$ as in (1.16), despite that neither P nor R is unique. This is shown in the following lemma:

Lemma 5. If L is an invertible $q \times q$ matrix with singular value decomposition $L = RSP^T$, then

$$PR^T = (L^T L)^{-\frac{1}{2}} L^T.$$

Proof. Given that

$$L^T L = PS^2 P^T,$$

we have

$$(L^T L)^{\frac{1}{2}} = PSP^T.$$

Therefore:

$$L = RP^T PSP^T = RP^T (L^T L)^{\frac{1}{2}}.$$

The result follows from inverting $(L^T L)^{\frac{1}{2}}$ and take transpose. $\square$

We end this section with a result concerning the limit of a sequence.

Proposition 6. Given a sequence $\{\mathbf{A}_n\}_{n=1}^\infty$ of real $p \times q$ matrices with the property $\mathbf{A}_n \mathbf{A}_n^T \rightarrow \mathbf{A} \mathbf{A}^T$ as $n \rightarrow \infty$ for a $p \times q$ matrix $\mathbf{A}$ with $\text{rank}(\mathbf{A}) = q$, then

$$\lim_{n \rightarrow \infty} \min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_n \mathbf{Q} - \mathbf{A}\|_F = 0.$$

In particular, if $\{(\mathbf{A}_n, s_n)\}_{n=1}^\infty$ is a sequence in Θ such that, for some $(\mathbf{A}, s) \in \Theta$, we have both $s_n \rightarrow s$ and $\Phi((\mathbf{A}_n, s_n)) \rightarrow \Phi((\mathbf{A}, s))$ as $n \rightarrow \infty$, then $[\mathbf{A}_n, s_n] \rightarrow [\mathbf{A}, s]$ in M as $n \rightarrow \infty$.

Proof. Assume, for the sake of contradiction, that there exists an $\epsilon > 0$ and a subsequence $\{n_k\}_{k=1}^\infty$ such that, for all k ,

$$\min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_{n_k} \mathbf{Q} - \mathbf{A}\|_F \geq \epsilon.$$

Given that $\|\mathbf{A}_n\|_F^2 = \text{tr}(\mathbf{A}_n \mathbf{A}_n^T)$ is convergent, in particular bounded, choose subsubsequence $\{n_{k_l}\}_{l=1}^\infty$ such that $\mathbf{A}_{n_{k_l}} \rightarrow \mathbf{B}$ as $l \rightarrow \infty$ for some $p \times q$ matrix $\mathbf{B}$. It follows that $\mathbf{B} \mathbf{B}^T = \mathbf{A} \mathbf{A}^T$. Therefore, $\text{rank}(\mathbf{A}) = \text{rank}(\mathbf{B}) = q$ and $\mathbf{B} = \mathbf{A} \mathbf{P}$ for some $\mathbf{P} \in O(q)$. We have

$$\min_{\mathbf{Q} \in O(q)} \|\mathbf{A}_{n_{k_l}} \mathbf{Q} - \mathbf{A}\|_F \leq \|\mathbf{A}_{n_{k_l}} \mathbf{P}^T - \mathbf{A}\|_F = \|\mathbf{A}_{n_{k_l}} \mathbf{P}^T - \mathbf{B} \mathbf{P}^T\|_F = \|\mathbf{A}_{n_{k_l}} - \mathbf{B}\|_F \rightarrow 0$$

as $l \rightarrow \infty$, a contradiction.

Now suppose that $\{(\mathbf{A}_n, s_n)\}_{n=1}^\infty$ is a sequence in Θ such that, for some $(\mathbf{A}, s) \in \Theta$, we have both $s_n \rightarrow s$ and $\Phi((\mathbf{A}_n, s_n)) \rightarrow \Phi((\mathbf{A}, s))$ as $n \rightarrow \infty$. It follows that $\mathbf{A}_n \mathbf{A}_n^T \rightarrow \mathbf{A} \mathbf{A}^T$ and the previous paragraph shows that there exists a sequence $\{\mathbf{Q}_n\}_{n=1}^\infty$ of $q \times q$ orthogonal matrices such that $\mathbf{A}_n \mathbf{Q}_n \rightarrow \mathbf{A}$ as $n \rightarrow \infty$. Thus,

$$\lim_{n \rightarrow \infty} [\mathbf{A}_n, s_n] = \lim_{n \rightarrow \infty} \pi((\mathbf{A}_n \mathbf{Q}_n, s_n)) = \pi((\mathbf{A}, s)) = [\mathbf{A}, s].$$

The proof is complete. □

1.3 Properties of the Probabilistic Principal Component Analysis Model

We now show that the statistical model $\mathcal{P}$ given by (1.7) is *differentiable in quadratic mean* and has *positive definite Fisher information operator* in the sense developed in [Sun et al. \(2026\)](#), definition III.1 and III.2.

Let us introduce a few notations. For each $[\mathbf{W}, \sigma^2] \in M$, let

$$p(\mathbf{x}; [\mathbf{W}, \sigma^2]) = (2\pi)^{-\frac{p}{2}} \det^{-\frac{1}{2}}(\Sigma([\mathbf{W}, \sigma^2])) \exp\left(-\frac{1}{2} \mathbf{x}^T (\Sigma([\mathbf{W}, \sigma^2]))^{-1} \mathbf{x}\right) \quad (1.19)$$

be the density of $N_p(\mathbf{0}, \Sigma([\mathbf{W}, \sigma^2]))$. For $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]} \subset T_{[\mathbf{W}, \sigma^2]} M \cong \mathbb{R}^d$, write $l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ for the log-likelihood function of the local experiment

$$l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \log\left(p(\mathbf{x}; \text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))\right). \quad (1.20)$$

Define the square root likelihood function $r_{[\mathbf{W}, \sigma^2]} : \mathcal{E}_{[\mathbf{W}, \sigma^2]} \rightarrow L^2(\mathbb{R}^p)$ by setting, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$, that

$$r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) = \sqrt{p(\mathbf{x}; \text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))}. \quad (1.21)$$

Theorem 7. $r_{[\mathbf{W}, \sigma^2]}$ is Fréchet differentiable at any $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$. In particular, $\mathcal{P}$ is differentiable in quadratic mean with score

$$s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h}) = \frac{1}{2} \text{tr} \left\{ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma([\mathbf{W}, \sigma^2]))] (\Sigma([\mathbf{W}, \sigma^2]))^{-1} (D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2])) \right\}, \quad (1.22)$$

where

$$D_{\mathbf{h}}\Sigma([\mathbf{W}, \sigma^2]) := \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(t\mathbf{h})) = \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \quad (1.23)$$

if $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ lifts to $(\mathbf{U}, b)$ at $(\mathbf{W}, \sigma^2)$. Furthermore, $\mathcal{P}$ is a regular model in the sense that the Fisher information operator

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}_1, \mathbf{h}_2) = \frac{1}{2} \text{tr} \left\{ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} D_{\mathbf{h}_1}\Sigma([\mathbf{W}, \sigma^2]) (\Sigma([\mathbf{W}, \sigma^2]))^{-1} D_{\mathbf{h}_2}\Sigma([\mathbf{W}, \sigma^2]) \right\} \quad (1.24)$$

is positive definite.

Proof. The proof for the Fréchet differentiability is long but routine. We compute the Gâteaux derivative of $r_{[\mathbf{W}, \sigma^2]}$ at $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ in a given direction $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ and show that the Gâteaux derivative is the Fréchet derivative by a direct analysis of the remainder term. At the very end, we specialize our computations to $\mathbf{v} = \mathbf{0}$ to prove (1.22) and (1.24).

Given $\mathbf{v}$ and $\mathbf{h}$, there exists an $\epsilon > 0$ such that $\mathbf{v} + t\mathbf{h} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$ whenever $t \in (-\epsilon, \epsilon)$. If they respectively lift to $(\mathbf{V}, c)$ and $(\mathbf{U}, b)$ at $(\mathbf{W}, \sigma^2)$, then $\mathbf{v} + t\mathbf{h}$ lifts to $(\mathbf{V} + t\mathbf{U}, c + tb)$ at $(\mathbf{W}, \sigma^2)$. First of all:

$$\begin{aligned} D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) &:= \left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \\ &= \left. \frac{d}{dt} \right|_{t=0} \{ (\mathbf{W} + \mathbf{V} + t\mathbf{U})(\mathbf{W} + \mathbf{V} + t\mathbf{U})^T + (\sigma^2 + c + tb)\mathbf{I}_p \} \\ &= \mathbf{U}\mathbf{V}^T + \mathbf{V}\mathbf{U}^T + \mathbf{U}\mathbf{W}^T + \mathbf{W}\mathbf{U}^T + b\mathbf{I}_p, \end{aligned} \quad (1.25)$$

proving (1.23) with special case $\mathbf{v} = \mathbf{0}$. Secondly, for all $\mathbf{x} \in \mathbb{R}^p$:

$$\begin{aligned} D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &:= \left. \frac{d}{dt} \right|_{t=0} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) \\ &= \left. \frac{d}{dt} \right|_{t=0} \left\{ -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \det \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) - \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right)^{-1} \mathbf{x} \right\} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(\left. \frac{d}{dt} \right|_{t=0} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &\quad + \frac{1}{2} \mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} [\mathbf{x}\mathbf{x}^T - (\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})))] \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} (D_{\mathbf{h}}\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) \right\}. \end{aligned} \quad (1.26)$$

Finally:

$$\begin{aligned} D_{\mathbf{h}}r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= D_{\mathbf{h}} \exp \left(\frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) = \frac{1}{2} \exp \left(\frac{1}{2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \right) D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) \\ &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}}l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}). \end{aligned} \quad (1.27)$$

To prove that (1.27) indeed defines a valid Gâteaux derivative, we must prove that $D_{\mathbf{h}}r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$. Luckily, the expectation and covariance of normal quadratic forms have already been very well studied in Gupta and Nagar (2000). For $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma)$, Theorem 7.7.1(i) in Gupta and Nagar (2000) states that

$$\mathbb{E} [\mathbf{X}^T \mathbf{Z} \mathbf{X}] = \text{tr}(\mathbf{Z} \Sigma)$$

for any $p \times p$ matrix $\mathbf{Z}$ and Theorem 7.7.2 in Gupta and Nagar (2000) states that

$$\text{Cov}(\mathbf{X}^T \mathbf{Z} \mathbf{X}, \mathbf{X}^T \mathbf{Y} \mathbf{X}) = \text{tr}(\mathbf{Z} \Sigma \mathbf{Y}^T \Sigma) + \text{tr}(\mathbf{Z}^T \Sigma \mathbf{Y}^T \Sigma)$$

for any two $p \times p$ matrices $\mathbf{Y}$ and $\mathbf{Z}$. Therefore, we can confirm without effort that the expectation of derivative of log density is zero:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= -\frac{1}{2} \text{tr} \left[\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right] \\ &+ \frac{1}{2} \int_{\mathbb{R}^p} \left[\mathbf{x}^T \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \mathbf{x} \right] p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = 0 \end{aligned}$$

and, whenever $\mathbf{h}_1$ and $\mathbf{h}_2 \in T_{[\mathbf{W}, \sigma^2]}M$:

$$\begin{aligned} & \int_{\mathbb{R}^p} D_{\mathbf{h}_1} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) D_{\mathbf{h}_2} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} \\ &= \frac{1}{4} \text{Cov}(\mathbf{X}^T \mathbf{Z}_1 \mathbf{X}, \mathbf{X}^T \mathbf{Z}_2 \mathbf{X}) = \frac{1}{4} \cdot 2 \text{tr} \left(\mathbf{Z}_1 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \mathbf{Z}_2 \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_1} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}_2} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right\}, \end{aligned} \quad (1.28)$$

where $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})))$ and

$$\mathbf{Z}_j := \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}_j} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \quad (1.29)$$

is symmetric for $j = 1, 2$, so that the two terms in Theorem 7.7.2 of Gupta and Nagar (2000) coincide. Thus, in the particular case $\mathbf{h} = \mathbf{h}_1 = \mathbf{h}_2$:

$$\begin{aligned} & \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|^2 = \int_{\mathbb{R}^p} [D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})]^2 d\mathbf{x} \\ &= \frac{1}{4} \int_{\mathbb{R}^p} (D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}))^2 p(\mathbf{x}; \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))) d\mathbf{x} = \frac{1}{8} \text{tr} \left\{ \left[\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right]^2 \right\}. \end{aligned} \quad (1.30)$$

Therefore, $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) \in L^2(\mathbb{R}^p)$ and (1.27) is indeed a valid Gâteaux derivative at $\mathbf{v}$ in the direction $\mathbf{h}$.

To prove that this is indeed the Fréchet derivative, let us look at the remainder term:

$$R(\mathbf{v}, \mathbf{h})(\mathbf{x}) := r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$$

and prove that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 = 0.$$

To start, the fundamental theorem of calculus shows that

$$\begin{aligned}
r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + \mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x}) &= \int_0^1 \frac{d}{dt} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt \\
&= \int_0^1 \lim_{s \rightarrow 0} \frac{r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + (t+s)\mathbf{h})(\mathbf{x}) - r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x})}{s} dt = \int_0^1 D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) dt.
\end{aligned}$$

Therefore, by Minkowski's integral inequality and the linearity of $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned}
\|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\| &= \left\| \int_0^1 (D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h})(\mathbf{x}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})) dt; L^2(\mathbb{R}^p) \right\| \\
&\leq \int_0^1 \|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = |\mathbf{h}| \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt
\end{aligned}$$

and

$$\begin{aligned}
\frac{1}{|\mathbf{h}|^2} \|R(\mathbf{v}, \mathbf{h}); L^2(\mathbb{R}^p)\|^2 &\leq \left\{ \int_0^1 \|D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\frac{\mathbf{h}}{|\mathbf{h}|}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2 \\
&\leq \left\{ \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt \right\}^2.
\end{aligned}$$

The proof will be complete once we can show that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \int_0^1 \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| dt = 0. \quad (1.31)$$

We must therefore examine the continuity of $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$, which is easy to see if we revisit the formulas. (1.25) shows that $D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Thus, (1.27) shows that, for each fixed $\mathbf{x} \in \mathbb{R}^p$, $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$ and (1.30) shows that $\|D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|$ is smooth in both $\mathbf{h}$ and $\mathbf{v}$. Scheffé's lemma shows that $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$, as a map to $L^2(\mathbb{R}^p)$, is continuous in $\mathbf{h}$ and in $\mathbf{v}$. The integrand inside (1.31) is thus bounded uniformly over all $t \in [0, 1]$ and all $\mathbf{h}$ in a compact neighborhood of $\mathbf{0}$ and it suffices to show that, for every $t \in [0, 1]$, that

$$\lim_{\mathbf{h} \rightarrow \mathbf{0}} \sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| = 0, \quad (1.32)$$

thanks to the bounded convergence theorem. To prove (1.32), let $\mathbf{e}_1, \dots, \mathbf{e}_d$ be a $g_{[\mathbf{W}, \sigma^2]}$ orthonormal basis. Every unit vector $\mathbf{u}$ can be expressed as $\mathbf{u} = \sum_{i=1}^d u_i \mathbf{e}_i$ with $\sum_{i=1}^d u_i^2 = 1$. By linearity of $D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ in $\mathbf{h}$, we have

$$\begin{aligned}
\|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| &\leq \sum_{i=1}^d |u_i| \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \\
&\leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.
\end{aligned}$$

Taking supremum yields

$$\sup_{|\mathbf{u}|=1} \|D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{u}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\| \leq \sqrt{d} \max_{i=1, \dots, d} \|D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) - D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v}); L^2(\mathbb{R}^p)\|.$$

Given that $D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v} + t\mathbf{h}) \rightarrow D_{\mathbf{e}_i} r_{[\mathbf{W}, \sigma^2]}(\mathbf{v})$ in $L^2(\mathbb{R}^p)$ as $\mathbf{h} \rightarrow \mathbf{0}$ for every i , every t , and every $\mathbf{v}$, the proof of Fréchet differentiability is complete.

Finally, the formulas (1.25), (1.26), and (1.27) simplifies when $\mathbf{v} = \mathbf{0}$:

$$\begin{aligned} D_{\mathbf{h}} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) &= \frac{1}{2} r_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x}) \\ &= \frac{1}{4} \sqrt{p(\mathbf{x}; [\mathbf{W}, \sigma^2])} \operatorname{tr} \left\{ (\Sigma([\mathbf{W}, \sigma^2]))^{-1} [\mathbf{x} \mathbf{x}^T - (\Sigma([\mathbf{W}, \sigma^2]))] (\Sigma([\mathbf{W}, \sigma^2]))^{-1} (D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2])) \right\}. \end{aligned}$$

The score $s(\mathbf{x}; [\mathbf{W}, \sigma^2])(\mathbf{h})$ therefore equals to $D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{0})(\mathbf{x})$ and is indeed given by (1.22). The Fisher information operator (1.24) is obtained by taking $\mathbf{v} = \mathbf{0}$ in (1.28). To show that $G_{[\mathbf{W}, \sigma^2]}$ is positive definite, let $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ be given. We show that $G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) \geq 0$ and equality holds only when $\mathbf{h} = \mathbf{0}$. The non-negative part is clear from (1.24):

$$G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left\{ \left[(\Sigma([\mathbf{W}, \sigma^2]))^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) \right]^2 \right\} \geq 0.$$

Assume from now on that the equality holds. A slight annoyance here is that the product of two symmetric matrices is not necessarily symmetric and we cannot immediately conclude that $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$. We easily overcome this difficulty by using the cyclic property of the trace to get a symmetric matrix

$$\mathbf{N} := (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}} D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) (\Sigma([\mathbf{W}, \sigma^2]))^{-\frac{1}{2}}$$

such that

$$0 = G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr}(\mathbf{N}^2).$$

Therefore, $\mathbf{N} = \mathbf{0}$ and $D_{\mathbf{h}} \Sigma([\mathbf{W}, \sigma^2]) = \mathbf{0}$. Now (1.23) reads the same as (1.9) in the proof of Proposition 1. We therefore get $(\mathbf{U}, b) \in \ker(d\Phi_{(\mathbf{W}, \sigma^2)}) = V_{(\mathbf{W}, \sigma^2)}$. However, being lifted from $\mathbf{h}$, $(\mathbf{U}, b)$ must also be a horizontal vector. Therefore, $\mathbf{h} = \mathbf{0}$. We have proven that $G_{[\mathbf{W}, \sigma^2]}$ is positive definite. $\square$

Proposition 8. Suppose that $[\mathbf{W}, \sigma^2] \in M$ is the true parameter so that each $\mathbf{X}_i$ has density $p(\cdot; [\mathbf{W}, \sigma^2])$ as in (1.19). We have the local asymptotic normality expansion

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{W}, \sigma^2]} \left(\frac{\mathbf{h}}{\sqrt{n}} \right)\right)}{p(\mathbf{X}_i; [\mathbf{W}, \sigma^2])} = \sum_{i=1}^n s(\mathbf{X}_i; [\mathbf{W}, \sigma^2]) \left(\frac{\mathbf{h}}{\sqrt{n}} \right) - \frac{1}{2} G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) + o_{\mathbb{P}}(1) \quad (1.33)$$

whenever n large enough that $\frac{\mathbf{h}}{\sqrt{n}} \in \mathcal{E}_{[\mathbf{W}, \sigma^2]}$. In particular,

$$\sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \operatorname{Exp}_{[\mathbf{W}, \sigma^2]} \left(\frac{\mathbf{h}}{\sqrt{n}} \right)\right)}{p(\mathbf{X}_i; [\mathbf{W}, \sigma^2])} \rightarrow N \left(-\frac{1}{2} G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}), G_{[\mathbf{W}, \sigma^2]}(\mathbf{h}, \mathbf{h}) \right) \quad (1.34)$$

in distribution as $n \rightarrow \infty$.

Proof. Given that $\mathcal{P}$ is differentiable in quadratic mean and has positive definite Fisher operator, this follows from Proposition III.1 in Sun et al. (2026). $\square$

1.4 Properties of the Maximum Likelihood Estimator of $\mathcal{P}$

In this section, we assume that there is a true parameter $[\mathbf{W}_0, \sigma_0^2] \in M$ for the model $\mathcal{P}$ given in (1.7) and we investigate the consistency and asymptotic normality of the maximum likelihood estimator towards this true parameter.

More concretely, given data points $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^p$, we consider them as realizations of independent identically distributed random variables $\mathbf{X}_1, \dots, \mathbf{X}_n$ whose true density is $p(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$, where p is given in (1.19). In what follows, we denote the true covariance matrix by

$$\Sigma_0 := \Sigma([\mathbf{W}_0, \sigma_0^2]) = \mathbf{W}_0 \mathbf{W}_0^T + \sigma_0^2 \mathbf{I}_p. \quad (1.35)$$

The eigenvalues of the true covariance matrix are

$$\lambda_j(\Sigma_0) = \begin{cases} \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2 & j = 1, \dots, q \\ \sigma_0^2 & j = q+1, \dots, p. \end{cases} \quad (1.36)$$

The sample covariance matrix is defined by

$$\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{x}_i \mathbf{x}_i^T. \quad (1.37)$$

Given that we will discuss $\mathbf{S}_n$ a lot, we shall write λ_j in place of $\lambda_j(\mathbf{S}_n)$. We need also to fix one choice of orthonormal basis of eigenvectors $\mathbf{v}_1, \dots, \mathbf{v}_p$ where $\mathbf{v}_j = \mathbf{v}_j(\mathbf{S}_n)$ corresponds to the eigenvalue λ_j . The facts we will use repeatedly is that, when p and q are both fixed, $\mathbf{S}_n \rightarrow \Sigma_0$ almost surely as $n \rightarrow \infty$ and, due to the fact that eigenvalues depends continuously on the matrices, $\lambda_j \rightarrow \lambda_j(\Sigma_0)$ almost surely as $n \rightarrow \infty$.

We now formulate the maximum likelihood estimator as computed in [Tipping and Bishop \(1999\)](#). Put $\mathbf{U} = [\mathbf{v}_1, \dots, \mathbf{v}_q]$, a $p \times q$ matrix, and Δ_q to be a $q \times q$ diagonal matrix with diagonal entries $\lambda_1, \dots, \lambda_q$. Then, the collection of all choices of $(\hat{\mathbf{W}}_n, \hat{\sigma}_n^2)$ maximizing the likelihood is given by

$$\hat{\mathbf{W}}_n = \mathbf{U} (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R} \quad (1.38)$$

and

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \quad (1.39)$$

ranged over all rotation matrices $\mathbf{R} \in O(q)$.

It is immediately clear from (1.36) that, almost surely, for all sufficiently large n , $\lambda_j > 0$ for all j and $\hat{\sigma}_n^2 > 0$ so that $\hat{\sigma}_n^2$ is well-defined. Also, $\hat{\sigma}_n^2 \rightarrow \sigma_0^2$ almost surely as $n \rightarrow \infty$. Another look at (1.36) shows that, almost surely, for all sufficiently large n , $\lambda_j - \hat{\sigma}_n^2 > 0$ for all $j \in \{1, \dots, q\}$, $\lambda_q > \lambda_{q+1}$, and $\text{rank}(\hat{\mathbf{W}}_n) = q$ so that $\hat{\mathbf{W}}_n$ is well-defined. Thus, almost surely, for all sufficiently large n , the same model parametrized by M has a unique maximum likelihood estimator $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]$.

First of all, I would like to argue that this quotient manifold parameter space allows us to settle consistency in an (arguably more straightforward) way that does not use [Datta and Chakrabarty \(2023\)](#) or [Redner \(1981\)](#).

Theorem 9. *The maximum likelihood estimator is consistent. That is, $[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \rightarrow [\mathbf{W}_0, \sigma_0^2]$ almost surely as $n \rightarrow \infty$.*

Proof. Put

$$\hat{\Sigma}_n := \Sigma([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2]) = \hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T + \hat{\sigma}_n^2 \mathbf{I}_p. \quad (1.40)$$

In light of Proposition 6 and the fact that $\hat{\sigma}_n^2$ is consistent, it suffices to prove that $\hat{\Sigma}_n \rightarrow \Sigma_0$ almost surely as $n \rightarrow \infty$. The fact that $\sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \mathbf{I}_p$ allows us to write

$$\hat{\Sigma}_n = \mathbf{U} (\Delta_q - \hat{\sigma}_n^2 \mathbf{I}_q) \mathbf{U}^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q \lambda_j \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\mathbf{S}_n = \mathbf{S}_n \sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \sum_{j=1}^p \lambda_j \mathbf{v}_j \mathbf{v}_j^T.$$

Thus,

$$\hat{\Sigma}_n - \mathbf{S}_n = \sum_{j=q+1}^p (\hat{\sigma}_n^2 - \lambda_j) \mathbf{v}_j \mathbf{v}_j^T$$

and

$$\|\hat{\Sigma}_n - \mathbf{S}_n\| \leq \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j| \|\mathbf{v}_j \mathbf{v}_j^T\| = \sum_{j=q+1}^p |\hat{\sigma}_n^2 - \lambda_j|.$$

Therefore, given that both $\hat{\sigma}_n^2$ and λ_j will converge almost surely to σ_0^2 when $j \in \{q+1, \dots, p\}$, $\|\hat{\Sigma}_n - \mathbf{S}_n\| \rightarrow 0$ almost surely as $n \rightarrow \infty$. The proof is complete. $\square$

We now move to prove that, locally at the true parameter, the maximum likelihood estimator is asymptotically normal in the tangent space of the true parameter by local linearity. Given that the manifold is locally Euclidean, this result can be attacked with the classical approach, namely by verifying conditions of Theorem 5.23 in [van der Vaart \(1998\)](#), without too much use of geometric analysis. We start with computational lemmas.

Lemma 10. Given $[\mathbf{W}, \sigma^2] \in M$ and any convex neighborhood U of $\mathbf{0}$ in $T_{[\mathbf{W}, \sigma^2]}M$ such that $\bar{U} \subset \mathcal{E}_{[\mathbf{W}, \sigma^2]}$, there exists a function $m : \mathbb{R}^p \rightarrow \mathbb{R}$ such that, whenever $\mathbf{v}_1, \mathbf{v}_2 \in U$ and $\mathbf{x} \in \mathbb{R}^p$, we always have

$$|l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_1)(\mathbf{x}) - l_{[\mathbf{W}, \sigma^2]}(\mathbf{v}_2)(\mathbf{x})| \leq m(\mathbf{x})|\mathbf{v}_1 - \mathbf{v}_2|.$$

This m has its second moment under any distribution μ on $\mathbb{R}^p$ that has the fourth moment. That is,

$$\int_{\mathbb{R}^p} (m(\mathbf{x}))^2 d\mu(\mathbf{x}) < \infty$$

whenever

$$\int_{\mathbb{R}^p} |\mathbf{x}|^4 d\mu(\mathbf{x}) < \infty.$$

Proof. Given $[\mathbf{W}, \sigma^2] \in M$ and a convex neighborhood U of $\mathbf{0}$ compactly contained in $\mathcal{E}_{[\mathbf{W}, \sigma^2]}$, the result will follow by the mean value inequality if we can find a function m such that

$$|D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| \leq m(\mathbf{x})$$

for all unit vector $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$, all $\mathbf{v} \in \bar{U}$, and all $\mathbf{x} \in \mathbb{R}^p$. Given that we have proved in Proposition 7 that $\left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))\right)^{-1}$ is continuous in $\mathbf{v}$ and that $D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v}))$ is continuous in both $\mathbf{h}$ and $\mathbf{v}$, we choose constants $C_1 \in (0, \infty)$ and $C_2 \in (0, \infty)$ such that

$$\frac{1}{2} \left\| \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \leq C_1$$

and

$$\frac{1}{2} \left\| \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_2$$

for all all unit vector $\mathbf{h} \in T_{[\mathbf{W}, \sigma^2]}M$ and all $\mathbf{v} \in \bar{U}$, thanks to the fact that $\bar{U}$ is compact. from Proposition 3. By (1.26), we have

$$\begin{aligned}
|D_{\mathbf{h}} l_{[\mathbf{W}, \sigma^2]}(\mathbf{v})(\mathbf{x})| &\leq \frac{1}{2} \|\mathbf{x} \mathbf{x}^T\|_F \left\| \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \right\|_F \\
&\quad + \frac{1}{2} \left\| \left(\Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right)^{-1} \left(D_{\mathbf{h}} \Sigma(\text{Exp}_{[\mathbf{W}, \sigma^2]}(\mathbf{v})) \right) \right\|_F \leq C_1 \mathbf{x}^T \mathbf{x} + C_2
\end{aligned}$$

Setting $m(\mathbf{x}) = C_1 \mathbf{x}^T \mathbf{x} + C_2$ will finish the proof. $\square$

Lemma 11. Define function $L : \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]} \rightarrow \mathbb{R}$ by setting, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$,

$$L(\mathbf{v}) = \int_{\mathbb{R}^p} l_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})(\mathbf{x}) p(\mathbf{x}; \Sigma_0) d\mathbf{x}.$$

Then $\mathbf{0}$ is the unique maximizer of L and, for all $\mathbf{v} \in \mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$:

$$L(\mathbf{v}) = L(\mathbf{0}) - \frac{1}{2} G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{v}) + R(\mathbf{v})$$

with the remainder of the second order Taylor expansion satisfying

$$\lim_{|\mathbf{v}| \rightarrow 0} \frac{R(\mathbf{v})}{|\mathbf{v}|^2} = 0.$$

Proof. One should immediately recognize that $L(\mathbf{0}) - L(\mathbf{v})$ equals to the Kullback-Leibler divergence of the true model $N_p(\mathbf{0}, \Sigma_0)$ with respect to the model $N_p(\mathbf{0}, \Sigma(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})))$. Therefore, $L(\mathbf{v})$ is maximized at all $\mathbf{v}$ with $\Sigma(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v})) = \Sigma_0$. Given that Σ is injective and that we have proven in Proposition 3 that $\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}$ is injective in $\mathcal{E}_{[\mathbf{W}_0, \sigma_0^2]}$, this only happens when $\mathbf{v} = \mathbf{0}$ and $\mathbf{0}$ is the unique maximizer of L .

To obtain the Taylor expansion, we basically have to repeat the computation part in the proof of Proposition 7 on a slightly different problem with the second derivative at $\mathbf{0}$ being computed as well. The first order derivative at $\mathbf{0}$ is automatically 0 because $\mathbf{0}$ is the maximizer.

Let $\mathbf{h} \in T_{[\mathbf{W}_0, \sigma_0^2]} M$ be given. We can easily compute that

$$L(t\mathbf{h}) = -\frac{p}{2} \log(2\pi) - \frac{1}{2} \log \left(\det \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) - \frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}.$$

Suppose that $\mathbf{h}$ lifts to (U, b) at $(\mathbf{W}_0, \sigma_0^2)$. (1.25) immediately shows that

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) = 2UU^T.$$

We now compute the second derivative L term by term. For the log determinant term, we have

$$\begin{aligned}
\frac{d}{dt} \left(-\frac{1}{2} \log \left(\det \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \text{tr} \left[\left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right], \\
\frac{d^2}{dt^2} \left(-\frac{1}{2} \log \left(\det \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) &= -\frac{1}{2} \frac{d}{dt} \text{tr} \left[\left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right] \\
&= -\frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right\} \\
&\quad + \frac{1}{2} \text{tr} \left\{ \left[\left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right]^2 \right\},
\end{aligned}$$

and therefore

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left(-\frac{1}{2} \log \left(\det \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \right) \right) = -\text{tr} \{ \Sigma_0^{-1} \mathbf{U} \mathbf{U}^T \} + \frac{1}{2} \text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\}.$$

Next, for the trace term, we have

$$\begin{aligned} & \frac{d}{dt} \left(-\frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \\ & \frac{d^2}{dt^2} \left(-\frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) \\ &= \frac{1}{2} \text{tr} \left\{ -2 \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \right. \right. \\ & \quad \left. \frac{d}{dt} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \left. \right\} \\ & \quad + \frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \frac{d^2}{dt^2} \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right) \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\}, \end{aligned}$$

and therefore

$$\left. \frac{d^2}{dt^2} \right|_{t=0} \left(-\frac{1}{2} \text{tr} \left\{ \left(\Sigma \left(\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}(t\mathbf{h}) \right) \right)^{-1} \Sigma_0 \right\} \right) = -\text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\} + \text{tr} (\Sigma_0^{-1} \mathbf{U} \mathbf{U}^T).$$

We end up the negative of the Fisher information operator, as computed in (1.24).

$$\left. \frac{d^2}{dt^2} \right|_{t=0} L(t\mathbf{h}) = -\frac{1}{2} \text{tr} \left\{ [\Sigma_0^{-1} D_{\mathbf{h}} \Sigma([\mathbf{W}_0, \sigma_0^2])]^2 \right\} = -G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{h}, \mathbf{h}).$$

This completes the proof. $\square$

Definition 12. We define the **inverse Fisher information operator** $G_{[\mathbf{W}, \sigma^2]}^{-1}$ as the inverse of the flat version of the Fisher information bilinear form. More concretely, instead of treating $G_{[\mathbf{W}, \sigma^2]}$ as a bilinear form on $T_{[\mathbf{W}, \sigma^2]}M$, we treat it as a map $G_{[\mathbf{W}, \sigma^2]} : T_{[\mathbf{W}, \sigma^2]}M \rightarrow T_{[\mathbf{W}, \sigma^2]}^*M$ by setting $G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}) = G_{[\mathbf{W}, \sigma^2]}(\mathbf{v}, \cdot)$ and we define $G_{[\mathbf{W}, \sigma^2]}^{-1} : T_{[\mathbf{W}, \sigma^2]}^*M \rightarrow T_{[\mathbf{W}, \sigma^2]}M$ to be the inverse.

Proposition 13. The maximum likelihood estimator is asymptotically linear at $[\mathbf{W}_0, \sigma_0^2]$ with the influence function $\text{IF} : \mathbb{R}^p \rightarrow T_{[\mathbf{W}_0, \sigma_0^2]}M$ given by

$$\text{IF}(\mathbf{x}) = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])). \quad (1.41)$$

As described in Sun et al. (2026) Lemma III.3, this means

$$\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) \rightarrow 0 \quad (1.42)$$

in probability as $n \rightarrow \infty$. Also, $\text{IF}(\mathbf{X})$ has mean $\mathbf{0}$ and is square integrable if $\mathbf{X}$ follows the true distribution $N_p(\mathbf{0}, \Sigma_0)$.

In particular, once we fix a $g_{[\mathbf{W}_0, \sigma_0^2]}$ orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ and we identify $T_{[\mathbf{W}_0, \sigma_0^2]}M$ canonically as $\mathbb{R}^d$ (in concrete terms, identify $\sum_{j=1}^d a_j \mathbf{e}_j$ to the point $(a_1, \dots, a_d) \in \mathbb{R}^d$), then $\sqrt{n} \text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1} \left([\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \right)$ converges in distribution to $N_d(\mathbf{0}, \mathbf{G}^{-1})$ as $n \rightarrow \infty$, where $\mathbf{G}$ is the $d \times d$ matrix defined by $\mathbf{G}_{ij} = G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j)$.

Proof. Fix a $g_{[\mathbf{W}_0, \sigma_0^2]}$ orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of $T_{[\mathbf{W}_0, \sigma_0^2]}M$. Define the matrix $\mathbf{G} \in \mathbb{R}^{d \times d}$ as in the statement of lemma and the vector $\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$ by setting $(\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j)$, for all $j = 1, \dots, d$. Identifying $\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right) \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ to a point in $\mathbb{R}^d$ through this basis, we immediately have, by combining the two previous lemmas and Theorem 5.23 in [van der Vaart \(1998\)](#), that

$$\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right) - \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{G}^{-1} \nabla l(\mathbf{X}_i; [\mathbf{W}_0, \sigma_0^2]) \rightarrow 0 \quad (1.43)$$

in probability as $n \rightarrow \infty$. In particular, $\sqrt{n}\text{Exp}_{[\mathbf{W}_0, \sigma_0^2]}^{-1}\left(\left[\hat{\mathbf{W}}_n, \hat{\sigma}_n^2\right]\right)$ converges in distribution to $N_d(\mathbf{0}, \mathbf{G}^{-1})$ as $n \rightarrow \infty$. Also, if $\mathbf{X}$ follows the true distribution $N_p(\mathbf{0}, \Sigma_0)$, then $\mathbb{E}[\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$ and thus $\mathbb{E}[\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])] = \mathbf{0}$. Given that $\nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$ has its second moment, so does $\mathbf{G}^{-1} \nabla l(\mathbf{X}; [\mathbf{W}_0, \sigma_0^2])$.

We remark that (1.42) is simply (1.43) rephrased in a way using only vectors in $T_{[\mathbf{W}_0, \sigma_0^2]}M$ defined without $\mathbf{e}_1, \dots, \mathbf{e}_d$ and hence without the need of identifying $T_{[\mathbf{W}_0, \sigma_0^2]}M$ with $\mathbb{R}^d$. Let us for simplicity define $\mathbf{y} := \mathbf{G}^{-1} \nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]) \in \mathbb{R}^d$. Put $\mathbf{v} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$ by setting $\mathbf{v} = \sum_{j=1}^d \mathbf{y}_j \mathbf{e}_j \in T_{[\mathbf{W}_0, \sigma_0^2]}M$. We claim that $\mathbf{v}$ is the unique vector such that, for all $\mathbf{z} \in T_{[\mathbf{W}_0, \sigma_0^2]}M$:

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{z}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{z}). \quad (1.44)$$

The uniqueness of $\mathbf{v}$ is immediate by the positive definiteness of $G_{[\mathbf{W}_0, \sigma_0^2]}$. Given that $(\mathbf{G}\mathbf{y})_j = (\nabla l(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))_j$ for all $j = 1, \dots, d$, we have, for all $j = 1, \dots, d$:

$$G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{e}_i, \mathbf{e}_j) = \sum_{i=1}^d \mathbf{y}_i \mathbf{G}_{ij} = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])(\mathbf{e}_j).$$

The claim is therefore proved. By looking at Definition 12, (1.44) reads $G_{[\mathbf{W}_0, \sigma_0^2]}(\mathbf{v}) = s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2])$ and hence $\mathbf{v} = G_{[\mathbf{W}_0, \sigma_0^2]}^{-1}(s(\mathbf{x}; [\mathbf{W}_0, \sigma_0^2]))$, giving (1.42). The results about the integral of $\text{IF}(\mathbf{X})$ translates to here as well. This completes the proof. $\square$

Chapter 2

Local Regularity, Optimality, and the Fisher Information in Closed Form

The first chapter produced three things: the quotient Riemannian manifold (M, g) together with its exponential and logarithmic maps; the verification that the probabilistic principal component analysis model is differentiable in quadratic mean with positive definite Fisher information operator, hence locally asymptotically normal; and, for the maximum likelihood estimator, consistency together with asymptotic linearity and asymptotic normality in the tangent space at the true parameter.

That last result is a statement made under a single law and about a single estimator. The present chapter adds what it does not contain. Section 2.2 computes the parallel transport of (M, g) and uses it to show that the asymptotics do not degrade when the truth is moved at the scale $n^{-\frac{1}{2}}$, which is *local regularity*. Section 2.3 turns this into statements about *every* estimator: a convolution theorem and a local asymptotic minimax theorem, both attained by the maximum likelihood estimator. Section 2.4 draws the inferential consequences, namely confidence sets with locally uniform coverage and the limit law of the fitted covariance matrix. Section 2.5 computes the Fisher information operator and its inverse in closed form, which is what makes any of the preceding usable in practice.

Because several results of the first chapter were stated there in the form best suited to their proofs, Section 2.1 records them once more in the notation used from here on, and supplies the few elementary complements that the later arguments need.

2.1 Notation and the Results of Chapter 1 in the Form Used Here

Notation

Throughout this chapter we abbreviate

$$\theta_0 := [\mathbf{W}_0, \sigma_0^2] \in M, \quad \hat{\theta}_n := [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in M, \quad \Sigma_0 := \Sigma(\theta_0), \quad G := G_{\theta_0}, \quad (2.1)$$

and we keep $d = pq + 1 - \frac{q(q-1)}{2}$ of (1.2) for the dimension of M . The letter θ , which in Section 1.1 denoted a point of Θ , was retired at the beginning of Section 1.4 when the coordinates on Θ were renamed $(\mathbf{W}, \sigma^2)$; from here on θ always denotes a point of M . We write $\mathbb{P}_\theta^n$ for the law of $(\mathbf{X}_1, \dots, \mathbf{X}_n)$ when the true parameter is θ , that is the n -fold product of $p(\cdot; \theta) d\mathbf{x}$ with p as in (1.19), and $\rightsquigarrow$ for convergence in distribution. Unless another law is named, $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ refer to $\mathbb{P}_{\theta_0}^n$. Finally $\text{Sym}(m)$ denotes the space of real symmetric $m \times m$ matrices, and $\preceq$ the Loewner order on it.

The horizontal lift, and a Lyapunov equation

The horizontal lift was used repeatedly in the first chapter but never named; we name it now. For $(\mathbf{A}, s) \in \Theta$ let

$$\iota_{(\mathbf{A},s)} : T_{[\mathbf{A},s]}M \longrightarrow H_{(\mathbf{A},s)} \quad (2.2)$$

be the inverse of the isomorphism $d\pi_{(\mathbf{A},s)}|_{H_{(\mathbf{A},s)}}$, so that $\iota_{(\mathbf{A},s)}(\mathbf{v})$ is the unique horizontal vector at $(\mathbf{A}, s)$ projecting to $\mathbf{v}$. By the very definition (1.5) of g , each $\iota_{(\mathbf{A},s)}$ is a linear isometry onto $H_{(\mathbf{A},s)}$. We abbreviate

$$\iota := \iota_{(\mathbf{W}_0, \sigma_0^2)}, \quad \iota(\mathbf{h}) =: (\mathbf{U}_h, b_h) \quad \text{for } \mathbf{h} \in T_{\theta_0}M, \quad (2.3)$$

so that, by part 3 of Proposition 1 and by (1.4),

$$\mathbf{U}_h^T \mathbf{W}_0 = \mathbf{W}_0^T \mathbf{U}_h \quad \text{and} \quad |\mathbf{h}|^2 = \|\mathbf{U}_h\|_F^2 + b_h^2. \quad (2.4)$$

The next lemma makes the orthogonal splitting $T_{(\mathbf{A},s)}\Theta = V_{(\mathbf{A},s)} \oplus H_{(\mathbf{A},s)}$ explicit. It is used twice: here, to see that the horizontal lift depends smoothly on its base point, and in Section 2.2, where the same equation turns out to govern the parallel transport.

Lemma 14. Let $\mathbf{A} \in \mathbb{R}^{p \times q}$ have rank q and put $\mathbf{P}_A := \mathbf{A}^T \mathbf{A}$, a positive definite $q \times q$ matrix.

1. For every $\mathbf{K} \in \mathbb{R}^{q \times q}$ the Lyapunov equation

$$\mathbf{P}_A \Omega + \Omega \mathbf{P}_A = \mathbf{K} \quad (2.5)$$

has a unique solution $\Omega =: \mathcal{L}_A(\mathbf{K})$. The map $\mathcal{L}_A$ is linear, the map $(\mathbf{A}, \mathbf{K}) \mapsto \mathcal{L}_A(\mathbf{K})$ is smooth, and $\mathcal{L}_A(\mathbf{K})$ is skew-symmetric whenever $\mathbf{K}$ is.

2. For every $(\mathbf{V}, c) \in T_{(\mathbf{A},s)}\Theta$ the orthogonal projection onto the vertical space is

$$\mathcal{V}\text{proj}_{(\mathbf{A},s)}((\mathbf{V}, c)) = (\mathbf{A} \mathcal{L}_A(\mathbf{A}^T \mathbf{V} - \mathbf{V}^T \mathbf{A}), 0), \quad (2.6)$$

and the projection $\mathcal{H}_{(\mathbf{A},s)}$ onto the horizontal space is $\text{id} - \mathcal{V}\text{proj}_{(\mathbf{A},s)}$.

3. Consequently $\iota_{(\mathbf{A},s)}(d\pi_{(\mathbf{A},s)}((\mathbf{V}, c))) = \mathcal{H}_{(\mathbf{A},s)}((\mathbf{V}, c))$, and this expression is smooth in $(\mathbf{A}, s, \mathbf{V}, c)$.

Proof. **1.** Diagonalize $\mathbf{P}_A = \mathbf{O} \text{diag}(p_1, \dots, p_q) \mathbf{O}^T$ with $\mathbf{O} \in O(q)$ and every $p_i > 0$. Conjugating (2.5) by $\mathbf{O}$ turns the linear operator $\Omega \mapsto \mathbf{P}_A \Omega + \Omega \mathbf{P}_A$ into the entrywise multiplication by $p_i + p_j > 0$, which is invertible; existence, uniqueness and linearity follow. Smoothness holds because the operator depends polynomially on $\mathbf{A}$ and the inverse of an invertible linear map depends smoothly on it. If $\mathbf{K}^T = -\mathbf{K}$, transposing (2.5) gives $\mathbf{P}_A(-\Omega^T) + (-\Omega^T)\mathbf{P}_A = -\mathbf{K}^T = \mathbf{K}$, so $-\Omega^T$ solves the same equation and uniqueness forces $\Omega^T = -\Omega$.

2. By Proposition 1 the vertical space is $\{(\mathbf{A}\Omega, 0) : \Omega^T = -\Omega\}$, so the orthogonal projection of $(\mathbf{V}, c)$ onto it is $(\mathbf{A}\Omega_*, 0)$, where Ω_* minimizes $\|\mathbf{V} - \mathbf{A}\Omega\|_F^2$ over skew-symmetric Ω ; the scalar component contributes the constant c^2 and plays no role. Differentiating this strictly convex quadratic in a skew-symmetric direction $\mathbf{K}$ gives the stationarity condition $\text{tr}(\mathbf{K}^T \mathbf{A}^T (\mathbf{V} - \mathbf{A}\Omega_*)) = 0$ for every skew-symmetric $\mathbf{K}$, which says exactly that the skew-symmetric part of $\mathbf{A}^T \mathbf{V} - \mathbf{P}_A \Omega_*$ vanishes:

$$\mathbf{A}^T \mathbf{V} - \mathbf{V}^T \mathbf{A} = \mathbf{P}_A \Omega_* - \Omega_*^T \mathbf{P}_A = \mathbf{P}_A \Omega_* + \Omega_* \mathbf{P}_A.$$

This is (2.5) with the skew-symmetric right hand side $\mathbf{K} = \mathbf{A}^T \mathbf{V} - \mathbf{V}^T \mathbf{A}$, and part 1 identifies $\Omega_* = \mathcal{L}_A(\mathbf{K})$ and confirms that it is skew-symmetric, as it must be.

3. The vector $\mathcal{H}_{(\mathbf{A},s)}((\mathbf{V}, c))$ is horizontal and differs from $(\mathbf{V}, c)$ by a vertical vector, so it has the same image under $d\pi_{(\mathbf{A},s)}$, whose kernel is $V_{(\mathbf{A},s)}$ by (1.8). Smoothness is that of (2.6). $\square$

The normal chart at the true parameter

Put

$$\rho := \min(\sigma_{\min}(\mathbf{W}_0), \sigma_0^2) > 0, \quad B_\epsilon := \{\mathbf{u} \in T_{\theta_0}M : |\mathbf{u}| < \epsilon\} \quad \text{for } 0 < \epsilon \leq \rho, \quad (2.7)$$

so that $B_\rho = \mathcal{E}_{\theta_0}$ is the domain of the exponential map supplied by Proposition 3, and put $\mathcal{U} := \text{Exp}_{\theta_0}(B_\rho)$. By that proposition $\text{Exp}_{\theta_0} : B_\rho \rightarrow \mathcal{U}$ is a diffeomorphism onto an open subset of M , and it is *affine* in the horizontal lift:

$$\text{Exp}_{\theta_0}(\mathbf{u}) = [\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u], \quad \mathbf{u} \in B_\rho. \quad (2.8)$$

We write $\Sigma_u := \Sigma(\text{Exp}_{\theta_0}(\mathbf{u}))$ for the covariance matrix at the corresponding point of M .

Lemma 15. Let $\mathcal{D}_0 : T_{\theta_0}M \rightarrow \text{Sym}(p)$ be the linear map

$$\mathcal{D}_0(\mathbf{h}) := D_{\mathbf{h}}\Sigma(\theta_0) = \mathbf{U}_h \mathbf{W}_0^T + \mathbf{W}_0 \mathbf{U}_h^T + b_h \mathbf{I}_p \quad (2.9)$$

of (1.23). Then, for every $\mathbf{u} \in B_\rho$, one has the *exact* identity

$$\Sigma_u = \Sigma_0 + \mathcal{D}_0(\mathbf{u}) + \mathbf{U}_u \mathbf{U}_u^T, \quad (2.10)$$

with no remainder. Moreover, for every $r \in (0, \rho)$ there are constants $0 < c_1 \leq c_2 < \infty$, depending only on θ_0 and r , such that

$$c_1 \mathbf{I}_p \preceq \Sigma_u \preceq c_2 \mathbf{I}_p \quad \text{for every } \mathbf{u} \in \overline{B_r}. \quad (2.11)$$

In particular $\mathbf{u} \mapsto \Sigma_u^{-1}$ is smooth and bounded on $\overline{B_r}$.

Proof. By (2.8) and (1.6),

$$\Sigma_u = (\mathbf{W}_0 + \mathbf{U}_u)(\mathbf{W}_0 + \mathbf{U}_u)^T + (\sigma_0^2 + b_u) \mathbf{I}_p = \Sigma_0 + \mathbf{U}_u \mathbf{W}_0^T + \mathbf{W}_0 \mathbf{U}_u^T + b_u \mathbf{I}_p + \mathbf{U}_u \mathbf{U}_u^T,$$

which is (2.10). For the bounds, the first summand of the middle expression is positive semidefinite, and $|b_u| \leq |\mathbf{u}| \leq r < \rho \leq \sigma_0^2$ by (2.4) and (2.7); hence $\Sigma_u \succeq (\sigma_0^2 - r) \mathbf{I}_p$ and we may take $c_1 := \sigma_0^2 - r > 0$. Similarly $\|\mathbf{U}_u\| \leq \|\mathbf{U}_u\|_F \leq r$, so that $\|\Sigma_u\| \leq (\|\mathbf{W}_0\| + r)^2 + \sigma_0^2 + r =: c_2$. $\square$

Lemma 16 (Spectrum of the true covariance). Fix a singular value decomposition $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$, in which $\mathbf{U}_0 \in \mathbb{R}^{p \times q}$ has orthonormal columns, $\mathbf{V}_0 \in O(q)$, and $\mathbf{D}_0 = \text{diag}(\sigma_1(\mathbf{W}_0), \dots, \sigma_q(\mathbf{W}_0))$ has strictly positive diagonal because $\text{rank}(\mathbf{W}_0) = q$. Put $\mathbf{P}_0 := \mathbf{U}_0 \mathbf{U}_0^T$, the orthogonal projection onto the column space of $\mathbf{W}_0$. Then

$$\Sigma_0 = \mathbf{U}_0 (\mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q) \mathbf{U}_0^T + \sigma_0^2 (\mathbf{I}_p - \mathbf{P}_0), \quad (2.12)$$

the eigenvalues of Σ_0 are those listed in (1.36), and the gap separating the signal eigenvalues from the noise eigenvalue is

$$\lambda_q(\Sigma_0) - \lambda_{q+1}(\Sigma_0) = \sigma_q(\mathbf{W}_0)^2 > 0. \quad (2.13)$$

Passing to another representative $(\mathbf{W}_0 \mathbf{Q}, \sigma_0^2)$ of θ_0 , with $\mathbf{Q} \in O(q)$, leaves $\mathbf{U}_0$, $\mathbf{D}_0$ and $\mathbf{P}_0$ unchanged and replaces $\mathbf{V}_0$ by $\mathbf{Q}^T \mathbf{V}_0$.

Proof. We have $\mathbf{W}_0 \mathbf{W}_0^T = \mathbf{U}_0 \mathbf{D}_0^2 \mathbf{U}_0^T$ and $\mathbf{I}_p = \mathbf{P}_0 + (\mathbf{I}_p - \mathbf{P}_0)$, whence (2.12). The two summands act on orthogonal complementary subspaces, so the eigenvalues are $\sigma_j(\mathbf{W}_0)^2 + \sigma_0^2$ for $j \leq q$ and σ_0^2 with multiplicity $p - q$, which is (1.36); (2.13) follows because $\sigma_q(\mathbf{W}_0) = \sigma_{\min}(\mathbf{W}_0) > 0$. The last sentence is immediate from $\mathbf{W}_0 \mathbf{Q} = \mathbf{U}_0 \mathbf{D}_0 (\mathbf{Q}^T \mathbf{V}_0)^T$. $\square$

The last piece of chart material is the standard fact that the logarithm may be taken at a *moving* base point, provided both points stay in a small enough neighbourhood of θ_0 . It is what makes the two-base-point comparisons of Section 2.2 possible at all.

Lemma 17 (A uniformly normal neighbourhood). There exists $r \in (0, \frac{\rho}{4})$ such that, writing

$$\mathcal{V} := \text{Exp}_{\theta_0}(B_r) \subseteq \mathcal{V}^\sharp := \text{Exp}_{\theta_0}(B_{2r}) \subseteq \mathcal{U}, \quad (2.14)$$

the following hold.

1. For all $x, y \in \mathcal{V}^\sharp$ there is exactly one $\mathbf{v} \in \mathcal{E}_x$ with $\text{Exp}_x(\mathbf{v}) = y$; we denote it by $\text{Exp}_x^{-1}(y)$, and $\text{Exp}_x^{-1}(x) = \mathbf{0}$.
2. The map $(\mathbf{u}, \mathbf{w}) \mapsto \text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w}))$ is smooth on $B_{2r} \times B_{2r}$, read through the horizontal lift $\iota_{(\mathbf{w}_0 + \mathbf{u}_u, \sigma_0^2 + b_u)}$ as a map into $\mathbb{R}^{p \times q} \times \mathbb{R}$.

Proof. Let $\mathcal{E} \subseteq TM$ be the set of pairs $(x, \mathbf{v})$ with $\mathbf{v} \in \mathcal{E}_x$, the ball of Proposition 3. Its radius at $x = [\mathbf{A}, s]$ is $\min(\sigma_{\min}(\mathbf{A}), s)$, which is a well defined function on M because the singular values of $\mathbf{A}$ are unchanged by $\mathbf{A} \mapsto \mathbf{A}\mathbf{Q}$ with $\mathbf{Q} \in O(q)$, and is continuous because its composition with the submersion π is. Hence $\mathcal{E}$ is open in TM and contains the zero section, and $\text{Exp} : \mathcal{E} \rightarrow M$ is smooth, being affine in the horizontal lift by (1.14) and part 3 of Lemma 14. Consider

$$E : \mathcal{E} \longrightarrow M \times M, \quad E(x, \mathbf{v}) := (x, \text{Exp}_x(\mathbf{v})).$$

Its differential at $(\theta_0, \mathbf{0})$ is invertible: in the splitting $T_{(\theta_0, \mathbf{0})}TM \cong T_{\theta_0}M \oplus T_{\theta_0}M$ it is $(\mathbf{a}, \mathbf{b}) \mapsto (\mathbf{a}, \mathbf{a} + \mathbf{b})$, because $d(\text{Exp}_{\theta_0})_{\mathbf{0}} = \text{id}$. The inverse function theorem therefore gives open sets $\mathcal{N} \ni (\theta_0, \mathbf{0})$ in $\mathcal{E}$ and $\mathcal{D} \ni (\theta_0, \theta_0)$ in $M \times M$ such that $E : \mathcal{N} \rightarrow \mathcal{D}$ is a diffeomorphism. This is the usual construction of uniformly normal neighbourhoods (Lee, 2018, Theorem 5.14 and its proof); note that completeness of M , which fails here, is nowhere used, only the openness of the domain $\mathcal{E}$ of Exp . Writing $E^{-1}(x, y) = (x, \Xi(x, y))$, the map Ξ is smooth on $\mathcal{D}$ and satisfies $\text{Exp}_x(\Xi(x, y)) = y$ and $\Xi(x, x) = \mathbf{0}$. Since Exp_{θ_0} is continuous and $(\theta_0, \theta_0) \in \mathcal{D}$, we may choose $r \in (0, \frac{\rho}{4})$ so small that $\mathcal{V}^\sharp \times \mathcal{V}^\sharp \subseteq \mathcal{D}$.

For part 1, existence is $\text{Exp}_x(\Xi(x, y)) = y$ together with $\Xi(x, y) \in \mathcal{E}_x$, and uniqueness is the injectivity of Exp_x on $\mathcal{E}_x$, which is part 2 of Proposition 3; so $\text{Exp}_x^{-1} = \Xi(x, \cdot)$ unambiguously, and $\text{Exp}_x^{-1}(x) = \mathbf{0}$ because $\Xi(x, x) = \mathbf{0}$. Part 2 follows by composing Ξ with the smooth map $(\mathbf{u}, \mathbf{w}) \mapsto (\text{Exp}_{\theta_0}(\mathbf{u}), \text{Exp}_{\theta_0}(\mathbf{w}))$ and with the horizontal lift, which is smooth in its base point by part 3 of Lemma 14. $\square$

We fix such an r once and for all, together with $\mathcal{V}$ and $\mathcal{V}^\sharp$ as in (2.14). On $\mathcal{V}^\sharp$ the map $\text{Exp}_{\theta_0}^{-1}$ in the sense of part 1 of Lemma 17 agrees with the logarithmic map (1.15) of Proposition 3, both being inverses of the injective map Exp_{θ_0} on B_ρ ; no ambiguity can therefore arise from the notation.

The maximum likelihood estimator

We recall the estimator itself. Let $\mathbf{S}_n = \frac{1}{n} \sum_{i=1}^n \mathbf{X}_i \mathbf{X}_i^T$ be the sample covariance matrix, $\lambda_1 \geq \dots \geq \lambda_p$ its eigenvalues and $\mathbf{v}_1, \dots, \mathbf{v}_p$ a corresponding orthonormal basis of eigenvectors, as in Section 1.4.

Definition 18 (Tipping and Bishop, 1999). Almost surely, for all n large enough, the likelihood of the model (1.7) has a unique maximizer $\hat{\theta}_n = [\hat{\mathbf{W}}_n, \hat{\sigma}_n^2] \in M$, whose representatives are

$$\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j=q+1}^p \lambda_j, \quad \hat{\mathbf{W}}_n = [\mathbf{v}_1 \cdots \mathbf{v}_q] (\text{diag}(\lambda_1, \dots, \lambda_q) - \hat{\sigma}_n^2 \mathbf{I}_q)^{\frac{1}{2}} \mathbf{R}, \quad \mathbf{R} \in O(q). \quad (2.15)$$

Proposition 19 (The fitted covariance matrix). With $\hat{\Sigma}_n := \Sigma(\hat{\theta}_n) = \hat{\mathbf{W}}_n \hat{\mathbf{W}}_n^T + \hat{\sigma}_n^2 \mathbf{I}_p$, one has, almost surely for all n large enough,

$$\hat{\Sigma}_n = \sum_{j=1}^q \lambda_j \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \mathbf{v}_j \mathbf{v}_j^T. \quad (2.16)$$

That is, $\hat{\Sigma}_n$ retains the q leading eigenpairs of $\mathbf{S}_n$ and replaces the trailing $p - q$ eigenvalues by their average.

Proof. Substituting (2.15) and using $\sum_{j=1}^p \mathbf{v}_j \mathbf{v}_j^T = \mathbf{I}_p$ gives

$$\hat{\Sigma}_n = \sum_{j=1}^q (\lambda_j - \hat{\sigma}_n^2) \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \mathbf{I}_p = \sum_{j=1}^q \lambda_j \mathbf{v}_j \mathbf{v}_j^T + \hat{\sigma}_n^2 \sum_{j=q+1}^p \mathbf{v}_j \mathbf{v}_j^T,$$

and the orthogonal matrix $\mathbf{R}$ cancels. □

Proposition 20 (Consistency; Section 1.4). $\hat{\theta}_n \rightarrow \theta_0$ almost surely as $n \rightarrow \infty$.

The next proposition is the bookkeeping that lets us speak of *the* error vector of $\hat{\theta}_n$ in the chart, on an event whose probability tends to one.

Proposition 21 (The error vector). Define

$$\mathcal{A}_n := \left\{ \hat{\theta}_n \in \mathcal{V} \right\}, \quad \hat{\mathbf{u}}_n := \begin{cases} \text{Exp}_{\hat{\theta}_n}^{-1}(\hat{\theta}_n) & \text{on } \mathcal{A}_n, \\ \mathbf{0} & \text{otherwise.} \end{cases} \quad (2.17)$$

Then $\mathcal{A}_n$ holds almost surely for all n large enough; in particular $\mathbb{P}_{\theta_0}^n(\mathcal{A}_n) \rightarrow 1$, and the same holds with $\mathcal{V}$ replaced by $\text{Exp}_{\theta_0}(B_\epsilon)$ for any $\epsilon \in (0, r]$. Moreover, on $\mathcal{A}_n$ the horizontal lift of $\hat{\mathbf{u}}_n$ is the *aligned error*

$$\iota(\hat{\mathbf{u}}_n) = \mathbf{e}_n := \left(\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2 \right), \quad \mathbf{Q}_n := \mathbf{Q}(\mathbf{W}_0, \hat{\mathbf{W}}_n) \in O(q), \quad (2.18)$$

with $\mathbf{Q}(\cdot, \cdot)$ the orthogonal matrix (1.16).

Proof. $\text{Exp}_{\theta_0}(B_\epsilon)$ is an open neighbourhood of θ_0 in M for every $\epsilon \in (0, \rho]$, by part 2 of Proposition 3; since $\hat{\theta}_n \rightarrow \theta_0$ almost surely by Proposition 20, almost every realization eventually enters it. The identity (2.18) is (1.15) applied with $[\mathbf{A}, s] = \theta_0$ and $[\mathbf{B}, t] = \hat{\theta}_n$, together with the fact that $d\pi_{(\mathbf{W}_0, \sigma_0^2)}$ and ι are mutually inverse on horizontal vectors. □

What Chapter 1 proves about the estimator

Two conventions are fixed once and for all. First, we fix a g_{θ_0} -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of $T_{\theta_0}M$ and use it to identify $T_{\theta_0}M$ with $\mathbb{R}^d$; under this identification the Riemannian norm $|\cdot|$ becomes the Euclidean norm. Second, we write

$$\mathbf{G} \in \text{Sym}(d), \quad \mathbf{G}_{jk} := G(\mathbf{e}_j, \mathbf{e}_k), \quad \mathbf{s}(\mathbf{x}) \in \mathbb{R}^d, \quad \mathbf{s}(\mathbf{x})_j := s(\mathbf{x}; \theta_0)(\mathbf{e}_j), \quad (2.19)$$

for the matrix of the Fisher information operator (1.24) and the coordinate vector of the score (1.22), so that $G(\mathbf{h}_1, \mathbf{h}_2) = \mathbf{h}_1^T \mathbf{G} \mathbf{h}_2$ and $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \mathbf{s}(\mathbf{x})^T \mathbf{h}$ by bilinearity, and $\mathbf{G}$ is positive definite by Theorem 7. Finally we abbreviate the normalized score

$$\Delta_n := \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathbf{s}(\mathbf{X}_i) \in \mathbb{R}^d. \quad (2.20)$$

By Theorem 7 the vectors $\mathbf{s}(\mathbf{X}_1), \mathbf{s}(\mathbf{X}_2), \dots$ are independent, identically distributed, centred and square integrable with covariance matrix $\mathbf{G}$; the multivariate central limit theorem therefore gives

$$\Delta_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}) \quad \text{under } \mathbb{P}_{\theta_0}^n. \quad (2.21)$$

Definition 22 (Asymptotic linearity). An M -valued estimator sequence T_n with $\mathbb{P}_{\theta_0}^n(T_n \in \mathcal{V}) \rightarrow 1$ is *asymptotically linear at θ_0 with influence operator* $\text{IF} : \mathbb{R}^p \rightarrow T_{\theta_0}M$ if $\text{IF}(\mathbf{X}_1)$ is centred and square integrable and

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.22)$$

where $\hat{\mathbf{u}}_n[T_n] := \text{Exp}_{\theta_0}^{-1}(T_n)$ on $\{T_n \in \mathcal{V}\}$ and $:= \mathbf{0}$ elsewhere, exactly as in (2.17).

Theorem 23 (Section 1.4). Let $1 \leq q < p$ and $\theta_0 \in M$. The maximum likelihood estimator is asymptotically linear at θ_0 in the sense of Definition 22, with influence operator

$$\text{IF}(\mathbf{x}) := G^{-1}(s(\mathbf{x}; \theta_0)) \in T_{\theta_0}M, \quad (2.23)$$

the inverse Fisher information operator of Definition 12 being applied to the score (1.22). That is,

$$\sqrt{n} \hat{\mathbf{u}}_n = \frac{1}{\sqrt{n}} \sum_{i=1}^n \text{IF}(\mathbf{X}_i) + o_{\mathbb{P}}(1), \quad \mathbb{E}[\text{IF}(\mathbf{X}_1)] = \mathbf{0}, \quad \mathbb{E}[|\text{IF}(\mathbf{X}_1)|^2] < \infty. \quad (2.24)$$

In the coordinates (2.19) the expansion reads

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \Delta_n + o_{\mathbb{P}}(1), \quad (2.25)$$

and consequently

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1}) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.26)$$

a statement which, read invariantly, says that $\sqrt{n} \text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$ converges in distribution to the centred Gaussian law $N(\mathbf{0}, G_{\theta_0}^{-1})$ on the vector space $T_{\theta_0}M$. Equivalently, in terms of the aligned error (2.18),

$$\sqrt{n} \mathbf{e}_n = \sqrt{n} (\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow \iota(\mathbf{Z}), \quad \mathbf{Z} \sim N(\mathbf{0}, G_{\theta_0}^{-1}), \quad (2.27)$$

a Gaussian law on the horizontal space $H_{(\mathbf{W}_0, \sigma_0^2)}$. Finally, the local asymptotic normality expansion (1.33) of Proposition 8 reads, in the same coordinates,

$$\Lambda_n(\mathbf{h}) := \sum_{i=1}^n \log \frac{p\left(\mathbf{X}_i; \text{Exp}_{\theta_0}\left(\frac{\mathbf{h}}{\sqrt{n}}\right)\right)}{p(\mathbf{X}_i; \theta_0)} = \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} + o_{\mathbb{P}}(1) \quad (\mathbf{h} \in T_{\theta_0}M). \quad (2.28)$$

Proof. Everything except the reformulations is the content of the last proposition of Section 1.4: (2.24) is (1.42) together with the moment statements proved there, (2.25) is (1.43), and (2.26) is the convergence to $N_d(\mathbf{0}, \mathbf{G}^{-1})$ established there; the convention (2.17) on $\mathcal{A}_n^c$ changes nothing, since $\mathbb{P}_{\theta_0}^n(\mathcal{A}_n) \rightarrow 1$ by Proposition 21 and two random vectors that agree on events of probability tending to one have the same weak limits. Statement (2.27) follows from (2.26) by the continuous mapping theorem applied to the linear isometry ι , using (2.18). For (2.28), the score $s(\mathbf{x}; \theta_0)$ is a linear functional on $T_{\theta_0}M$, so $\sum_{i=1}^n s(\mathbf{X}_i; \theta_0) \left(\frac{\mathbf{h}}{\sqrt{n}}\right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n s(\mathbf{X}_i)^T \mathbf{h} = \Delta_n^T \mathbf{h}$, while $G(\mathbf{h}, \mathbf{h}) = \mathbf{h}^T \mathbf{G} \mathbf{h}$. $\square$

Remark 24. Three comments on Theorem 23 orient everything that follows.

1. It is an intrinsic statement. The quantity that converges is the point $\hat{\theta}_n$ of M , read through the logarithm at θ_0 . The orthogonal alignment $\mathbf{Q}_n$ appearing in (2.27) is not an ad hoc Procrustes device introduced to make a limit exist: by Proposition 21 it is the logarithmic map of Chapter 1, written in a representative.

2. It concerns one law and one estimator. The expansion (2.24) and the limit (2.26) hold under the single law $\mathbb{P}_{\theta_0}^n$, and they describe a single sequence of estimators. They say nothing about what happens when the truth is perturbed, and nothing about competing procedures. Sections 2.2 and 2.3 supply exactly these two missing statements, and Remark 34 shows that the first of them does not follow from (2.26) alone, not even if (2.26) is assumed at every point of M .

3. No assumption is made on the multiplicities of the $\sigma_j(W_0)$. The theorem is valid when some, or all, of $\sigma_1(W_0), \dots, \sigma_q(W_0)$ coincide, a case in which the individual eigenvectors of S_n are *not* asymptotically normal. There is no contradiction: an individual eigenvector is not a function of the point $\hat{\theta}_n \in M$, whereas everything in Theorem 23 is. Section 2.5 makes this quantitative by exhibiting coordinates in which the limiting covariance is computed without ever dividing by a difference of singular values.

2.2 Parallel Transport and Local Regularity

Local regularity is the assertion that the conclusion of Theorem 23 survives a perturbation of the true parameter at the estimation scale $n^{-\frac{1}{2}}$. Stating it requires comparing the estimation error measured at θ_0 with the error measured at a nearby point $\theta_{n,h}$, and these two errors live in *different* tangent spaces. Something is therefore needed to identify $T_{\theta_{n,h}}M$ with $T_{\theta_0}M$, and the canonical choice is the parallel transport of (M, g) , which we compute first.

Parallel transport along the geodesics of M

The metric g was constructed in Section 1.1 so that π is a Riemannian submersion from the *flat* manifold $\Theta \subseteq \mathbb{R}^{pq+1}$, and Proposition 3 exploited this to make the exponential map affine in the horizontal lift. The same structure turns the parallel transport into an explicit linear ordinary differential equation, governed by the very Lyapunov equation (2.5) that described the vertical projection. We first record the submersion identity in the form we need.

Lemma 25. Let $c : I \rightarrow \Theta$ be a smooth curve with horizontal velocity at every time, let $\gamma := \pi \circ c$, and let $\mathbf{x}$ be a smooth vector field along γ , with horizontal lift $\tilde{\mathbf{x}}(t) := \iota_{c(t)}(\mathbf{x}(t))$. Then the horizontal lift of the covariant derivative $D_t \mathbf{x}$ of (M, g) is the horizontal part of the ordinary derivative of $\tilde{\mathbf{x}}$:

$$\iota_{c(t)}(D_t \mathbf{x}(t)) = \mathcal{H}_{c(t)}(\dot{\tilde{\mathbf{x}}}(t)). \quad (2.29)$$

In particular $\mathbf{x}$ is parallel along γ if and only if $\dot{\tilde{\mathbf{x}}}(t) \in V_{c(t)}$ for every t .

Proof. Write $A(\mathbf{x}) := D_t \mathbf{x}$ and $B(\mathbf{x}) := d\pi_{c(t)}(\mathcal{H}_{c(t)}(\dot{\tilde{\mathbf{x}}}))$; both are $\mathbb{R}$ -linear operators on smooth vector fields along γ . Both satisfy the Leibniz rule $L(f\mathbf{x}) = f' \mathbf{x} + fL(\mathbf{x})$ for $f \in C^\infty(I)$: for A this is part of the definition of the covariant derivative along a curve, and for B it follows from $\iota_c(f\mathbf{x}) = f\tilde{\mathbf{x}}$, from $\frac{d}{dt}(f\tilde{\mathbf{x}}) = f'\tilde{\mathbf{x}} + f\dot{\tilde{\mathbf{x}}}$ and from $\mathcal{H}_{c(t)}(\tilde{\mathbf{x}}) = \tilde{\mathbf{x}}$. They agree on extendible fields. Indeed, let Y be a smooth vector field on an open subset of M and $\mathbf{x} = Y \circ \gamma$; then $\tilde{\mathbf{x}} = \tilde{Y} \circ c$ with $\tilde{Y}$ the basic field lifting Y , so $\dot{\tilde{\mathbf{x}}}$ is the Euclidean covariant derivative $\tilde{\nabla}_{\dot{c}} \tilde{Y}$ of Θ , whose horizontal part is the lift of $\nabla_{d\pi(\dot{c})} Y = \nabla_{\dot{\gamma}} Y = D_t \mathbf{x}$ by the fundamental identity for Riemannian submersions (O'Neill, 1967, Section 2; Valencia, 2021, Section 3), which applies because $\dot{c}$ is horizontal. A covariant derivative along a curve is uniquely determined by $\mathbb{R}$ -linearity, the Leibniz rule and its values on extendible fields (see Lee, 2018, Chapter 4, the uniqueness clause of the existence theorem for covariant derivatives along curves), so $A = B$, which is (2.29). The last sentence follows because $\iota_{c(t)}$ is injective and $\ker \mathcal{H}_{c(t)} = V_{c(t)}$. $\square$

Proposition 26 (Parallel transport). Let $[A, s] \in M$, let $\mathbf{v} \in \mathcal{E}_{[A,s]}$ with horizontal lift $(V, c) := \iota_{(A,s)}(\mathbf{v})$, and let

$$\gamma(t) := \text{Exp}_{[A,s]}(t\mathbf{v}) = [A_t, s + tc], \quad A_t := A + tV, \quad t \in [0, 1], \quad (2.30)$$

be the geodesic of Proposition 3.

1. $\text{rank}(\mathbf{A}_t) = q$ for every $t \in [0, 1]$, so Lemma 14 applies to $\mathbf{A}_t$; we abbreviate $\mathcal{L}_t := \mathcal{L}_{\mathbf{A}_t}$.

2. For every $(\mathbf{X}_0, \beta_0) \in H_{(\mathbf{A}, s)}$ the linear initial value problem

$$\dot{\mathbf{X}}(t) = \mathbf{A}_t \Omega(t), \quad \Omega(t) := \mathcal{L}_t (\mathbf{X}(t)^T \mathbf{V} - \mathbf{V}^T \mathbf{X}(t)), \quad \dot{\beta}(t) = 0, \quad (2.31)$$

with initial value $(\mathbf{X}(0), \beta(0)) = (\mathbf{X}_0, \beta_0)$, has a unique solution on $[0, 1]$. It stays horizontal, that is $(\mathbf{X}(t), \beta(t)) \in H_{(\mathbf{A}_t, s+tc)}$ for every t , and it preserves the Euclidean norm: $\|\mathbf{X}(t)\|_F^2 + \beta(t)^2$ is constant.

3. The field $\mathbf{x}(t) := d\pi_{(\mathbf{A}_t, s+tc)}((\mathbf{X}(t), \beta(t)))$ is the parallel field along γ with initial value $d\pi_{(\mathbf{A}, s)}((\mathbf{X}_0, \beta_0))$. Consequently parallel transport along γ from time 0 to time t exists, is a linear isometry, and, read in horizontal lifts, is the time- t flow of (2.31), which depends smoothly on $(t, \mathbf{A}, s, \mathbf{V}, c)$.

Proof. 1. Since $\iota_{(\mathbf{A}, s)}$ is an isometry, $\|\mathbf{V}\| \leq \|\mathbf{V}\|_F \leq |\mathbf{v}| < \min(\sigma_{\min}(\mathbf{A}), s)$, so Weyl's perturbation inequality, as quoted in the proof of Proposition 3, gives $\sigma_{\min}(\mathbf{A}_t) \geq \sigma_{\min}(\mathbf{A}) - t\|\mathbf{V}\| > 0$ for $t \in [0, 1]$; hence $\mathbf{A}_t$ has rank q .

2. By part 1 of Lemma 14 the right hand side of (2.31) is linear in $(\mathbf{X}, \beta)$ with coefficients that are smooth in t and in $(\mathbf{A}, s, \mathbf{V}, c)$; solutions of linear systems therefore exist and are unique on all of $[0, 1]$ and depend smoothly on the initial condition and on the parameters (Lee, 2013, Theorem D.1 and Theorem D.6). By the same lemma each $\Omega(t)$ is skew-symmetric, because $\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}$ is.

For horizontality, put $\mathbf{S}(t) := \mathbf{A}_t^T \mathbf{X}(t) - \mathbf{X}(t)^T \mathbf{A}_t$, which by part 3 of Proposition 1 vanishes precisely when $(\mathbf{X}(t), \beta(t))$ is horizontal at $(\mathbf{A}_t, s + tc)$. Using $\dot{\mathbf{A}}_t = \mathbf{V}$, $\dot{\mathbf{X}} = \mathbf{A}_t \Omega$ and $\Omega^T = -\Omega$,

$$\dot{\mathbf{S}}(t) = \mathbf{V}^T \mathbf{X} + \mathbf{A}_t^T \mathbf{A}_t \Omega - \Omega^T \mathbf{A}_t^T \mathbf{A}_t - \mathbf{X}^T \mathbf{V} = (\mathbf{A}_t^T \mathbf{A}_t \Omega + \Omega \mathbf{A}_t^T \mathbf{A}_t) - (\mathbf{X}^T \mathbf{V} - \mathbf{V}^T \mathbf{X}) = 0, \quad (2.32)$$

the last equality being exactly (2.5) for $\Omega = \Omega(t)$. Since $\mathbf{S}(0) = 0$ by hypothesis, $\mathbf{S} \equiv 0$.

For the norm, horizontality just proved makes $\mathbf{A}_t^T \mathbf{X}(t)$ symmetric, so

$$\frac{d}{dt} [\text{tr}(\mathbf{X} \mathbf{X}^T) + \beta^2] = 2 \text{tr}(\mathbf{X}^T \dot{\mathbf{X}}) + 0 = 2 \text{tr}(\mathbf{X}^T \mathbf{A}_t \Omega) = 2 \text{tr}((\mathbf{A}_t^T \mathbf{X}) \Omega) = 0,$$

the last equality because the trace of the product of a symmetric and a skew-symmetric matrix vanishes.

3. The curve $t \mapsto (\mathbf{A}_t, s+tc)$ is a straight line in Θ with horizontal initial velocity $(\mathbf{V}, c)$, hence has horizontal velocity at every time, and it projects to γ ; this is the observation used in the proof of Proposition 3. By part 2 the pair $(\mathbf{X}(t), \beta(t))$ is the horizontal lift of $\mathbf{x}(t)$, and its derivative $(\mathbf{A}_t \Omega(t), 0)$ is vertical by Proposition 1, $\Omega(t)$ being skew-symmetric. Lemma 25 therefore gives $D_t \mathbf{x} = 0$. Uniqueness of parallel fields with a given initial value identifies the flow of (2.31) with parallel transport, which is a linear isometry by part 2 and by the fact that each ι is one; the smooth dependence was recorded in part 2. $\square$

Definition 27. For $\mathbf{u} \in B_\rho$ we write

$$\Gamma_{\mathbf{u}} : T_{\text{Exp}_{\theta_0}(\mathbf{u})} M \longrightarrow T_{\theta_0} M \quad (2.33)$$

for the parallel transport from time 1 back to time 0 along the radial geodesic $t \mapsto \text{Exp}_{\theta_0}(t\mathbf{u})$ of the normal chart, that is, for the inverse of the linear isometry supplied by Proposition 26 applied with $[\mathbf{A}, s] = \theta_0$ and $\mathbf{v} = \mathbf{u}$. This is the map denoted Γ in Definition III.5 of Sun et al. (2026), the radial geodesic being the canonical curve joining the two points inside a normal neighbourhood. By Proposition 26, $\Gamma_{\mathbf{u}}$ is a linear isometry and $\Gamma_0 = \text{id}$; and, after both tangent spaces are identified with subspaces of $\mathbb{R}^{p \times q} \times \mathbb{R}$ through the horizontal lifts ι and $\iota_{(\mathbf{W}_0 + \mathbf{U}_u, \sigma_0^2 + b_u)}$, the map $(\mathbf{u}, \mathbf{z}) \mapsto \Gamma_{\mathbf{u}}(\mathbf{z})$ is smooth on $B_\rho \times (\mathbb{R}^{p \times q} \times \mathbb{R})$.

The transport comparison

Definition III.5 of [Sun et al. \(2026\)](#) compares $\text{Exp}_{\theta_n, h}^{-1}(\hat{\theta}_n)$, transported back to $T_{\theta_0}M$, with $\text{Exp}_{\theta_0}^{-1}(\hat{\theta}_n)$. On a flat space the two would differ by exactly the vector joining the two base points; on M they differ by that vector plus a curvature correction, and the whole content of this subsection is that the correction is quadratically small. Define, using part 1 of Lemma 17 and Definition 27,

$$F : B_{2r} \times B_{2r} \longrightarrow T_{\theta_0}M, \quad F(\mathbf{u}, \mathbf{w}) := \Gamma_{\mathbf{u}} \left(\text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u})}^{-1} (\text{Exp}_{\theta_0}(\mathbf{w})) \right). \quad (2.34)$$

Lemma 28 (Transport comparison). There is a constant $C < \infty$, depending only on θ_0 and r , such that

$$|F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u})| \leq C |\mathbf{u}| |\mathbf{w} - \mathbf{u}| \quad \text{for all } \mathbf{u}, \mathbf{w} \in B_r. \quad (2.35)$$

Proof. By part 2 of Lemma 17 and Definition 27, F is smooth on $B_{2r} \times B_{2r}$; put

$$R(\mathbf{u}, \mathbf{w}) := F(\mathbf{u}, \mathbf{w}) - (\mathbf{w} - \mathbf{u}),$$

a smooth $T_{\theta_0}M$ -valued function on $B_{2r} \times B_{2r}$. It vanishes on two slices:

- $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$ for every $\mathbf{u} \in B_{2r}$, because $\text{Exp}_x^{-1}(x) = \mathbf{0}$ and $\Gamma_{\mathbf{u}}$ is linear, so $F(\mathbf{u}, \mathbf{u}) = \mathbf{0} = \mathbf{u} - \mathbf{u}$;
- $R(\mathbf{0}, \mathbf{w}) = \mathbf{0}$ for every $\mathbf{w} \in B_{2r}$, because $\Gamma_{\mathbf{0}} = \text{id}$ and $\text{Exp}_{\theta_0}^{-1}(\text{Exp}_{\theta_0}(\mathbf{w})) = \mathbf{w}$, so $F(\mathbf{0}, \mathbf{w}) = \mathbf{w} = \mathbf{w} - \mathbf{0}$.

Set

$$C := \sup_{(\mathbf{u}, \mathbf{w}) \in \overline{B_r} \times \overline{B_r}} \|\partial_1 \partial_2 R(\mathbf{u}, \mathbf{w})\| < \infty,$$

the norm being the one induced on bilinear maps $T_{\theta_0}M \times T_{\theta_0}M \rightarrow T_{\theta_0}M$ by $|\cdot|$; the supremum is finite because $\partial_1 \partial_2 R$ is continuous and $\overline{B_r} \times \overline{B_r}$ is a compact subset of the open set $B_{2r} \times B_{2r}$ on which R is smooth.

Let $\mathbf{u}, \mathbf{w} \in B_r$. The ball B_r being convex, the segment from $\mathbf{u}$ to $\mathbf{w}$ lies in it, and the fundamental theorem of calculus in the second variable, together with $R(\mathbf{u}, \mathbf{u}) = \mathbf{0}$, gives

$$R(\mathbf{u}, \mathbf{w}) = \int_0^1 \partial_2 R(\mathbf{u}, \mathbf{u} + t(\mathbf{w} - \mathbf{u})) [\mathbf{w} - \mathbf{u}] dt. \quad (2.36)$$

Fix $\mathbf{z} \in B_r$. The segment from $\mathbf{0}$ to $\mathbf{u}$ also lies in B_r , and $\partial_2 R(\mathbf{0}, \mathbf{z}) = \mathbf{0}$ because $R(\mathbf{0}, \cdot)$ vanishes identically; the same argument in the first variable therefore gives

$$\partial_2 R(\mathbf{u}, \mathbf{z}) = \int_0^1 \partial_1 \partial_2 R(\varsigma \mathbf{u}, \mathbf{z}) [\mathbf{u}] d\varsigma, \quad \text{whence} \quad \|\partial_2 R(\mathbf{u}, \mathbf{z})\| \leq C |\mathbf{u}|. \quad (2.37)$$

Substituting (2.37) into (2.36) yields (2.35). $\square$

Remark 29. Lemma 28 is the only place where the curvature of (M, g) enters the asymptotic theory, and (2.35) says exactly how much it costs. In the applications below $\mathbf{u}$ is a local perturbation of size $n^{-\frac{1}{2}}$ and $\mathbf{w} - \mathbf{u}$ is an estimation error of size $O_{\mathbb{P}}(n^{-\frac{1}{2}})$, so the right hand side is $O_{\mathbb{P}}(n^{-1})$; after the multiplication by $\sqrt{n}$ that all these statements carry, it is $O_{\mathbb{P}}(n^{-\frac{1}{2}})$ and disappears. The two vanishing slices used in the proof are the two trivial cases of the comparison, namely no displacement of the base point and no error to compare, and it is their combination, not either alone, that produces the product $|\mathbf{u}| |\mathbf{w} - \mathbf{u}|$.

Local regularity

We can now state the definition. For $\mathbf{h} \in T_{\theta_0}M$ and n large enough that $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$ we write

$$\theta_{n,\mathbf{h}} := \text{Exp}_{\theta_0} \left(\frac{\mathbf{h}}{\sqrt{n}} \right) \in \mathcal{V}, \quad (2.38)$$

the *geodesic local alternative* in the direction $\mathbf{h}$; note $\theta_{n,0} = \theta_0$. Given an M -valued estimator sequence T_n we put, the first of the two repeating Definition 22,

$$\hat{\mathbf{u}}_n[T_n] := \begin{cases} \text{Exp}_{\theta_0}^{-1}(T_n) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] := \begin{cases} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left(\text{Exp}_{\theta_{n,\mathbf{h}}}^{-1}(T_n) \right) & \text{on } \{T_n \in \mathcal{V}\}, \\ \mathbf{0} & \text{otherwise,} \end{cases} \quad (2.39)$$

both taking values in the single vector space $T_{\theta_0}M$; the second is well defined as soon as $\frac{\mathbf{h}}{\sqrt{n}} \in B_r$, by part 1 of Lemma 17. For the maximum likelihood estimator the first of these is exactly the vector $\hat{\mathbf{u}}_n$ of (2.17), and we keep that notation for it. Following Sun et al. (2026) we call an estimator sequence T_n *localizable at θ_0* when $\mathbb{P}_{\theta_0}^n(T_n \in \mathcal{V}) \rightarrow 1$, which by Proposition 21 the maximum likelihood estimator is; only for such sequences do the quantities (2.39) carry any information in the limit.

Definition 30. A localizable estimator sequence T_n is *locally regular at θ_0 with limit law L* , where L is a Borel probability measure on $T_{\theta_0}M$, if for every $\mathbf{h} \in T_{\theta_0}M$

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n, \quad (2.40)$$

the limit L being the same for every $\mathbf{h}$. This is Definition III.5 of Sun et al. (2026) with $\psi = \text{id}$ and $\Psi = M$.

Lemma 31 (Contiguity). For every $\mathbf{h} \in T_{\theta_0}M$ the sequences $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$ and $\mathbb{P}_{\theta_0}^n$ are mutually contiguous. Consequently, if a sequence of random vectors is $o_{\mathbb{P}}(1)$, respectively $O_{\mathbb{P}}(1)$, under $\mathbb{P}_{\theta_0}^n$, then it is so under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$ as well.

Proof. By (1.34) of Proposition 8 the log-likelihood ratio $\Lambda_n(\mathbf{h})$ of (2.28) converges in distribution under $\mathbb{P}_{\theta_0}^n$ to $N(-\frac{1}{2}\zeta^2, \zeta^2)$ with $\zeta^2 := G(\mathbf{h}, \mathbf{h})$. If $Z \sim N(-\frac{1}{2}\zeta^2, \zeta^2)$ then $\mathbb{E}[e^Z] = e^{-\frac{1}{2}\zeta^2 + \frac{1}{2}\zeta^2} = 1$, so Le Cam's first lemma in the form of Example 6.5 of van der Vaart (1998) gives mutual contiguity. The transfer of $o_{\mathbb{P}}(1)$ is immediate from the definition of contiguity. For $O_{\mathbb{P}}(1)$, suppose $\mathbf{Z}_n$ were not bounded in probability under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$: there would be $\varepsilon > 0$, a subsequence n_k and constants $M_k \rightarrow \infty$ with $\mathbb{P}_{\theta_{n_k,\mathbf{h}}}^{n_k}(|\mathbf{Z}_{n_k}| > M_k) > \varepsilon$ for all k , whereas $\mathbb{P}_{\theta_0}^{n_k}(|\mathbf{Z}_{n_k}| > M_k) \rightarrow 0$ because $\mathbf{Z}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$ and $M_k \rightarrow \infty$; contiguity makes these two incompatible. $\square$

The next lemma is the bridge: on this manifold, regularity in the intrinsic sense of Definition 30 is the same thing as regularity of the ordinary Euclidean kind, read in the normal chart at θ_0 . It is what allows the classical theory to be applied verbatim in Section 2.3.

Lemma 32 (Chart form of regularity). Let T_n be a localizable estimator sequence with $\sqrt{n} \hat{\mathbf{u}}_n[T_n] = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$. Then, for every $\mathbf{h} \in T_{\theta_0}M$,

$$\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}}[T_n] = \sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} + o_{\mathbb{P}}(1) \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n. \quad (2.41)$$

Consequently T_n is locally regular at θ_0 with limit law L if and only if

$$\sqrt{n} \hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow L \quad \text{under } \mathbb{P}_{\theta_{n,\mathbf{h}}}^n \text{ for every } \mathbf{h} \in T_{\theta_0}M. \quad (2.42)$$

Moreover the boundedness hypothesis is automatic: every locally regular sequence satisfies it.

Proof. Fix $\mathbf{h}$ and abbreviate $\hat{\mathbf{u}}_n := \hat{\mathbf{u}}_n[T_n]$, $\mathbf{u}_n := \frac{\mathbf{h}}{\sqrt{n}}$ and $\mathcal{B}_n := \{T_n \in \mathcal{V}\}$. By hypothesis $\mathbb{P}_{\theta_0}^n(\mathcal{B}_n) \rightarrow 1$, hence $\mathbb{P}_{\theta_{n,h}}^n(\mathcal{B}_n) \rightarrow 1$ by Lemma 31, and by the same lemma $\sqrt{n}\hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,h}}^n$ as well. On $\mathcal{B}_n$, and for n large enough that $\mathbf{u}_n \in B_r$, we have $T_n = \text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)$ with $\hat{\mathbf{u}}_n \in B_r$, so (2.34) and (2.39) give

$$\hat{\mathbf{u}}_n^h[T_n] = \Gamma_{\mathbf{u}_n} \left(\text{Exp}_{\text{Exp}_{\theta_0}(\mathbf{u}_n)}^{-1} \left(\text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n) \right) \right) = F(\mathbf{u}_n, \hat{\mathbf{u}}_n).$$

Lemma 28 therefore gives, on $\mathcal{B}_n$,

$$\left| \sqrt{n} \hat{\mathbf{u}}_n^h[T_n] - (\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}) \right| = \sqrt{n} |F(\mathbf{u}_n, \hat{\mathbf{u}}_n) - (\hat{\mathbf{u}}_n - \mathbf{u}_n)| \leq \sqrt{n} C \frac{|\mathbf{h}|}{\sqrt{n}} \left| \hat{\mathbf{u}}_n - \frac{\mathbf{h}}{\sqrt{n}} \right| = \frac{C|\mathbf{h}|}{\sqrt{n}} |\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}|. \quad (2.43)$$

The right hand side is $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) = o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,h}}^n$, and $\mathbb{P}_{\theta_{n,h}}^n(\mathcal{B}_n^c) \rightarrow 0$, which proves (2.41). The equivalence of (2.40) and (2.42) is then Slutsky's lemma.

For the last assertion, let T_n be locally regular with limit law L and take $\mathbf{h} = \mathbf{0}$ in (2.40): since $\theta_{n,0} = \theta_0$ and $\Gamma_0 = \text{id}$, this reads $\sqrt{n}\hat{\mathbf{u}}_n[T_n] \rightsquigarrow L$ under $\mathbb{P}_{\theta_0}^n$, and a weakly convergent sequence is bounded in probability by Prohorov's theorem. $\square$

Theorem 33 (Local regularity of the maximum likelihood estimator). *Let $1 \leq q < p$ and let $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$. Then the maximum likelihood estimator $\hat{\theta}_n$ of Definition 18 is locally regular at θ_0 in the sense of Definition 30, with limit law*

$$L = N(\mathbf{0}, G_{\theta_0}^{-1}) \quad \text{on } T_{\theta_0}M, \quad (2.44)$$

which does not depend on $\mathbf{h}$. Explicitly, for every $\mathbf{h} \in T_{\theta_0}M$,

$$\sqrt{n} \Gamma_{\frac{\mathbf{h}}{\sqrt{n}}} \left(\text{Exp}_{\theta_{n,h}}^{-1} \left(\hat{\theta}_n \right) \right) \rightsquigarrow N(\mathbf{0}, G_{\theta_0}^{-1}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n. \quad (2.45)$$

Proof. We work in the coordinates (2.19) and fix $\mathbf{h} \in T_{\theta_0}M$.

Step 1: a single linear statistic drives everything. By (2.25) and (2.28),

$$\sqrt{n} \hat{\mathbf{u}}_n = \mathbf{G}^{-1} \Delta_n + o_{\mathbb{P}}(1), \quad \Lambda_n(\mathbf{h}) = \Delta_n^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} + o_{\mathbb{P}}(1), \quad (2.46)$$

both under $\mathbb{P}_{\theta_0}^n$. The two expansions are linear in the *same* vector Δ_n of (2.20), so no separate joint central limit theorem is needed: by (2.21), $\Delta_n \rightsquigarrow N(\mathbf{0}, \mathbf{G})$, and the continuous mapping theorem applied to the linear map $\delta \mapsto (\mathbf{G}^{-1} \delta, \delta^T \mathbf{h} - \frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h})$, together with Slutsky's lemma, yields

$$(\sqrt{n} \hat{\mathbf{u}}_n, \Lambda_n(\mathbf{h})) \rightsquigarrow N \left(\begin{pmatrix} \mathbf{0} \\ -\frac{1}{2} \mathbf{h}^T \mathbf{G} \mathbf{h} \end{pmatrix}, \begin{pmatrix} \mathbf{G}^{-1} & \mathbf{h} \\ \mathbf{h}^T & \mathbf{h}^T \mathbf{G} \mathbf{h} \end{pmatrix} \right) \quad \text{under } \mathbb{P}_{\theta_0}^n, \quad (2.47)$$

the off-diagonal block being $\text{Cov}(\mathbf{G}^{-1} \Delta, \Delta^T \mathbf{h}) = \mathbf{G}^{-1} \mathbf{G} \mathbf{h} = \mathbf{h}$ for $\Delta \sim N(\mathbf{0}, \mathbf{G})$.

Step 2: Le Cam's third lemma. The limit in (2.47) has exactly the form required by Example 6.7 of van der Vaart (1998), the Gaussian case of Le Cam's third lemma (Theorem 6.6 there): the last coordinate is the log-likelihood ratio and is $N(-\frac{1}{2} \varsigma^2, \varsigma^2)$ with $\varsigma^2 = \mathbf{h}^T \mathbf{G} \mathbf{h} = G(\mathbf{h}, \mathbf{h})$. The cited result therefore gives

$$\sqrt{n} \hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{h}, \mathbf{G}^{-1}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n : \quad (2.48)$$

the limit under the contiguous alternative is the limit under the null shifted by the covariance $\mathbf{h}$ between the statistic and the log-likelihood ratio. Equivalently $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,h}}^n$, a law free of $\mathbf{h}$.

Step 3: from the chart to the transported error. The hypotheses of Lemma 32 hold for $T_n = \hat{\theta}_n$: it is localizable by Proposition 21, and $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_0}^n$ by (2.26). The equivalence (2.42) of that lemma therefore turns (2.48) into (2.45), with limit law $N(\mathbf{0}, \mathbf{G}^{-1})$, which in the g_{θ_0} -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ represents $N(\mathbf{0}, G_{\theta_0}^{-1})$. $\square$

Remark 34. It is worth being explicit about what (2.45) excludes, since Theorem 23 alone excludes nothing of the kind. Fix a point $\theta_* \in M$, known to the statistician, and let $\hat{\theta}_n^H$ be the Hodges-type modification of $\hat{\theta}_n$ that replaces it by θ_* whenever $\hat{\theta}_n \in \text{Exp}_{\theta_*}(B)$, with B the ball of radius $n^{-\frac{1}{4}}$ in $T_{\theta_*}M$, and leaves it unchanged otherwise.

For any fixed true parameter $\theta_0 \neq \theta_*$ the modification occurs on an event of probability tending to zero, by Proposition 20, so $\hat{\theta}_n^H$ satisfies the exact analogue of (2.26) there. For $\theta_0 = \theta_*$ it does better: since $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$ by (2.26), we have $\hat{\theta}_n^H = \theta_*$ with probability tending to one, so that $\sqrt{n}\hat{\mathbf{u}}_n \left[\hat{\theta}_n^H \right] \rightsquigarrow \delta_0$, a limit strictly more concentrated than $N(\mathbf{0}, G_{\theta_0}^{-1})$. This is the superefficiency phenomenon, and it shows that no bound whatsoever can be extracted from pointwise statements of the type of (2.26) alone.

And yet, at $\theta_0 = \theta_*$ and $\mathbf{h} \neq \mathbf{0}$, the same computation gives $\sqrt{n}\hat{\mathbf{u}}_n \left[\hat{\theta}_n^H \right] \rightsquigarrow \delta_0$ under $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ as well — the estimator is pinned to θ_* on an event of probability tending to one, since $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) \ll n^{-\frac{1}{4}}$ under the alternative too, by Lemma 31 — so by (2.41) its transported error converges to $\delta_{-\mathbf{h}}$, which *does* depend on $\mathbf{h}$. Hence $\hat{\theta}_n^H$ is superefficient at θ_* and is not locally regular there. Local regularity is precisely the statement that the asymptotics of Theorem 23 do not degrade as the truth is moved at the scale $n^{-\frac{1}{2}}$, and it is a strictly stronger property.

2.3 Asymptotic Optimality

Theorem 33 is a statement about the maximum likelihood estimator alone. Local regularity is, however, exactly the hypothesis under which the classical lower bounds of asymptotic statistics become available, and this section records the two of them. Both are the Euclidean theorems of van der Vaart (1998) applied to the local experiments read in the normal chart, the reduction being supplied by Lemma 32; and both are, for this model, instances of the manifold statements of Sun et al. (2026).

The setting is the same throughout. The family $\left(\mathbb{P}_{\theta_n, \mathbf{h}}^n \right)_{\mathbf{h} \in T_{\theta_0}M}$ of local experiments is, by (2.28) and (2.21), locally asymptotically normal at $\mathbf{h} = \mathbf{0}$ with the nonsingular information matrix $\mathbf{G}$, in the parametrization $\mathbf{u} \mapsto \text{Exp}_{\theta_0}(\mathbf{u})$ of the neighbourhood $\mathcal{V}$ of θ_0 . Estimating $\theta \in \mathcal{V}$ is therefore the same problem as estimating the local parameter $\mathbf{u} \in B_r$, and an estimator sequence T_n of the former yields the estimator sequence $\hat{\mathbf{u}}_n[T_n]$ of the latter. We call a function $\ell : T_{\theta_0}M \rightarrow [0, \infty)$ a *bowl-shaped loss* if $\ell(-\mathbf{x}) = \ell(\mathbf{x})$ for all $\mathbf{x}$ and the sublevel set $\{\ell \leq c\}$ is convex and closed for every $c \geq 0$; the squared Riemannian norm $\ell(\mathbf{x}) = |\mathbf{x}|^2$ and its bounded truncations $\min(|\mathbf{x}|^2, K)$ are the examples to keep in mind.

Theorem 35 (Convolution theorem). *Let $1 \leq q < p$, let $\theta_0 \in M$, and let T_n be an estimator sequence that is locally regular at θ_0 with limit law L , in the sense of Definition 30. Then there is a Borel probability measure Λ on $T_{\theta_0}M$ such that*

$$L = N(\mathbf{0}, G_{\theta_0}^{-1}) * \Lambda. \quad (2.49)$$

Consequently $\int \ell dL \geq \int \ell dN(\mathbf{0}, G_{\theta_0}^{-1})$ for every bowl-shaped loss ℓ , and the bound is attained, with $\Lambda = \delta_0$, by the maximum likelihood estimator.

Proof. By Lemma 32, local regularity of T_n is the same as (2.42), that is, the estimator sequence $\hat{\mathbf{u}}_n[T_n]$ of the local parameter satisfies $\sqrt{n}\hat{\mathbf{u}}_n[T_n] - \mathbf{h} \rightsquigarrow L$ under $\mathbb{P}_{\theta_n, \mathbf{h}}^n$ for every $\mathbf{h} \in \mathbb{R}^d$. In the coordinates (2.19) this is precisely the classical definition of a regular estimator sequence at $\mathbf{u} = \mathbf{0}$ in the experiments $\left(\mathbb{P}_{\theta_n, \mathbf{h}}^n \right)_{\mathbf{h} \in \mathbb{R}^d}$, which are locally asymptotically normal with nonsingular information matrix $\mathbf{G}$ by (2.28), (2.21) and Theorem 7. Theorem 8.8 of van der Vaart (1998), applied with ψ the identity, therefore provides a probability measure Λ on $\mathbb{R}^d$ with $L = N(\mathbf{0}, \mathbf{G}^{-1}) * \Lambda$, and $N(\mathbf{0}, \mathbf{G}^{-1})$ is the coordinate expression of $N(\mathbf{0}, G_{\theta_0}^{-1})$; this is (2.49). The inequality for bowl-shaped ℓ is Anderson's lemma (van der Vaart, 1998, Lemma 8.5), which gives $\int \ell(\mathbf{x} + \mathbf{a}) dN(\mathbf{x}) \geq \int \ell dN$ for every $\mathbf{a}$ and every centred Gaussian N ; integrating this over Λ gives the claim. Attainment is Theorem 33. $\square$

The convolution theorem is sharp and distributional, but it says nothing about estimator sequences that are not regular, and Remark 34 exhibited one that beats the bound at a single point. The local asymptotic minimax theorem removes the hypothesis altogether, at the price of replacing the limit law by a maximal risk over a shrinking neighbourhood.

Theorem 36 (Local asymptotic minimax). *Let $1 \leq q < p$, let $\theta_0 \in M$, and let ℓ be a bowl-shaped loss. Then, for every sequence S_n of $T_{\theta_0}M$ -valued statistics, that is every sequence of measurable functions $S_n = S_n(\mathbf{X}_1, \dots, \mathbf{X}_n)$ with values in $T_{\theta_0}M$,*

$$\sup_I \liminf_{n \rightarrow \infty} \max_{\mathbf{h} \in I} \mathbb{E}_{\theta_{n,\mathbf{h}}} [\ell(\sqrt{n} S_n - \mathbf{h})] \geq \int \ell \, dN(\mathbf{0}, G_{\theta_0}^{-1}), \quad (2.50)$$

the supremum being taken over all finite subsets $I \subset T_{\theta_0}M$ and $\mathbb{E}_{\theta_{n,\mathbf{h}}}$ denoting expectation under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$. If moreover ℓ is bounded and continuous, then the maximum likelihood estimator attains the bound:

$$\sup_I \liminf_{n \rightarrow \infty} \max_{\mathbf{h} \in I} \mathbb{E}_{\theta_{n,\mathbf{h}}} [\ell(\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h})] = \int \ell \, dN(\mathbf{0}, G_{\theta_0}^{-1}), \quad (2.51)$$

and the same value is obtained if $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h}$ is replaced by the intrinsic transported error $\sqrt{n} \hat{\mathbf{u}}_n^{\mathbf{h}} [\hat{\theta}_n]$ of (2.39).

Proof. As in the proof of Theorem 35, the experiments $(\mathbb{P}_{\theta_{n,\mathbf{h}}}^n)_{\mathbf{h} \in \mathbb{R}^d}$ are locally asymptotically normal at $\mathbf{h} = \mathbf{0}$ with nonsingular information matrix $\mathbf{G}$, and S_n is an arbitrary estimator sequence for the local parameter. The bound (2.50) is then Theorem 8.11 of van der Vaart (1998) with ψ the identity, read in the coordinates (2.19). Note that no property of S_n is used, and that no reference to M survives: this half of the statement needs only Theorem 7 and Proposition 8.

For (2.51), take $S_n := \hat{\mathbf{u}}_n$. By (2.48), $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$ for each fixed $\mathbf{h}$, so for bounded continuous ℓ

$$\mathbb{E}_{\theta_{n,\mathbf{h}}} [\ell(\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h})] \longrightarrow \int \ell \, dN(\mathbf{0}, \mathbf{G}^{-1}),$$

a limit that does not depend on $\mathbf{h}$. For a *finite* set I the maximum of finitely many convergent sequences converges to the maximum of their limits, so $\liminf_n \max_{\mathbf{h} \in I} (\cdot) = \int \ell \, dN(\mathbf{0}, \mathbf{G}^{-1})$ for every I , and the supremum over I is the same number; this is (2.51). Finally, by (2.41) the two errors differ by $o_{\mathbb{P}}(1)$ under $\mathbb{P}_{\theta_{n,\mathbf{h}}}^n$, so by Slutsky's lemma they have the same weak limit and, ℓ being bounded and continuous, the same limiting risk. $\square$

Remark 37. Three comments. First, the two theorems are complementary rather than redundant: the convolution bound is sharp and distributional but restricted to regular sequences, whereas (2.50) holds for every estimator sequence whatsoever but only in the weaker, maximal-risk form. Second, it is worth noting where each hypothesis is used. The lower bound (2.50) needs nothing beyond Theorem 7 and Proposition 8, so it was already within reach at the end of Section 1.4; what needs Theorem 33 is the *attainment* (2.51), because the supremum there is taken over a neighbourhood that shrinks at the estimation scale and therefore probes exactly the local alternatives. Third, the restriction to bounded continuous ℓ in (2.51) is only a matter of uniform integrability: weak convergence alone does not control $\mathbb{E}[\ell(\cdot)]$ for unbounded ℓ , and we have not established the moment bounds that would be needed to remove it. The lower bound (2.50) carries no such restriction.

We close the section with a uniqueness statement that costs nothing and tidies up Definition 22: the influence operator of an asymptotically linear sequence is determined by the sequence, with no regularity assumed.

Proposition 38 (Uniqueness of the influence operator). *Let T_n be an estimator sequence that is asymptotically linear at θ_0 in the sense of Definition 22 with influence operator IF , and suppose IF' is another influence operator for which (2.22) holds. Then $\text{IF}'(\mathbf{X}_1) = \text{IF}(\mathbf{X}_1)$ almost surely. In particular the influence operator (2.23) of the maximum likelihood estimator is the only one, and, by Theorems 23 and 33, $\hat{\theta}_n$ is a *regular asymptotically linear* sequence, so that Lemma III.3 of Sun et al. (2026) applies to it with all its hypotheses verified.*

Proof. Subtracting the two expansions (2.22) gives $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{D}(\mathbf{X}_i) = o_{\mathbb{P}}(1)$ with $\mathbf{D} := \mathbf{IF} - \mathbf{IF}'$, which is centred and square integrable by Definition 22. The multivariate central limit theorem gives $n^{-\frac{1}{2}} \sum_{i=1}^n \mathbf{D}(\mathbf{X}_i) \rightsquigarrow N(\mathbf{0}, \text{Cov}(\mathbf{D}(\mathbf{X}_1)))$, and a sequence converging in distribution to a limit and also to $\mathbf{0}$ in probability forces that limit to be δ_0 ; hence $\text{Cov}(\mathbf{D}(\mathbf{X}_1)) = \mathbf{0}$ and, $\mathbf{D}(\mathbf{X}_1)$ being centred, $\mathbf{D}(\mathbf{X}_1) = \mathbf{0}$ almost surely. $\square$

2.4 Consequences

The first consequence is the inferential payoff, and it is the one that distinguishes local regularity from Theorem 23 in practice. We first record that the information operator may be evaluated at the estimator.

Lemma 39. For $\mathbf{u} \in B_r$ let $\mathbf{G}(\mathbf{u})$ be the $d \times d$ matrix

$$\mathbf{G}(\mathbf{u})_{jk} := G_{\text{Exp}_{\theta_0}(\mathbf{u})}(\Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_j), \Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_k)), \quad j, k = 1, \dots, d, \quad (2.52)$$

the matrix of the Fisher information operator at $\text{Exp}_{\theta_0}(\mathbf{u})$ in the frame obtained by transporting $\mathbf{e}_1, \dots, \mathbf{e}_d$ along the radial geodesic. Then $\mathbf{u} \mapsto \mathbf{G}(\mathbf{u})$ is smooth on B_r , positive definite everywhere, and $\mathbf{G}(\mathbf{0}) = \mathbf{G}$.

Proof. Write $(\mathbf{W}_{\mathbf{u}}, \sigma_{\mathbf{u}}^2) := (\mathbf{W}_0 + \mathbf{U}_{\mathbf{u}}, \sigma_0^2 + b_{\mathbf{u}})$ for the representative of $\text{Exp}_{\theta_0}(\mathbf{u})$ used in (2.8), which is affine in $\mathbf{u}$, and let $(\tilde{\mathbf{U}}_j(\mathbf{u}), \tilde{b}_j(\mathbf{u})) := \iota_{(\mathbf{W}_{\mathbf{u}}, \sigma_{\mathbf{u}}^2)}(\Gamma_{\mathbf{u}}^{-1}(\mathbf{e}_j))$ be the horizontal lift there. By (1.24) and (1.23) applied at the point $\text{Exp}_{\theta_0}(\mathbf{u})$,

$$\mathbf{G}(\mathbf{u})_{jk} = \frac{1}{2} \text{tr} \{ \Sigma_{\mathbf{u}}^{-1} \mathbf{N}_j(\mathbf{u}) \Sigma_{\mathbf{u}}^{-1} \mathbf{N}_k(\mathbf{u}) \}, \quad \mathbf{N}_j(\mathbf{u}) := \tilde{\mathbf{U}}_j(\mathbf{u}) \mathbf{W}_{\mathbf{u}}^T + \mathbf{W}_{\mathbf{u}} \tilde{\mathbf{U}}_j(\mathbf{u})^T + \tilde{b}_j(\mathbf{u}) \mathbf{I}_p.$$

Each ingredient is smooth in $\mathbf{u}$: the matrix $\Sigma_{\mathbf{u}}$ is polynomial in $\mathbf{u}$ by (2.10) and is bounded below by $c_1 \mathbf{I}_p$ by (2.11), so $\Sigma_{\mathbf{u}}^{-1}$ is smooth; $\Gamma_{\mathbf{u}}^{-1}$ is smooth by Proposition 26 and Definition 27; and $\iota_{(\mathbf{W}_{\mathbf{u}}, \sigma_{\mathbf{u}}^2)}$ is smooth in $\mathbf{u}$ by part 3 of Lemma 14. That $\mathbf{G}(\mathbf{0}) = \mathbf{G}$ is $\Gamma_0 = \text{id}$, and positive definiteness is Theorem 7 at the point $\text{Exp}_{\theta_0}(\mathbf{u})$, the change of frame being invertible. $\square$

Corollary 40 (Locally uniform Wald confidence sets). Let $\alpha \in (0, 1)$, let $\chi_{d, 1-\alpha}^2$ be the $(1 - \alpha)$ quantile of the chi-squared law with d degrees of freedom, and define the random subset of M

$$\mathcal{C}_{n, 1-\alpha} := \text{Exp}_{\hat{\theta}_n}(\{ \mathbf{v} \in T_{\hat{\theta}_n} M : n G_{\hat{\theta}_n}(\mathbf{v}, \mathbf{v}) \leq \chi_{d, 1-\alpha}^2 \}), \quad (2.53)$$

which is well defined on an event of probability tending to one. Then, for every $\mathbf{h} \in T_{\theta_0} M$,

$$\mathbb{P}_{\theta_n, \mathbf{h}}^n(\theta_{n, \mathbf{h}} \in \mathcal{C}_{n, 1-\alpha}) \longrightarrow 1 - \alpha. \quad (2.54)$$

In particular the coverage is asymptotically correct not only at θ_0 , which is the case $\mathbf{h} = \mathbf{0}$ and already follows from Theorem 23, but along every sequence of local alternatives at the estimation scale $n^{-\frac{1}{2}}$.

Proof. Fix $\mathbf{h}$ and work under $\mathbb{P}_{\theta_n, \mathbf{h}}^n$, which by Lemma 31 is contiguous to $\mathbb{P}_{\theta_0}^n$; all the $o_{\mathbb{P}}(1)$ and $O_{\mathbb{P}}(1)$ statements of Theorem 23 and Proposition 21 therefore remain valid. In particular $\mathbb{P}_{\theta_n, \mathbf{h}}^n(\mathcal{A}_n) \rightarrow 1$, $\hat{\mathbf{u}}_n \rightarrow \mathbf{0}$ in probability and $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$.

Well definedness. By Lemma 39 and the continuous mapping theorem, $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$ in probability, so that $\lambda_{\min}(\mathbf{G}(\hat{\mathbf{u}}_n))$ is bounded away from zero in probability. For any $\mathbf{v} \in T_{\hat{\theta}_n} M$, the vector $\mathbf{z} := \Gamma_{\hat{\mathbf{u}}_n}(\mathbf{v})$ has the same norm as $\mathbf{v}$, $\Gamma_{\hat{\mathbf{u}}_n}$ being a linear isometry, and its coordinates in $\mathbf{e}_1, \dots, \mathbf{e}_d$ are the coordinates of $\mathbf{v}$ in the frame transported along the radial geodesic; hence, by (2.52),

$$G_{\hat{\theta}_n}(\mathbf{v}, \mathbf{v}) = \mathbf{z}^T \mathbf{G}(\hat{\mathbf{u}}_n) \mathbf{z} \geq \lambda_{\min}(\mathbf{G}(\hat{\mathbf{u}}_n)) |\mathbf{v}|^2. \quad (2.55)$$

The set inside (2.53) is therefore contained in a ball of radius $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ about $\mathbf{0}$. Meanwhile the radius of the domain $\mathcal{E}_{\hat{\theta}_n}$ of $\text{Exp}_{\hat{\theta}_n}$, namely $\min\left(\sigma_{\min}(\hat{\mathbf{W}}_n), \hat{\sigma}_n^2\right)$, converges in probability to $\rho > 0$ by Proposition 20. Hence, with probability tending to one, that set is a subset of $\mathcal{E}_{\hat{\theta}_n}$ and $\mathcal{C}_{n,1-\alpha}$ is defined.

Reduction to a quadratic form. Both $\hat{\theta}_n$ and $\theta_{n,h}$ lie in $\mathcal{V}$ with probability tending to one, so part 1 of Lemma 17 makes $\mathbf{v}_n := \text{Exp}_{\hat{\theta}_n}^{-1}(\theta_{n,h})$ well defined there, and the injectivity of $\text{Exp}_{\hat{\theta}_n}$ on $\mathcal{E}_{\hat{\theta}_n}$ shows that, on an event of probability tending to one, membership $\theta_{n,h} \in \mathcal{C}_{n,1-\alpha}$ is equivalent to

$$n G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) \leq \chi_{d,1-\alpha}^2. \quad (2.56)$$

Put $\mathbf{z}_n := \Gamma_{\hat{\mathbf{u}}_n}(\mathbf{v}_n) \in T_{\theta_0}M$. The first equality of (2.55), applied with $\mathbf{v} = \mathbf{v}_n$, reads

$$G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) = \mathbf{z}_n^T \mathbf{G}(\hat{\mathbf{u}}_n) \mathbf{z}_n. \quad (2.57)$$

The limit. Apply Lemma 28 with $\mathbf{u} = \hat{\mathbf{u}}_n$ and $\mathbf{w} = \frac{\mathbf{h}}{\sqrt{n}}$, both in B_r with probability tending to one, noting that $F\left(\hat{\mathbf{u}}_n, \frac{\mathbf{h}}{\sqrt{n}}\right) = \mathbf{z}_n$ by (2.34):

$$\left|\sqrt{n}\mathbf{z}_n - (\mathbf{h} - \sqrt{n}\hat{\mathbf{u}}_n)\right| \leq \sqrt{n}C|\hat{\mathbf{u}}_n| \left|\frac{\mathbf{h}}{\sqrt{n}} - \hat{\mathbf{u}}_n\right| = \frac{C}{\sqrt{n}}|\sqrt{n}\hat{\mathbf{u}}_n| |\mathbf{h} - \sqrt{n}\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right),$$

so that $\sqrt{n}\mathbf{z}_n = -(\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h}) + o_{\mathbb{P}}(1)$. By (2.48), $\sqrt{n}\hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,h}}^n$, and this law is symmetric, so $\sqrt{n}\mathbf{z}_n \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$. Combining with $\mathbf{G}(\hat{\mathbf{u}}_n) \rightarrow \mathbf{G}$ in probability, Slutsky's lemma and the continuous mapping theorem applied to $(\mathbf{z}, \mathbf{A}) \mapsto \mathbf{z}^T \mathbf{A} \mathbf{z}$ give, through (2.57),

$$n G_{\hat{\theta}_n}(\mathbf{v}_n, \mathbf{v}_n) = (\sqrt{n}\mathbf{z}_n)^T \mathbf{G}(\hat{\mathbf{u}}_n) (\sqrt{n}\mathbf{z}_n) \rightsquigarrow \mathbf{Z}^T \mathbf{G} \mathbf{Z}, \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{G}^{-1}),$$

and $\mathbf{Z}^T \mathbf{G} \mathbf{Z} = \|\mathbf{G}^{\frac{1}{2}} \mathbf{Z}\|^2 \sim \chi_d^2$ because $\mathbf{G}^{\frac{1}{2}} \mathbf{Z} \sim N(\mathbf{0}, \mathbf{I}_d)$. The distribution function of χ_d^2 being continuous, the probability of (2.56) converges to $\mathbb{P}(\chi_d^2 \leq \chi_{d,1-\alpha}^2) = 1 - \alpha$, which is (2.54). $\square$

Remark 41. Corollary 40 is stated along each fixed sequence of local alternatives, which is the exact strength of Definition 30 and is already enough to rule out the failure exhibited in Remark 34: for the Hodges-type estimator the analogous set is a single point with probability tending to one, and its coverage of $\theta_{n,h}$ tends to 0 for every $\mathbf{h} \neq \mathbf{0}$. The genuinely uniform statement, $\sup_{|\mathbf{h}| \leq c} \left| \mathbb{P}_{\theta_{n,h}}^n(\theta_{n,h} \in \mathcal{C}_{n,1-\alpha}) - (1 - \alpha) \right| \rightarrow 0$, would require the expansion (2.28) along arbitrary bounded sequences $\mathbf{h}_n$ rather than along fixed $\mathbf{h}$, which Proposition 8 as stated does not provide; we do not need it anywhere.

The second consequence records what all of this says about the object the model is really about, namely the fitted covariance matrix $\hat{\Sigma}_n = \Sigma(\hat{\theta}_n)$ of Proposition 19, which, unlike $\hat{\theta}_n$ itself, is an honest element of the fixed vector space $\text{Sym}(p)$. Recall from (2.9) the linear map $\mathcal{D}_0$; it is the differential $d\Sigma_{\theta_0}$ of the map Σ of (1.6), read through the exponential chart.

Corollary 42 (Covariance level central limit theorem). Let $1 \leq q < p$ and $\theta_0 \in M$.

1. Under $\mathbb{P}_{\theta_0}^n$ the fitted covariance matrix is asymptotically linear in $\text{Sym}(p)$ with influence function $\mathcal{D}_0 \circ \text{IF}$,

$$\sqrt{n} \left(\hat{\Sigma}_n - \Sigma_0 \right) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \mathcal{D}_0(\text{IF}(\mathbf{X}_i)) + o_{\mathbb{P}}(1), \quad (2.58)$$

and consequently

$$\sqrt{n} \left(\hat{\Sigma}_n - \Sigma_0 \right) \rightsquigarrow \mathcal{D}_0(\mathbf{Z}), \quad \mathbf{Z} \sim N(\mathbf{0}, \mathbf{G}_{\theta_0}^{-1}) \text{ on } T_{\theta_0}M. \quad (2.59)$$

2. The map $\mathcal{D}_0$ is injective, and, in the notation of Lemma 16, its image is

$$\mathcal{D}_0(T_{\theta_0}M) = \left\{ \mathbf{J}_0 \begin{pmatrix} \mathbf{M}_{11} & \mathbf{M}_{21}^T \\ \mathbf{M}_{21} & \beta \mathbf{I}_{p-q} \end{pmatrix} \mathbf{J}_0^T : \mathbf{M}_{11} \in \text{Sym}(q), \mathbf{M}_{21} \in \mathbb{R}^{(p-q) \times q}, \beta \in \mathbb{R} \right\}, \quad (2.60)$$

where $\mathbf{J}_0 := [\mathbf{U}_0 \ \mathbf{U}_0^\perp] \in O(p)$ as in (2.65) below. Hence the limit in (2.59) is a centred Gaussian law carried by a d -dimensional subspace of $\text{Sym}(p)$, namely the tangent space at Σ_0 to the immersed submanifold $\Sigma(M)$ of Proposition 1; it is degenerate in $\text{Sym}(p)$ exactly when $p - q \geq 2$, the codimension being

$$\frac{p(p+1)}{2} - d = \frac{(p-q)(p-q+1)}{2} - 1 = \frac{(p-q+2)(p-q-1)}{2}. \quad (2.61)$$

3. The convergence (2.59) is locally regular: for every $\mathbf{h} \in T_{\theta_0}M$,

$$\sqrt{n} \left(\hat{\Sigma}_n - \Sigma(\theta_{n,h}) \right) \rightsquigarrow \mathcal{D}_0(\mathbf{Z}) \quad \text{under } \mathbb{P}_{\theta_{n,h}}^n, \quad (2.62)$$

with the same limit law, free of $\mathbf{h}$.

Proof. 1. On $\mathcal{A}_n$ we have $\hat{\theta}_n = \text{Exp}_{\theta_0}(\hat{\mathbf{u}}_n)$, so (2.10) at $\mathbf{u} = \hat{\mathbf{u}}_n$ is the exact identity

$$\hat{\Sigma}_n - \Sigma_0 = \mathcal{D}_0(\hat{\mathbf{u}}_n) + \mathbf{U}_{\hat{\mathbf{u}}_n} \mathbf{U}_{\hat{\mathbf{u}}_n}^T \quad \text{on } \mathcal{A}_n, \quad (2.63)$$

with no remainder, because Exp_{θ_0} is affine in the horizontal lift. Since ι is a linear isometry, $\|\mathbf{U}_{\hat{\mathbf{u}}_n} \mathbf{U}_{\hat{\mathbf{u}}_n}^T\|_F \leq \|\mathbf{U}_{\hat{\mathbf{u}}_n}\| \|\mathbf{U}_{\hat{\mathbf{u}}_n}\|_F \leq \|\mathbf{U}_{\hat{\mathbf{u}}_n}\|_F^2 \leq |\hat{\mathbf{u}}_n|^2$, and $|\hat{\mathbf{u}}_n| = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ by (2.26); hence $\sqrt{n} \|\mathbf{U}_{\hat{\mathbf{u}}_n} \mathbf{U}_{\hat{\mathbf{u}}_n}^T\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right) = o_{\mathbb{P}}(1)$. Multiplying (2.63) by $\sqrt{n}$ therefore gives

$$\sqrt{n} \left(\hat{\Sigma}_n - \Sigma_0 \right) = \mathcal{D}_0(\sqrt{n} \hat{\mathbf{u}}_n) + o_{\mathbb{P}}(1) \quad \text{on } \mathcal{A}_n, \quad (2.64)$$

and the restriction to $\mathcal{A}_n$ is harmless: the two sides of (2.64) differ only on $\mathcal{A}_n^c$, whose probability tends to zero by Proposition 21, so their difference is $o_{\mathbb{P}}(1)$ and they have the same weak limits. Now (2.58) follows from (2.24) and the linearity of $\mathcal{D}_0$, and (2.59) from (2.26), Slutsky's lemma and the continuous mapping theorem applied to the linear, hence continuous, map $\mathcal{D}_0$.

2. If $\mathcal{D}_0(\mathbf{h}) = \mathbf{0}$ then $G(\mathbf{h}, \mathbf{h}) = 0$ by (1.24), whence $\mathbf{h} = \mathbf{0}$ by Theorem 7; so $\mathcal{D}_0$ is injective and its image has dimension d . That image is the one described in (2.60) by (2.68) of Lemma 45 below, together with part 3 of that lemma and Lemma 46: as $\mathbf{h}$ ranges over $T_{\theta_0}M$, the triple $(2\Psi + b_h \mathbf{I}_q, \mathbf{C}\mathbf{D}_0, b_h)$ ranges over all of $\text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}$, the matrix $\mathbf{D}_0$ being invertible. The image is that of $d\Sigma_{\theta_0}$ because $\mathcal{D}_0(\mathbf{h}) = \frac{d}{dt} \big|_{t=0} \Sigma(\text{Exp}_{\theta_0}(t\mathbf{h}))$ by (1.23) and $t \mapsto \text{Exp}_{\theta_0}(t\mathbf{h})$ has velocity $\mathbf{h}$ at $t = 0$. Reading (2.60) off, the only constraint that $\mathcal{D}_0(T_{\theta_0}M)$ imposes on a symmetric matrix is that its trailing $(p-q) \times (p-q)$ block, in the basis $\mathbf{J}_0$, be a multiple of the identity; the codimension is therefore $\dim \text{Sym}(p-q) - 1 = \frac{(p-q)(p-q+1)}{2} - 1$. Writing $m := p-q$, this is $\frac{m^2+m-2}{2} = \frac{(m+2)(m-1)}{2}$, and it vanishes if and only if $m = 1$, since $m+2 \geq 3$. The same number is $\frac{p(p+1)}{2} - d$: by (1.2),

$$\frac{p(p+1)}{2} - d = \frac{p(p+1)}{2} - pq - 1 + \frac{q(q-1)}{2} = \frac{p^2 - 2pq + q^2 + p - q - 2}{2} = \frac{m^2 + m - 2}{2}.$$

3. Fix $\mathbf{h}$ and work under $\mathbb{P}_{\theta_{n,h}}^n$, which is contiguous to $\mathbb{P}_{\theta_0}^n$ by Lemma 31; in particular $\mathbb{P}_{\theta_{n,h}}^n(\mathcal{A}_n) \rightarrow 1$ and the $O_{\mathbb{P}}$ statements above persist, as in the proof of Corollary 40. Applying (2.10) at $\mathbf{u} = \frac{\mathbf{h}}{\sqrt{n}}$ gives the second exact identity

$$\Sigma(\theta_{n,h}) - \Sigma_0 = \frac{1}{\sqrt{n}} \mathcal{D}_0(\mathbf{h}) + \frac{1}{n} \mathbf{U}_h \mathbf{U}_h^T,$$

so subtracting it from (2.63) and multiplying by $\sqrt{n}$,

$$\sqrt{n} \left(\hat{\Sigma}_n - \Sigma(\theta_{n,h}) \right) = \mathcal{D}_0 \left(\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h} \right) + \sqrt{n} \mathbf{U}_{\hat{\mathbf{u}}_n} \mathbf{U}_{\hat{\mathbf{u}}_n}^T - \frac{1}{\sqrt{n}} \mathbf{U}_h \mathbf{U}_h^T \quad \text{on } \mathcal{A}_n.$$

The last term is deterministic and $O\left(n^{-\frac{1}{2}}\right)$. By (2.48), $\sqrt{n} \hat{\mathbf{u}}_n - \mathbf{h} \rightsquigarrow N(\mathbf{0}, \mathbf{G}^{-1})$ under $\mathbb{P}_{\theta_{n,h}}^n$; in particular $\sqrt{n} \hat{\mathbf{u}}_n = O_{\mathbb{P}}(1)$ there, so the middle term is again $O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ by the bound of part 1. The continuous mapping theorem applied to $\mathcal{D}_0$ now gives (2.62), the limit being the image of $N(\mathbf{0}, \mathbf{G}^{-1})$, that is $\mathcal{D}_0(\mathbf{Z})$, and free of $\mathbf{h}$. $\square$

Remark 43 (What part 1 does and does not use). Part 1 of Corollary 42 uses nothing from Section 2.2. Its proof invokes Theorem 23, the exact expansion (2.10), Proposition 21 and the continuous mapping theorem, and no more; neither the parallel transport of Proposition 26, nor the transport comparison of Lemma 28, nor the contiguity of Lemma 31, nor the local regularity of Theorem 33 appears anywhere in it. It could have been stated at the end of Section 1.4; it is placed here only because part 3 belongs here, and because it is the covariance level reading of the whole chapter.

The reason no transport is needed is worth isolating, because it is the converse of the phenomenon that made Section 2.2 necessary in the first place. The intrinsic error $\text{Exp}_{\theta_{n,h}}^{-1}(\hat{\theta}_n)$ of Definition 30 lives in $T_{\theta_{n,h}}M$, a vector space that *moves with* n , which is exactly why $\Gamma_{h/\sqrt{n}}$ has to be inserted before any limit can be taken. The covariance level error $\hat{\Sigma}_n - \Sigma(\theta)$, by contrast, is a difference of two elements of the single fixed vector space $\text{Sym}(p)$, for every θ at once; there is nothing to transport, and the delta method applies verbatim. Curvature is invisible at the covariance level because $\Sigma(M) \subseteq \text{Sym}(p)$ is being viewed extrinsically.

What part 3 costs is correspondingly small: given (2.48), which is the substance of Theorem 33 and does use Lemma 31 and Le Cam's third lemma, the passage to (2.62) is three lines of exact algebra. One should resist reading more into part 3 than it says: it asserts that the *maximum likelihood* fitted covariance is locally regular as an estimator of $\Sigma(\theta)$, and nothing about other estimators. The corresponding lower bound would require the convolution theorem for the differentiable functional $\theta \mapsto \Sigma(\theta)$, which is Theorem IV.1 of Sun et al. (2026) rather than Theorem 35, and which we do not develop.

Remark 44. It may be useful to summarize what this chapter has added to Theorem 23, since the three groups of results answer three different questions.

Theorem 33 is about *other parameters*: it says that the conclusion of Theorem 23 is stable under perturbations of the truth at the scale $n^{-\frac{1}{2}}$. Nothing in (2.24)–(2.27) implies this, as Remark 34 shows by exhibiting an estimator that satisfies the analogue of (2.26) at every point of M and is regular nowhere.

Theorems 35 and 36 are about *other estimators*: they are lower bounds over a class of procedures, and are therefore statements of a different kind from anything in Section 1.4, which concerns one estimator only.

Finally, Lemma 14, Proposition 26 and Lemmas 17 and 28 are geometric results about (M, g) in their own right, independent of any statistics, while Lemma 31 is a probabilistic one. They are what the two-base-point comparison costs, and Theorem 23, which lives entirely inside the single vector space $T_{\theta_0}M$, needed none of them.

In the terminology of Sun et al. (2026), Theorem 33 verifies their Definition III.5 for the probabilistic principal component analysis model, Theorem 35 is the instance of their Theorem III.1, Theorem 36 of their Theorem III.2, and the two together are their Theorem V.1 for this model. We have proved them here through the chart reduction of Lemma 32 and the classical Euclidean theorems of van der Vaart (1998), rather than by citing the manifold versions, because Lemma 28 makes the reduction exact and because it isolates precisely where the curvature of M is felt: in the $O(\|\mathbf{u}\|\|\mathbf{w} - \mathbf{u}\|)$ term of (2.35), and nowhere else.

2.5 The Fisher Information Operator in Closed Form

Theorem 23 identifies the limiting covariance of the maximum likelihood estimator as $G_{\theta_0}^{-1}$, and Theorems 35 and 36 identify the same operator as the efficiency bound; but nothing so far says what $G_{\theta_0}^{-1}$ is. This section computes

it. The only inputs are (1.23), (1.22) and (1.24) of Section 1.3, the spectral decomposition (2.12) of Lemma 16 and the horizontal lift (2.2); no asymptotics are used until Theorem 49, which merely reads Theorem 23 in the coordinates constructed here.

Two features of the answer deserve to be announced. First, the whole calculation is a 2×2 block computation in the singular basis of $\mathbf{W}_0$, and it produces a set of coordinates in which G_{θ_0} is block diagonal with blocks of size 1, 1 and $q + 1$; the only genuinely coupled block is the one that mixes the q leading directions with the noise variance, and it is inverted by a one-line Schur complement. Second, no assumption whatsoever is made on the multiplicities of the singular values of $\mathbf{W}_0$: the formulas below are valid when $\sigma_1(\mathbf{W}_0) = \dots = \sigma_q(\mathbf{W}_0)$, a case in which individual eigenvectors of $\mathbf{S}_n$ are not asymptotically normal at all. This is the quantitative form of the third comment of Remark 24.

Block coordinates at the true parameter

Throughout this section we keep the notation (2.1), we fix the singular value decomposition $\mathbf{W}_0 = \mathbf{U}_0 \mathbf{D}_0 \mathbf{V}_0^T$ of Lemma 16, and we complete $\mathbf{U}_0$ to an orthogonal matrix: we let $\mathbf{U}_0^\perp \in \mathbb{R}^{p \times (p-q)}$ have orthonormal columns spanning the orthogonal complement of the column space of $\mathbf{W}_0$, so that

$$\mathbf{J}_0 := [\mathbf{U}_0 \ \mathbf{U}_0^\perp] \in O(p), \quad \mathbf{I}_p - \mathbf{P}_0 = \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T. \quad (2.65)$$

We abbreviate the signal eigenvalues of Σ_0 supplied by Lemma 16 as

$$\tau_j := \lambda_j(\Sigma_0) = \sigma_j(\mathbf{W}_0)^2 + \sigma_0^2 > \sigma_0^2 \quad (j = 1, \dots, q), \quad \mathbf{T}_0 := \mathbf{D}_0^2 + \sigma_0^2 \mathbf{I}_q = \text{diag}(\tau_1, \dots, \tau_q). \quad (2.66)$$

The letters $\mathbf{B}, \mathbf{C}, \Psi, \mathbf{y}, \mathbf{z}, \mathbf{J}_0, \mathbf{T}_0, \tau_j, \kappa, \mathbf{K}$ and $\mathcal{G}$ are local to this section.

Lemma 45 (Block form of the differential). Let $\mathbf{h} \in T_{\theta_0} M$ have horizontal lift $\iota(\mathbf{h}) = (\mathbf{U}_h, b_h)$ as in (2.3), and set

$$\mathbf{B} := \mathbf{U}_0^T \mathbf{U}_h \mathbf{V}_0 \in \mathbb{R}^{q \times q}, \quad \mathbf{C} := (\mathbf{U}_0^\perp)^T \mathbf{U}_h \mathbf{V}_0 \in \mathbb{R}^{(p-q) \times q}, \quad \Psi := \frac{1}{2} (\mathbf{B} \mathbf{D}_0 + \mathbf{D}_0 \mathbf{B}^T) \in \text{Sym}(q). \quad (2.67)$$

Then:

1. $\mathbf{J}_0^T \Sigma_0 \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0 & \mathbf{0} \\ \mathbf{0} & \sigma_0^2 \mathbf{I}_{p-q} \end{pmatrix}$ and $\mathbf{J}_0^T \Sigma_0^{-1} \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} & \mathbf{0} \\ \mathbf{0} & \sigma_0^{-2} \mathbf{I}_{p-q} \end{pmatrix}$.
2. $\mathbf{U}_h = \mathbf{U}_0 \mathbf{B} \mathbf{V}_0^T + \mathbf{U}_0^\perp \mathbf{C} \mathbf{V}_0^T$ and $|\mathbf{h}|^2 = \|\mathbf{B}\|_F^2 + \|\mathbf{C}\|_F^2 + b_h^2$.
3. $\mathbf{D}_0 \mathbf{B} \in \text{Sym}(q)$; conversely every triple $(\mathbf{B}, \mathbf{C}, b)$ with $\mathbf{D}_0 \mathbf{B}$ symmetric arises from exactly one $\mathbf{h} \in T_{\theta_0} M$.
4. In the basis (2.65),

$$\mathbf{J}_0^T \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0 = \begin{pmatrix} 2\Psi + b_h \mathbf{I}_q & \mathbf{D}_0 \mathbf{C}^T \\ \mathbf{C} \mathbf{D}_0 & b_h \mathbf{I}_{p-q} \end{pmatrix}, \quad (2.68)$$

with $\mathcal{D}_0(\mathbf{h}) = D_h \Sigma(\theta_0)$ as in (2.9).

Proof. 1. Substitute $\mathbf{P}_0 = \mathbf{U}_0 \mathbf{U}_0^T$ and $\mathbf{I}_p - \mathbf{P}_0 = \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T$ into (2.12) and use $\mathbf{U}_0^T \mathbf{U}_0 = \mathbf{I}_q$, $(\mathbf{U}_0^\perp)^T \mathbf{U}_0^\perp = \mathbf{I}_{p-q}$ and $\mathbf{U}_0^T \mathbf{U}_0^\perp = \mathbf{0}$.

2. Since $\mathbf{J}_0 \in O(p)$ and $\mathbf{V}_0 \in O(q)$, the matrix $\mathbf{J}_0^T \mathbf{U}_h \mathbf{V}_0$ has $\mathbf{B}$ and $\mathbf{C}$ stacked as its blocks, which is the first identity; the Frobenius norm is invariant under multiplication by orthogonal matrices on either side, so $\|\mathbf{U}_h\|_F^2 = \|\mathbf{B}\|_F^2 + \|\mathbf{C}\|_F^2$, and the second identity is (2.4).

3. By part 3 of Proposition 1 the pair (U_h, b_h) is horizontal at (W_0, σ_0^2) if and only if $U_h^T W_0 = W_0^T U_h$. Using part 2 and $W_0 = U_0 D_0 V_0^T$,

$$W_0^T U_h = V_0 D_0 U_0^T (U_0 B V_0^T + U_0^\perp C V_0^T) = V_0 (D_0 B) V_0^T,$$

which is symmetric if and only if $D_0 B$ is. Conversely, given such a triple, part 2 defines U_h , the displayed identity makes (U_h, b) horizontal, and $h := d\pi_{(W_0, \sigma_0^2)}((U_h, b))$ is the unique preimage because ι is a bijection onto $H_{(W_0, \sigma_0^2)}$.

4. By (2.9), $\mathcal{D}_0(h) = U_h W_0^T + W_0 U_h^T + b_h I_p$, and by part 2,

$$U_h W_0^T = (U_0 B V_0^T + U_0^\perp C V_0^T) V_0 D_0 U_0^T = U_0 B D_0 U_0^T + U_0^\perp C D_0 U_0^T,$$

so that $J_0^T U_h W_0^T J_0 = \begin{pmatrix} B D_0 & 0 \\ C D_0 & 0 \end{pmatrix}$. Adding the transpose of this matrix and $b_h I_p$, and using $B D_0 + (B D_0)^T = 2\Psi$, gives (2.68). $\square$

The three blocks have direct meanings, and it is worth recording them before any computation. By (2.68) the matrix $2\Psi + b_h I_q$ is the first order change of Σ inside the signal block, the matrix $C D_0$ is its change in the signal-noise block, and $b_h I_{p-q}$ is its change in the noise block — where the model forces the change to be a *multiple of the identity*, which is the whole source of the efficiency gain computed below, and which is the only constraint, as (2.60) recorded. Note also that Ψ , not B , is the natural unknown: the map $B \mapsto \Psi$ is exactly the operation that quotients out the $\frac{q(q-1)}{2}$ directions along which Σ does not move.

Lemma 46 (Linear coordinates on $T_{\theta_0} M$). The map

$$\kappa : T_{\theta_0} M \longrightarrow \text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}, \quad \kappa(h) := (\Psi, C, b_h) \quad (2.69)$$

of (2.67) is a linear isomorphism.

Proof. Linearity is clear: ι is linear by (2.2), and $U_h \mapsto (B, C) \mapsto \Psi$ is a composition of linear maps.

For injectivity, suppose $\kappa(h) = 0$. Then $b_h = 0$ and $C = 0$, while $\Psi = 0$ reads $B D_0 = -D_0 B^T$. Since $\text{rank}(W_0) = q$, all $\sigma_j(W_0)$ are strictly positive and D_0 is invertible, so part 3 of Lemma 45, that is $D_0 B = B^T D_0$, gives $B^T = D_0 B D_0^{-1}$. Substituting this into $B D_0 = -D_0 B^T$ yields $B D_0 = -D_0^2 B D_0^{-1}$, and multiplying on the right by D_0 ,

$$B D_0^2 + D_0^2 B = 0.$$

Both D_0^2 factors are diagonal, so the (i, j) entry of the left hand side is $(\sigma_i(W_0)^2 + \sigma_j(W_0)^2) B_{ij}$, and the bracket is strictly positive; hence $B = 0$. By part 2 of Lemma 45 this forces $U_h = 0$, so $\iota(h) = (0, 0)$ and $h = 0$.

For surjectivity it suffices to compare dimensions, κ being injective and linear:

$$\dim(\text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}) = \frac{q(q+1)}{2} + (p-q)q + 1 = pq + 1 - \frac{q(q-1)}{2} = d$$

by (1.2). $\square$

Lemma 46 lets us treat (Ψ, C, b) as coordinates on $T_{\theta_0} M$. We use throughout the *scalar* coordinates

$$(\Psi_{ij})_{1 \leq i \leq j \leq q}, \quad (C_{ki})_{1 \leq k \leq p-q, 1 \leq i \leq q}, \quad b, \quad (2.70)$$

which identify the target of κ with $\mathbb{R}^d$, and we let $\mathcal{G}$ denote the $d \times d$ symmetric matrix representing the Fisher form in them, that is

$$G_{\theta_0}(h_1, h_2) = \kappa(h_1)^T \mathcal{G} \kappa(h_2) \quad \text{for all } h_1, h_2 \in T_{\theta_0} M. \quad (2.71)$$

Note that κ is *not* a g_{θ_0} -isometry, so $\mathcal{G}$ is not the matrix G of (2.19); the relation between the two is recorded in the proof of Theorem 49 and is the only place where it is needed.

The Fisher form and its inverse

Proposition 47 (Explicit Fisher form). For every $\mathbf{h} \in T_{\theta_0}M$, in the notation (2.67),

$$G_{\theta_0}(\mathbf{h}, \mathbf{h}) = \frac{1}{2} \operatorname{tr} \left[\left(\mathbf{T}_0^{-1} (2\Psi + b_{\mathbf{h}} \mathbf{I}_q) \right)^2 \right] + \frac{1}{\sigma_0^2} \operatorname{tr} \left(\mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C} \right) + \frac{(p-q)b_{\mathbf{h}}^2}{2\sigma_0^4}, \quad (2.72)$$

and, in the scalar coordinates (2.70),

$$G_{\theta_0}(\mathbf{h}, \mathbf{h}) = \underbrace{\sum_{i=1}^q \frac{(2\Psi_{ii} + b_{\mathbf{h}})^2}{2\tau_i^2}}_{\text{coupled block}} + \underbrace{\frac{(p-q)b_{\mathbf{h}}^2}{2\sigma_0^4}}_{\text{}} + \underbrace{\sum_{1 \leq i < j \leq q} \frac{4\Psi_{ij}^2}{\tau_i \tau_j}}_{\Psi \text{ off-diagonal}} + \underbrace{\sum_{i=1}^q \sum_{k=1}^{p-q} \left(\frac{1}{\sigma_0^2} - \frac{1}{\tau_i} \right) \mathbf{C}_{ki}^2}_{\mathbf{C}\text{-block}}. \quad (2.73)$$

In particular $\mathcal{G}$ is block diagonal with respect to the partition of (2.70) into the three groups $\{\Psi_{ij}\}_{i < j}$, $\{\mathbf{C}_{ki}\}$ and $(\Psi_{11}, \dots, \Psi_{qq}, b)$, with

$$\mathcal{G}_{\Psi_{ij}\Psi_{ij}} = \frac{4}{\tau_i \tau_j} \quad (i < j), \quad \mathcal{G}_{\mathbf{C}_{ki}\mathbf{C}_{ki}} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i} = \frac{\sigma_i(\mathbf{W}_0)^2}{\sigma_0^2 \tau_i}, \quad \mathcal{G}_c = \begin{pmatrix} \Pi & \nu \\ \nu^T & \eta \end{pmatrix}, \quad (2.74)$$

the last being the coupled block in the ordering $(\Psi_{11}, \dots, \Psi_{qq}, b)$, where

$$\Pi := \operatorname{diag} (2\tau_1^{-2}, \dots, 2\tau_q^{-2}), \quad \nu := (\tau_1^{-2}, \dots, \tau_q^{-2})^T \in \mathbb{R}^q, \quad \eta := \frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4}. \quad (2.75)$$

Proof. Write $\mathbf{M} := 2\Psi + b_{\mathbf{h}} \mathbf{I}_q$ and $b := b_{\mathbf{h}}$. Conjugating by $\mathbf{J}_0$ does not change traces, so by (1.24) with $\mathbf{h}_1 = \mathbf{h}_2 = \mathbf{h}$ it suffices to compute the trace of the square of $\mathbf{J}_0^T \Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0$. Multiplying the two matrices of Lemma 45,

$$\mathbf{J}_0^T \Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}) \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} \mathbf{M} & \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T \\ \sigma_0^{-2} \mathbf{C} \mathbf{D}_0 & \sigma_0^{-2} b \mathbf{I}_{p-q} \end{pmatrix}. \quad (2.76)$$

The two diagonal blocks of the square of (2.76) are

$$(\mathbf{T}_0^{-1} \mathbf{M})^2 + \sigma_0^{-2} \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T \mathbf{C} \mathbf{D}_0 \quad \text{and} \quad \sigma_0^{-2} \mathbf{C} \mathbf{D}_0 \mathbf{T}_0^{-1} \mathbf{D}_0 \mathbf{C}^T + \sigma_0^{-4} b^2 \mathbf{I}_{p-q},$$

so, using the cyclic property of the trace and the fact that the diagonal matrices $\mathbf{T}_0$ and $\mathbf{D}_0$ commute,

$$\operatorname{tr} \left[(\Sigma_0^{-1} \mathcal{D}_0(\mathbf{h}))^2 \right] = \operatorname{tr} \left[(\mathbf{T}_0^{-1} \mathbf{M})^2 \right] + \frac{2}{\sigma_0^2} \operatorname{tr} \left(\mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C} \right) + \frac{(p-q)b^2}{\sigma_0^4}.$$

Halving gives (2.72).

For (2.73), both $\mathbf{T}_0^{-1}$ and $\mathbf{D}_0^2$ are diagonal, so

$$\operatorname{tr} (\mathbf{T}_0^{-1} \mathbf{D}_0^2 \mathbf{C}^T \mathbf{C}) = \sum_{i=1}^q \frac{\sigma_i(\mathbf{W}_0)^2}{\tau_i} (\mathbf{C}^T \mathbf{C})_{ii} = \sum_{i=1}^q \frac{\sigma_i(\mathbf{W}_0)^2}{\tau_i} \sum_{k=1}^{p-q} \mathbf{C}_{ki}^2,$$

and $\frac{\sigma_i(\mathbf{W}_0)^2}{\sigma_0^2 \tau_i} = \frac{\tau_i - \sigma_0^2}{\sigma_0^2 \tau_i} = \frac{1}{\sigma_0^2} - \frac{1}{\tau_i}$ by (2.66). For the first term, $\mathbf{M}$ is symmetric, so

$$\operatorname{tr} \left[(\mathbf{T}_0^{-1} \mathbf{M})^2 \right] = \sum_{i,j=1}^q \frac{\mathbf{M}_{ij} \mathbf{M}_{ji}}{\tau_i \tau_j} = \sum_{i=1}^q \frac{\mathbf{M}_{ii}^2}{\tau_i^2} + 2 \sum_{i < j} \frac{\mathbf{M}_{ij}^2}{\tau_i \tau_j},$$

and $\mathbf{M}_{ii} = 2\Psi_{ii} + b$, $\mathbf{M}_{ij} = 2\Psi_{ij}$ for $i \neq j$; halving gives the first two groups of terms in (2.73).

Finally, (2.73) is a sum of three quadratic forms in *disjoint* sets of the scalar coordinates (2.70), which is exactly the assertion of block diagonality; reading off the coefficients gives the first two entries of (2.74). For the third, expanding the coupled part of (2.73),

$$\sum_{i=1}^q \frac{(2\Psi_{ii} + b)^2}{2\tau_i^2} + \frac{(p-q)b^2}{2\sigma_0^4} = \sum_{i=1}^q \frac{2\Psi_{ii}^2}{\tau_i^2} + 2b \sum_{i=1}^q \frac{\Psi_{ii}}{\tau_i^2} + b^2 \left(\frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} + \frac{p-q}{2\sigma_0^4} \right),$$

whose matrix in the ordering $(\Psi_{11}, \dots, \Psi_{qq}, b)$ is $\mathcal{G}_c$ of (2.74): the coefficient of Ψ_{ii}^2 is $2\tau_i^{-2} = \Pi_{ii}$, that of $b\Psi_{ii}$ is $2\tau_i^{-2} = 2\nu_i$, and that of b^2 is η . $\square$

Proposition 48 (Explicit inverse). The matrix $\mathcal{G}$ of (2.71) is invertible, and $\mathcal{G}^{-1}$ is block diagonal with respect to the same partition, with

$$(\mathcal{G}^{-1})_{\Psi_{ij}\Psi_{ij}} = \frac{\tau_i\tau_j}{4} \quad (i < j), \quad (\mathcal{G}^{-1})_{C_{ki}C_{ki}} = \frac{\sigma_0^2\tau_i}{\sigma_i(\mathbf{W}_0)^2} = \left(\frac{1}{\sigma_0^2} - \frac{1}{\tau_i} \right)^{-1}, \quad (2.77)$$

these two groups being diagonal, and, on the coupled block,

$$(\mathcal{G}^{-1})_{bb} = \frac{2\sigma_0^4}{p-q}, \quad (\mathcal{G}^{-1})_{\Psi_{ii}b} = -\frac{\sigma_0^4}{p-q}, \quad (\mathcal{G}^{-1})_{\Psi_{ii}\Psi_{jj}} = \delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)}. \quad (2.78)$$

Proof. $\mathcal{G}$ is positive definite because G_{θ_0} is, by Theorem 7, and κ is a bijection; so it is invertible, and the inverse of a block diagonal matrix is block diagonal with the blocks inverted. The first two groups are diagonal by (2.74), and their inverses are the reciprocals, which is (2.77).

For the coupled block, $\Pi^{-1} = \text{diag} \left(\frac{\tau_1^2}{2}, \dots, \frac{\tau_q^2}{2} \right)$ and

$$\Pi^{-1}\boldsymbol{\nu} = \left(\frac{\tau_i^2}{2} \cdot \frac{1}{\tau_i^2} \right)_{i=1}^q = \frac{1}{2}\mathbf{1}_q, \quad (2.79)$$

so the Schur complement of Π in $\mathcal{G}_c$ is

$$\zeta := \eta - \boldsymbol{\nu}^T \Pi^{-1} \boldsymbol{\nu} = \eta - \frac{1}{2} \sum_{i=1}^q \frac{1}{\tau_i^2} = \frac{p-q}{2\sigma_0^4} \quad (2.80)$$

by (2.75); it is strictly positive because $q < p$, which is where the standing assumption $q < p$ enters the inversion. The standard block inversion formula

$$\begin{pmatrix} \Pi & \boldsymbol{\nu} \\ \boldsymbol{\nu}^T & \eta \end{pmatrix}^{-1} = \begin{pmatrix} \Pi^{-1} + \zeta^{-1} \Pi^{-1} \boldsymbol{\nu} \boldsymbol{\nu}^T \Pi^{-1} & -\zeta^{-1} \Pi^{-1} \boldsymbol{\nu} \\ -\zeta^{-1} \boldsymbol{\nu}^T \Pi^{-1} & \zeta^{-1} \end{pmatrix}$$

together with (2.79) and (2.80) gives $\zeta^{-1} = \frac{2\sigma_0^4}{p-q}$, then $-\zeta^{-1} (\Pi^{-1} \boldsymbol{\nu})_i = -\frac{2\sigma_0^4}{p-q} \cdot \frac{1}{2} = -\frac{\sigma_0^4}{p-q}$, and finally

$$(\Pi^{-1})_{ij} + \zeta^{-1} (\Pi^{-1} \boldsymbol{\nu})_i (\Pi^{-1} \boldsymbol{\nu})_j = \delta_{ij} \frac{\tau_i^2}{2} + \frac{2\sigma_0^4}{p-q} \cdot \frac{1}{2} \cdot \frac{1}{2} = \delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)},$$

which is (2.78). $\square$

Theorem 49 (Theorem 23 in block coordinates). Let $1 \leq q < p$ and $\theta_0 = [\mathbf{W}_0, \sigma_0^2] \in M$. Define, on the event $\mathcal{A}_n$ of (2.17), the aligned error blocks

$$\begin{aligned} B_n &:= U_0^T \left(\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0 \right) \mathbf{V}_0, & C_n &:= (U_0^\perp)^T \left(\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0 \right) \mathbf{V}_0, \\ \Psi_n &:= \frac{1}{2} (B_n D_0 + D_0 B_n^T), & b_n &:= \hat{\sigma}_n^2 - \sigma_0^2, \end{aligned} \quad (2.81)$$

with $\mathbf{Q}_n$ as in (2.18), and set $(\Psi_n, C_n, b_n) := (\mathbf{0}, \mathbf{0}, 0)$ off $\mathcal{A}_n$. Then

$$\sqrt{n}(\Psi_n, \mathbf{C}_n, b_n) \rightsquigarrow N(\mathbf{0}, \mathcal{G}^{-1}) \quad \text{in } \text{Sym}(q) \times \mathbb{R}^{(p-q) \times q} \times \mathbb{R}, \quad (2.82)$$

with $\mathcal{G}^{-1}$ given by Proposition 48. Explicitly, writing $\mathbf{Z} = (\mathbf{Z}_\Psi, \mathbf{Z}_C, Z_b)$ for the limit:

1. the $(p-q)q$ entries of $\mathbf{Z}_C$ are mutually independent, independent of $(\mathbf{Z}_\Psi, Z_b)$, and $\text{Var}((\mathbf{Z}_C)_{ki}) = \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2}$;
2. the $\frac{q(q-1)}{2}$ strictly upper triangular entries of $\mathbf{Z}_\Psi$ are mutually independent, independent of everything else, and $\text{Var}((\mathbf{Z}_\Psi)_{ij}) = \frac{\tau_i \tau_j}{4}$ for $i < j$;
3. the remaining $q+1$ coordinates $((\mathbf{Z}_\Psi)_{11}, \dots, (\mathbf{Z}_\Psi)_{qq}, Z_b)$ are jointly Gaussian with the covariances (2.78); in particular

$$\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) \rightsquigarrow N\left(0, \frac{2\sigma_0^4}{p-q}\right). \quad (2.83)$$

Proof. By (2.18) we have $\iota(\hat{\mathbf{u}}_n) = (\hat{\mathbf{W}}_n \mathbf{Q}_n - \mathbf{W}_0, \hat{\sigma}_n^2 - \sigma_0^2)$ on $\mathcal{A}_n$, so by (2.67) the triple $(\Psi_n, \mathbf{C}_n, b_n)$ of (2.81) is exactly $\kappa(\hat{\mathbf{u}}_n)$ there; and both vanish off $\mathcal{A}_n$ because $\hat{\mathbf{u}}_n = \mathbf{0}$ there by (2.17). Hence $(\Psi_n, \mathbf{C}_n, b_n) = \kappa(\hat{\mathbf{u}}_n)$ identically.

Let $\mathbf{K}$ denote the matrix of κ with respect to the g_{θ_0} -orthonormal basis $\mathbf{e}_1, \dots, \mathbf{e}_d$ of (2.19) on the source and the scalar coordinates (2.70) on the target. Comparing (2.71) with $\mathbf{G}_{jk} = G_{\theta_0}(\mathbf{e}_j, \mathbf{e}_k)$ gives $\mathbf{G} = \mathbf{K}^T \mathcal{G} \mathbf{K}$, so

$$\mathbf{K} \mathbf{G}^{-1} \mathbf{K}^T = \mathbf{K} (\mathbf{K}^T \mathcal{G} \mathbf{K})^{-1} \mathbf{K}^T = \mathbf{K} \mathbf{K}^{-1} \mathcal{G}^{-1} \mathbf{K}^{-T} \mathbf{K}^T = \mathcal{G}^{-1}. \quad (2.84)$$

By (2.26) we have $\sqrt{n}\hat{\mathbf{u}}_n \rightsquigarrow N(\mathbf{0}, \mathcal{G}^{-1})$ in the coordinates of that basis, so the continuous mapping theorem applied to the linear map $\mathbf{K}$ and the identity (2.84) give (2.82).

The three assertions are then read off Proposition 48: a Gaussian vector whose covariance matrix is block diagonal has independent blocks, and within the first two groups the blocks are diagonal, so those coordinates are mutually independent. Statement (2.83) is the (b, b) entry of (2.78) together with $b_n = \hat{\sigma}_n^2 - \sigma_0^2$. $\square$

The score and the influence function in coordinates

The same block decomposition makes the score (1.22) and the influence operator (2.23) completely explicit. Throughout, for $\mathbf{x} \in \mathbb{R}^p$ we write

$$\mathbf{y} := \mathbf{U}_0^T \mathbf{x} \in \mathbb{R}^q, \quad \mathbf{z} := (\mathbf{U}_0^\perp)^T \mathbf{x} \in \mathbb{R}^{p-q}, \quad (2.85)$$

so that, by part 1 of Lemma 45, when $\mathbf{X} \sim N_p(\mathbf{0}, \Sigma_0)$ the vectors $\mathbf{U}_0^T \mathbf{X}$ and $(\mathbf{U}_0^\perp)^T \mathbf{X}$ are independent, the first being $N_q(\mathbf{0}, \mathbf{T}_0)$ and the second $N_{p-q}(\mathbf{0}, \sigma_0^2 \mathbf{I}_{p-q})$.

Proposition 50 (Score and influence function in coordinates). Let $\tilde{\mathbf{s}}(\mathbf{x}) \in \mathbb{R}^d$ be the coordinate vector of the score in the scalar coordinates (2.70), defined by $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \kappa(\mathbf{h})$ for all $\mathbf{h} \in T_{\theta_0} M$; it is related to the vector $\mathbf{s}$ of (2.19) by $\mathbf{s} = \mathbf{K}^T \tilde{\mathbf{s}}$. Then

$$\tilde{\mathbf{s}}_{\Psi_{ii}} = \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2}, \quad \tilde{\mathbf{s}}_{\Psi_{ij}} = \frac{2\mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j} \quad (i < j), \quad \tilde{\mathbf{s}}_{C_{ki}} = \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i \mathbf{z}_k}{\sigma_0^2 \tau_i}, \quad (2.86)$$

$$\tilde{\mathbf{s}}_b = \frac{1}{2} \left[\sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} + \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{\sigma_0^4} \right]. \quad (2.87)$$

Moreover $\mathbb{E}[\tilde{\mathbf{s}}(\mathbf{X})] = \mathbf{0}$ and $\mathbb{E}[\tilde{\mathbf{s}}(\mathbf{X}) \tilde{\mathbf{s}}(\mathbf{X})^T] = \mathcal{G}$, the influence operator (2.23) has coordinate vector $\kappa(\text{IF}(\mathbf{x})) = \mathcal{G}^{-1} \tilde{\mathbf{s}}(\mathbf{x})$, and explicitly

$$\text{IF}_b(\mathbf{x}) = \frac{\mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x} - (p-q)\sigma_0^2}{p-q}, \quad \text{IF}_{C_{ki}}(\mathbf{x}) = \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)}, \quad (2.88)$$

$$\text{IF}_{\Psi_{ij}}(\mathbf{x}) = \frac{\mathbf{y}_i \mathbf{y}_j}{2} \quad (i < j), \quad \text{IF}_{\Psi_{ii}}(\mathbf{x}) = \frac{1}{2} [\mathbf{y}_i^2 - \tau_i - \text{IF}_b(\mathbf{x})]. \quad (2.89)$$

Proof. The relation $\mathbf{s} = \mathbf{K}^T \tilde{\mathbf{s}}$ is immediate from $\mathbf{s}(\mathbf{x})^T \mathbf{h} = s(\mathbf{x}; \theta_0)(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \mathbf{K} \mathbf{h}$, valid for all $\mathbf{h}$.

Abbreviate $\mathbf{E} := \Sigma_0^{-1} [\mathbf{x} \mathbf{x}^T - \Sigma_0] \Sigma_0^{-1}$, so that (1.22) reads $s(\mathbf{x}; \theta_0)(\mathbf{h}) = \frac{1}{2} \text{tr} \{ \mathbf{E} \mathbf{D}_0(\mathbf{h}) \}$. Since $\mathbf{J}_0^T \mathbf{x} = (\mathbf{y}^T, \mathbf{z}^T)^T$ by (2.85), part 1 of Lemma 45 gives

$$\mathbf{J}_0^T \mathbf{E} \mathbf{J}_0 = \begin{pmatrix} \mathbf{T}_0^{-1} (\mathbf{y} \mathbf{y}^T - \mathbf{T}_0) \mathbf{T}_0^{-1} & \sigma_0^{-2} \mathbf{T}_0^{-1} \mathbf{y} \mathbf{z}^T \\ \sigma_0^{-2} \mathbf{z} \mathbf{y}^T \mathbf{T}_0^{-1} & \sigma_0^{-4} (\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}) \end{pmatrix} =: \begin{pmatrix} \mathbf{E}_{11} & \mathbf{E}_{12} \\ \mathbf{E}_{12}^T & \mathbf{E}_{22} \end{pmatrix}. \quad (2.90)$$

Pairing (2.90) with (2.68), the two off-diagonal contributions to the trace are equal, because $\text{tr} (\mathbf{E}_{12}^T \mathbf{D}_0 \mathbf{C}^T) = \text{tr} ((\mathbf{E}_{12}^T \mathbf{D}_0 \mathbf{C}^T)^T) = \text{tr} (\mathbf{C} \mathbf{D}_0 \mathbf{E}_{12}) = \text{tr} (\mathbf{E}_{12} \mathbf{C} \mathbf{D}_0)$, and therefore

$$2 s(\mathbf{x}; \theta_0)(\mathbf{h}) = \text{tr} (\mathbf{E}_{11} [2\Psi + b_h \mathbf{I}_q]) + \frac{2}{\sigma_0^2} \mathbf{z}^T \mathbf{C} \mathbf{D}_0 \mathbf{T}_0^{-1} \mathbf{y} + b_h \text{tr} \mathbf{E}_{22}. \quad (2.91)$$

Now read off the coefficient of each scalar coordinate. The matrix Ψ is symmetric, so $\text{tr} (\mathbf{E}_{11} \Psi) = \sum_i (\mathbf{E}_{11})_{ii} \Psi_{ii} + 2 \sum_{i < j} (\mathbf{E}_{11})_{ij} \Psi_{ij}$, while $(\mathbf{E}_{11})_{ii} = \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2}$ and $(\mathbf{E}_{11})_{ij} = \frac{\mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j}$ for $i \neq j$; dividing the resulting $2 \text{tr} (\mathbf{E}_{11} \Psi)$ by the factor 2 in (2.91) gives the first two formulas of (2.86). The second term of (2.91) equals $\frac{2}{\sigma_0^2} \sum_{k,i} \mathbf{C}_{ki} \mathbf{z}_k \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i}{\tau_i}$, which after division by 2 gives the third formula of (2.86). The coefficient of b_h is $\frac{1}{2} [\text{tr} \mathbf{E}_{11} + \text{tr} \mathbf{E}_{22}]$, which is (2.87).

The identities $\mathbb{E} [\tilde{\mathbf{s}}] = \mathbf{0}$ and $\mathbb{E} [\tilde{\mathbf{s}} \tilde{\mathbf{s}}^T] = \mathcal{G}$ are Theorem 7 read through (2.71), since for all $\mathbf{h}_1, \mathbf{h}_2 \in T_{\theta_0} M$

$$\mathbb{E} [s(\mathbf{X}; \theta_0)(\mathbf{h}_1) s(\mathbf{X}; \theta_0)(\mathbf{h}_2)] = G_{\theta_0}(\mathbf{h}_1, \mathbf{h}_2) = \kappa(\mathbf{h}_1)^T \mathcal{G} \kappa(\mathbf{h}_2)$$

and κ is onto. They can also be checked directly from (2.86): with $\mathbf{y}$ and $\mathbf{z}$ independent centred Gaussians, $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ii}}^2] = \text{Var}(\mathbf{y}_i^2) / \tau_i^4 = 2 / \tau_i^2$, $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ij}}^2] = 4 / (\tau_i \tau_j)$, $\mathbb{E} [\tilde{\mathbf{s}}_{C_{ki}}^2] = \sigma_i(\mathbf{W}_0)^2 \tau_i \sigma_0^2 / (\sigma_0^4 \tau_i^2) = \sigma_i(\mathbf{W}_0)^2 / (\sigma_0^2 \tau_i)$, $\mathbb{E} [\tilde{\mathbf{s}}_{\Psi_{ii}} \tilde{\mathbf{s}}_b] = 1 / \tau_i^2$ and $\mathbb{E} [\tilde{\mathbf{s}}_b^2] = \eta$, every product containing an odd power of a coordinate of $\mathbf{z}$ having mean zero; these are exactly the entries (2.74).

By Definition 12 the vector $\text{IF}(\mathbf{x})$ of (2.23) is characterized by $G_{\theta_0}(\text{IF}(\mathbf{x}), \mathbf{h}) = s(\mathbf{x}; \theta_0)(\mathbf{h})$ for all $\mathbf{h}$, that is, by (2.71), by $\kappa(\text{IF}(\mathbf{x}))^T \mathcal{G} \kappa(\mathbf{h}) = \tilde{\mathbf{s}}(\mathbf{x})^T \kappa(\mathbf{h})$ for all $\mathbf{h}$; since κ is onto, $\mathcal{G} \kappa(\text{IF}(\mathbf{x})) = \tilde{\mathbf{s}}(\mathbf{x})$.

It remains to compute $\mathcal{G}^{-1} \tilde{\mathbf{s}}$ block by block. For $i < j$, by (2.77), $\text{IF}_{\Psi_{ij}} = \frac{\tau_i \tau_j}{4} \cdot \frac{2 \mathbf{y}_i \mathbf{y}_j}{\tau_i \tau_j} = \frac{\mathbf{y}_i \mathbf{y}_j}{2}$, and $\text{IF}_{C_{ki}} = \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2} \cdot \frac{\sigma_i(\mathbf{W}_0) \mathbf{y}_i \mathbf{z}_k}{\sigma_0^2 \tau_i} = \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)}$, which are the second formula of (2.88) and the first of (2.89). On the coupled block, (2.78) and (2.87) give

$$\begin{aligned} \text{IF}_b &= \frac{2\sigma_0^4}{p-q} \tilde{\mathbf{s}}_b - \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \tilde{\mathbf{s}}_{\Psi_{ii}} \\ &= \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} + \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{p-q} - \frac{\sigma_0^4}{p-q} \sum_{i=1}^q \frac{\mathbf{y}_i^2 - \tau_i}{\tau_i^2} = \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{p-q}, \end{aligned}$$

the $\mathbf{y}$ terms cancelling identically; since $\|\mathbf{z}\|^2 = \mathbf{x}^T \mathbf{U}_0^\perp (\mathbf{U}_0^\perp)^T \mathbf{x} = \mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x}$ by (2.65), this is the first formula of (2.88). Similarly

$$\begin{aligned} \text{IF}_{\Psi_{ii}} &= \sum_{j=1}^q \left[\delta_{ij} \frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)} \right] \tilde{\mathbf{s}}_{\Psi_{jj}} - \frac{\sigma_0^4}{p-q} \tilde{\mathbf{s}}_b \\ &= \frac{\mathbf{y}_i^2 - \tau_i}{2} + \frac{\sigma_0^4}{2(p-q)} \sum_{j=1}^q \frac{\mathbf{y}_j^2 - \tau_j}{\tau_j^2} - \frac{\sigma_0^4}{2(p-q)} \sum_{j=1}^q \frac{\mathbf{y}_j^2 - \tau_j}{\tau_j^2} - \frac{\|\mathbf{z}\|^2 - (p-q)\sigma_0^2}{2(p-q)}, \end{aligned}$$

which is the second formula of (2.89). □

Corollary 51 (The noise variance). The maximum likelihood estimator of the noise variance satisfies (2.83), and it is asymptotically linear with influence function

$$\text{IF}_{\sigma^2}(\mathbf{x}) = \frac{\mathbf{x}^T (\mathbf{I}_p - \mathbf{P}_0) \mathbf{x} - (p - q) \sigma_0^2}{p - q}, \quad \mathbb{E} [\text{IF}_{\sigma^2}(\mathbf{X})] = 0, \quad \mathbb{E} [\text{IF}_{\sigma^2}(\mathbf{X})^2] = \frac{2\sigma_0^4}{p - q}. \quad (2.92)$$

Consequently $\frac{2\sigma_0^4}{p-q}$ cannot be improved: by Theorem 35 the limit law of any locally regular estimator sequence of θ_0 is $N(\mathbf{0}, G_{\theta_0}^{-1}) * \Lambda$, so the noise variance component of such a sequence has limit law $N\left(0, \frac{2\sigma_0^4}{p-q}\right) * \Lambda_b$ for some probability measure Λ_b on $\mathbb{R}$.

Proof. The weak limit is (2.83). The influence function is the first formula of (2.88), because $b_n = \hat{\sigma}_n^2 - \sigma_0^2$ is the b coordinate of $\kappa(\hat{\mathbf{u}}_n)$ by (2.81), so that the b coordinate of (2.24) reads $\sqrt{n}(\hat{\sigma}_n^2 - \sigma_0^2) = n^{-\frac{1}{2}} \sum_{i=1}^n \text{IF}_b(\mathbf{X}_i) + o_{\mathbb{P}}(1)$. For the moments, $(\mathbf{U}_0^\perp)^T \mathbf{X} \sim N_{p-q}(\mathbf{0}, \sigma_0^2 \mathbf{I}_{p-q})$ gives $\mathbb{E} \|\mathbf{z}\|^2 = (p - q) \sigma_0^2$ and $\text{Var}(\|\mathbf{z}\|^2) = 2(p - q) \sigma_0^4$, whence $\mathbb{E} [\text{IF}_{\sigma^2}^2] = \frac{2(p-q)\sigma_0^4}{(p-q)^2} = \frac{2\sigma_0^4}{p-q}$, consistent with (2.83) as it must be, since the asymptotic covariance of an asymptotically linear sequence is the covariance of its influence function. For the last assertion, a linear map carries a convolution of two laws to the convolution of their images, so it suffices to push $L = N(\mathbf{0}, G_{\theta_0}^{-1}) * \Lambda$ forward by the linear map $\mathbf{h} \mapsto b_{\mathbf{h}}$, which by (2.84) sends $N(\mathbf{0}, G_{\theta_0}^{-1})$ to $N(0, (\mathcal{G}^{-1})_{bb}) = N\left(0, \frac{2\sigma_0^4}{p-q}\right)$ by (2.78). $\square$

Remark 52 (Where the model helps, and where it does not). Formulas (2.88)–(2.89) answer a question the asymptotic theory alone does not: in which directions is the constrained estimator $\hat{\Sigma}_n$ actually better than the sample covariance matrix $\mathbf{S}_n$? The answer is: in exactly one block, and the reason is Proposition 19.

By (2.58) the influence function of $\hat{\Sigma}_n$ is $\mathcal{D}_0 \circ \text{IF}$, whose blocks are, by (2.68), $2\text{IF}_{\Psi} + \text{IF}_b \mathbf{I}_q$, $\text{IF}_C \mathbf{D}_0$ and $\text{IF}_b \mathbf{I}_{p-q}$. By (2.88)–(2.89) these are

$$(2\text{IF}_{\Psi} + \text{IF}_b \mathbf{I}_q)_{ij} = \mathbf{y}_i \mathbf{y}_j - \delta_{ij} \tau_i, \quad (\text{IF}_C \mathbf{D}_0)_{ki} = \sigma_i(\mathbf{W}_0) \frac{\mathbf{y}_i \mathbf{z}_k}{\sigma_i(\mathbf{W}_0)} = \mathbf{z}_k \mathbf{y}_i, \quad \text{IF}_b \mathbf{I}_{p-q}, \quad (2.93)$$

the diagonal case of the first identity being exactly the cancellation of IF_b in the second formula of (2.89). Now the influence function of $\mathbf{S}_n$, read in the same basis (2.65), is $\mathbf{J}_0^T (\mathbf{x} \mathbf{x}^T - \Sigma_0) \mathbf{J}_0$, whose blocks are $\mathbf{y} \mathbf{y}^T - \mathbf{T}_0$, $\mathbf{z} \mathbf{y}^T$ and $\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}$. The first two coincide with (2.93): *to first order the maximum likelihood fit reproduces the sample covariance matrix in the signal block and in the signal–noise block, and gains nothing there.* Only the trailing block differs, where $\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q}$ is replaced by its average

$$\frac{\text{tr}(\mathbf{z} \mathbf{z}^T - \sigma_0^2 \mathbf{I}_{p-q})}{p - q} \mathbf{I}_{p-q} = \text{IF}_b \mathbf{I}_{p-q} :$$

the isotropy constraint of the model averages the $p - q$ noise directions, dividing the variance by $p - q$. This is the infinitesimal shadow of the exact identity (2.16), which says the same thing non-asymptotically: $\hat{\Sigma}_n$ differs from $\mathbf{S}_n$ only through the trailing eigenvalues, which it replaces by their mean. It also explains why the leading blocks cannot be improved: by (2.60) the model imposes no constraint there at all, the image of $\mathcal{D}_0$ being everything except for the isotropy of the trailing block.

Remark 53 (Cross-checks against classical principal component analysis). Three consistency checks, each of which recovers a classical formula, and none of which is used elsewhere.

1. Pooled noise eigenvalues. The variance $\frac{2\sigma_0^4}{p-q}$ of (2.83) is the classical asymptotic variance of the average of the $p - q$ sample eigenvalues attached to a population eigenvalue of multiplicity $p - q$. Indeed $\hat{\sigma}_n^2 = \frac{1}{p-q} \sum_{j>q} \lambda_j(\mathbf{S}_n)$ exactly, by (2.15), and

$$\sum_{j>q} \lambda_j(\mathbf{S}_n) = \text{tr}(\mathbf{S}_n) - \max \{ \text{tr}(\mathbf{P} \mathbf{S}_n) : \mathbf{P} \text{ an orthogonal projection of rank } q \}$$

by Ky Fan's variational principle. Write $\mathbf{P}_n$ for the maximizer, which is the projection onto the span of $\mathbf{v}_1, \dots, \mathbf{v}_q$. Since $\mathbf{P}_0$ is the maximizer at Σ_0 , by (2.13), we have on the one hand $\text{tr}(\mathbf{P}_n \mathbf{S}_n) \geq \text{tr}(\mathbf{P}_0 \mathbf{S}_n)$ and on the other

$$\text{tr}(\mathbf{P}_n \mathbf{S}_n) - \text{tr}(\mathbf{P}_0 \mathbf{S}_n) \leq \text{tr}((\mathbf{P}_n - \mathbf{P}_0)(\mathbf{S}_n - \Sigma_0)) \leq \|\mathbf{P}_n - \mathbf{P}_0\|_F \|\mathbf{S}_n - \Sigma_0\|_F,$$

because $\text{tr}(\mathbf{P}_n \Sigma_0) \leq \text{tr}(\mathbf{P}_0 \Sigma_0)$. The multivariate central limit theorem applied to the independent, identically distributed and square integrable matrices $\mathbf{X}_i \mathbf{X}_i^T$ gives $\|\mathbf{S}_n - \Sigma_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$, and the Davis–Kahan $\sin \Theta$ theorem, with the eigengap (2.13), gives $\|\mathbf{P}_n - \mathbf{P}_0\|_F = O_{\mathbb{P}}\left(n^{-\frac{1}{2}}\right)$ as well (see for example Chen et al., 2021, Chapter 2, or Kato, 1995, Chapter II). Hence $\mathbf{P}_n$ may be replaced by $\mathbf{P}_0$ at a cost of $O_{\mathbb{P}}(n^{-1})$, and the statistic that remains is $\frac{1}{p-q} \text{tr}((\mathbf{I}_p - \mathbf{P}_0) \mathbf{S}_n)$, whose centred and $\sqrt{n}$ -scaled version is exactly the average of the influence functions (2.92).

2. The leading block is not improved. By (2.78), and writing $\text{Var}(2(\mathbf{Z}_{\Psi})_{ii} + Z_b) = 4\text{Var}((\mathbf{Z}_{\Psi})_{ii}) + 4\text{Cov}((\mathbf{Z}_{\Psi})_{ii}, Z_b) + \text{Var}(Z_b)$,

$$\text{Var}(2(\mathbf{Z}_{\Psi})_{ii} + Z_b) = 4 \left(\frac{\tau_i^2}{2} + \frac{\sigma_0^4}{2(p-q)} \right) + 4 \left(-\frac{\sigma_0^4}{p-q} \right) + \frac{2\sigma_0^4}{p-q} = 2\tau_i^2 = \text{Var}\left((\mathbf{U}_0^T \mathbf{X})_i^2\right),$$

and likewise $\text{Var}(2(\mathbf{Z}_{\Psi})_{ij}) = \tau_i \tau_j = \text{Var}(\mathbf{y}_i \mathbf{y}_j)$ for $i < j$ — the sample covariance variances for those coordinates, as Remark 52 requires.

3. Anderson's eigenvector variances. Suppose the $\sigma_i(\mathbf{W}_0)$ are simple. By part 2 of Lemma 45, the component of the aligned loading error orthogonal to the column space of $\mathbf{W}_0$ is $\mathbf{U}_0^\perp \mathbf{C}_n \mathbf{V}_0^T$; since the i -th column of $\mathbf{W}_0 \mathbf{V}_0$ is $\sigma_i(\mathbf{W}_0)$ times the i -th column of $\mathbf{U}_0$, the tilt of the estimated i -th direction out of the signal subspace is $\mathbf{U}_0^\perp$ applied to the i -th column of $\mathbf{C}_n$, divided by $\sigma_i(\mathbf{W}_0)$, with limiting per-coordinate variance

$$\frac{1}{\sigma_i(\mathbf{W}_0)^2} \cdot \frac{\sigma_0^2 \tau_i}{\sigma_i(\mathbf{W}_0)^2} = \frac{\sigma_0^2 \tau_i}{(\tau_i - \sigma_0^2)^2} = \frac{\lambda_i(\Sigma_0) \lambda_j(\Sigma_0)}{(\lambda_i(\Sigma_0) - \lambda_j(\Sigma_0))^2} \quad \text{for any } j > q,$$

which is Anderson's formula for the noise directions. When the $\sigma_i(\mathbf{W}_0)$ are repeated this eigenvector-by-eigenvector reading breaks down — individual eigenvectors are then not asymptotically normal — but nothing in Propositions 47 and 48 is affected: no formula in this section divides by a difference of singular values, and the only inverted quantities are τ_i , σ_0^2 , $\sigma_i(\mathbf{W}_0)$ and $p - q$, all strictly positive under the standing assumptions $\text{rank}(\mathbf{W}_0) = q$, $\sigma_0^2 > 0$ and $q < p$. This is the promised quantitative form of the third comment of Remark 24.

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